

MA2108S Mathematical Analysis I (S)

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AY25/26 Sem 2

The objective of this course is to develop the learning capabilities and hone the problem solving skills of talented students at a mathematically deeper and more rigorous level. In addition to the classes of the regular course, one extra special hour each week will be devoted to solving challenging problems and studying some additional topics and those topics briefly mentioned in the regular course. The contents of this course will consist of those in the regular course (MA2108) and the following additional topics: conditions equivalent to the completeness axiom, rearrangement of series, trigonometric series.

Prerequisite

If undertaking an Undergraduate Degree THEN ((must have completed 1 of CS1231 / CS1231S / MA1100 / MA1100T at a grade of at least D) AND (must have completed 1 of MA1505 / MA1511 / MA1521 / MA2002 at a grade of at least D))

Preclusion

If undertaking an Undergraduate Degree THEN (must not have completed 1 of MA2108 / YSC3206 at a grade of at least D)

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1 Preliminaries

1.1 Sets

Notation:

- $\emptyset$: The empty set;
- $\mathbb{Z} = \{0, 1, -1, 2, -2, \dots\}$: The set of integers;
- $\mathbb{N} = \{1, 2, \dots\}$: The set of natural numbers;
- $\mathbb{Q} = \left\{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\right\}$: The set of rational numbers;
- $\mathbb{R}$: The set of real numbers;
- $\mathbb{C}$: The set of complex numbers.

A set can be defined by listing all elements, e.g., $A = \{0, 2, 5, -2\}$, or particular properties satisfied by its elements, e.g., $A = \{x : P(x)\}$.

Notation for intervals:

- $(a, b) := \{x \in \mathbb{R} : a < x < b\}$;
- $[a, b) := \{x \in \mathbb{R} : a \leq x < b\}$;
- $(a, +\infty) := \{x \in \mathbb{R} : x > a\}$;
- $(-\infty, a) := \{x \in \mathbb{R} : x < a\}$.

More notation:

- $x \in A$: Element x is in set A ;
- $x \notin A$: Element x is not in set A ;
- $A \subseteq B$: A is a subset of B (including $A = B$);
- $A \subset B$: A is a proper subset of B (excluding $A = B$);
- Fact: For any set A , $\emptyset \subseteq A$.

Operations of sets:

- The union of A and B is $A \cup B := \{x : x \in A \text{ or } x \in B\}$;
- The intersection of A and B is $A \cap B := \{x : x \in A \text{ and } x \in B\}$;
- The complement of B relative to A is $A \setminus B := \{x : x \in A \text{ and } x \notin B\}$.

Given sets $A_1, A_2, \dots, A_n, \dots$ (finite or infinite),

- $\bigcup_{i=1}^n A_i := \{x : \exists 1 \leq i \leq n \text{ such that } x \in A_i\}$;
- $\bigcup_{i=1}^{\infty} A_i := \{x : \exists i \geq 1 \text{ such that } x \in A_i\}$;
- $\bigcap_{i=1}^n A_i := \{x : x \in A_i \forall 1 \leq i \leq n\}$;
- $\bigcap_{i=1}^{\infty} A_i := \{x : x \in A_i \forall i \geq 1\}$.

Theorem 1.1.

- $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$;

$$\bullet A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

Definition 1.2. Given sets A and B , the *Cartesian product* of A and B is

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

More generally, we can consider the Cartesian product of finitely many sets $A_1 \times A_2 \times \cdots \times A_n$, and the Cartesian product of countably many sets $\prod_{i=1}^{\infty} A_i$.

Definition 1.3. Given two sets A and B , a *function* from A to B is a rule f that assigns every $a \in A$ to $f(a) \in B$. We denote $f: A \rightarrow B$.

Definition 1.4. For $f: A \rightarrow B$, A is called the *domain* of f .

For $A' \subseteq A$, the *image* of A' is defined as

$$f(A') := \{f(a) : a \in A'\}.$$

$f(A)$ is called the *range* of f . For $B' \subseteq B$, the *pre-image* of B' is defined as

$$f^{-1}(B') := \{a \in A : f(a) \in B'\}.$$

Definition 1.5.

- (1) $f: A \rightarrow B$ is called *injective* if $f(a) \neq f(a')$ if $a \neq a'$;
- (2) $f: A \rightarrow B$ is called *surjective* if $f(A) = B$;
- (3) $f: A \rightarrow B$ is called *bijective* if it is both injective and surjective.

Definition 1.6. If $f: A \rightarrow B$ is bijective, the *inverse* $f^{-1}: B \rightarrow A$ is defined as follows:

$$f^{-1}(b) = a \quad \text{if } f(a) = b.$$

Definition 1.7. Let $f: A \rightarrow B$, $g: B \rightarrow C$. The *composition* of f and g , $g \circ f: A \rightarrow C$, is defined as follows:

$$g \circ f(a) = g(f(a)) \quad \text{for } a \in A.$$

1.2 Mathematical Induction

Axiom 1.8 (Well-ordering Principle of $\mathbb{N}$). Any non-empty subset $S \subseteq \mathbb{N}$ has a minimal element; that is, there exists $m \in S$ such that $m \leq s$ for any $s \in S$.

Theorem 1.9 (Principle of Mathematical Induction). Suppose $S \subseteq \mathbb{N}$ satisfies: (1) $1 \in S$; (2) If $k \in S$, then $k + 1 \in S$. Then $S = \mathbb{N}$.

Proof. For a contradiction, we assume $S \neq \mathbb{N}$. Then $T = \mathbb{N} \setminus S \neq \emptyset$. By Well-ordering Principle, there exists $t_0 \in T$ such that $t_0 \leq t$ for all $t \in T$.

First note that $t \neq 1$. Then $t - 1 \in S$. However, by the property of S , if $t - 1 \in S$ then $t \in S$. That is a contradiction! This proves that $S = \mathbb{N}$. $\square$

Theorem 1.10 (Principle of Mathematical Induction in practice). Suppose we want to prove a statement $P(n)$ (depending on $n \in \mathbb{N}$) for every $n \in \mathbb{N}$. We only need to prove the following two statements:

- (1) $P(1)$ holds;
- (2) If $P(k)$ holds, then $P(k + 1)$ holds.

Proof. Define $S = \{n \in \mathbb{N} : P(n) \text{ holds}\}$. Then S has the following properties:

- (1) $1 \in S$
- (2) If $k \in S$ then $k + 1 \in S$

Then by the previous statement, $S = \mathbb{N}$. This implies $P(n)$ holds for all $n \in \mathbb{N}$. □

Example. Prove

$$\sum_{j=1}^n j^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{1}{6}n(n+1)(2n+1).$$

Proof. For the base case $n = 1$, $\sum_{j=1}^1 j^2 = 1^2 = 1$ and $\frac{1}{6}(1)(2)(3) = 1$. Hence, the statement holds for $n = 1$. Then, we assume the statement holds for some $n = k$, i.e.,

$$\sum_{j=1}^k j^2 = \frac{1}{6}k(k+1)(2k+1).$$

Our goal is to prove the statement for $n = k + 1$, i.e.,

$$\begin{aligned} \sum_{j=1}^{k+1} j^2 &= \frac{1}{6}(k+1)((k+1)+1)(2(k+1)+1) \\ &= \frac{1}{6}(k+1)(k+2)(2k+3). \end{aligned}$$

By inductive assumption, note that

$$\begin{aligned} \sum_{j=1}^{k+1} j^2 &= \sum_{j=1}^k j^2 + (k+1)^2 \\ &= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2 \\ &= (k+1) \left[\frac{1}{6}k(2k+1) + (k+1) \right] \\ &= \frac{1}{6}(k+1)(2k^2 + 7k + 6) \\ &= \frac{1}{6}(k+1)(k+2)(2k+3). \end{aligned}$$

By principle of induction, the proof is complete. □

Example. Prove

$$2^n \leq (n+1)!.$$

Proof. For the base case $n = 1$, $2^1 = 2$ and $(n+1)! = 2! = 2$. Since $2 \leq 2$, the statement holds for $n = 1$. Assume that $2^k \leq (k+1)!$ for some $n = k$. The goal is to show $2^{k+1} \leq (k+2)!$. Note that

$$(k+2)! = (k+2)(k+1)! \geq (k+2)2^k.$$

Since $k+2 \geq 2$, we have

$$(k+2)2^k \geq 2 \cdot 2^k = 2^{k+1}.$$

By principle of induction, the proof is complete. □

Example. Prove Bernoulli's inequality: for $x > -1$ and $n \in \mathbb{N}$,

$$(1+x)^n \geq 1+nx.$$

Proof. For the base case $n = 1$, LHS = $1 + x$, RHS = $1 + x$, so the statement holds. Assume that $(1 + x)^k \geq 1 + kx$ for some $n = k$. Our goal is to show $(1 + x)^{k+1} \geq 1 + (k + 1)x$. Note that

$$\begin{aligned} (1 + x)^{k+1} &= (1 + x)^k(1 + x) \\ &\geq (1 + kx)(1 + x) \\ &= 1 + (k + 1)x + kx^2 \\ &\geq 1 + (k + 1)x. \end{aligned}$$

By principle of induction, the proof is complete. $\square$

Theorem 1.11 (Principle of Mathematical Induction (2nd version)). *Suppose we want to prove a statement $P(n)$ (depending on $n \in \mathbb{N}$) for every $n \geq n_0$. We only need to prove the following two statements:*

- (1) $P(n_0)$ holds;
- (2) If $P(k)$ holds, then $P(k + 1)$ holds.

Example. Prove that for $n \in \mathbb{N}$ and $n \geq 3$,

$$2^n \geq 2n + 1.$$

Theorem 1.12 (Principle of Strong Induction). *Suppose we want to prove a statement $P(n)$ (depending on $n \in \mathbb{N}$) for every $n \in \mathbb{N}$. We only need to prove the following two statements:*

- (1) $P(1)$ holds;
- (2) If for any $k' \leq k$, $P(k')$ holds, then $P(k + 1)$ holds.

Example. Prove that for any $n \in \mathbb{N}$, n can be written as sum of distinct non-negative integer powers of 2, namely,

$$n = 2^{r_1} + 2^{r_2} + \cdots + 2^{r_l},$$

where $0 \leq r_1 < r_2 < \cdots < r_l$ are integers.

Proof. The base case where $n = 1$ is obviously true. Assume the statement holds for all $k \leq n$. We will show the statement holds for $n + 1$. Consider cases:

- Case 1: $n + 1$ is odd. Then $n + 1 = 2l + 1$ for some $l \leq n$, so $l = 2^{r_1} + \cdots + 2^{r_p}$. Thus,

$$n + 1 = 2(2^{r_1} + \cdots + 2^{r_p}) + 1 = 2^0 + 2^{r_1+1} + \cdots + 2^{r_p+1},$$

where $0 < r_1 + 1 < \cdots < r_p + 1$.

- Case 2: $n + 1$ is even. Then $n + 1 = 2l$ for some $l \leq n$, so $l = 2^{r_1} + \cdots + 2^{r_p}$. Thus,

$$n + 1 = 2^{r_1+1} + \cdots + 2^{r_p+1},$$

where $0 < r_1 + 1 < \cdots < r_p + 1$.

$\square$

1.3 Finite and Infinite Sets

Definition 1.13.

- (1) $\emptyset$ has 0 elements and is *finite*;
- (2) A non-empty set S is *finite* if there exists a maximal $n \in \mathbb{N}$, called the *cardinality* of S , denoted by $|S|$, such that S has n elements.

(3) A set S is *infinite* if it is not finite.

Remark. S is finite if there exists $n \in \mathbb{N}$ and a bijection $f: \{1, 2, \dots, n\} \rightarrow S$.

Theorem 1.14. For $n \in \mathbb{N}$, suppose we have n finite sets $S_1, \dots, S_n$. Then $S = \bigcup_{i=1}^n S_i$ is also finite, and

$$|S| = \sum_{i=1}^n |S_i|.$$

Moreover, the equality holds if and only if $S_i \cap S_j = \emptyset$ for any $1 \leq i, j \leq n$ with $i \neq j$.

Definition 1.15. An infinite set S is *countable* if there exists a bijection $f: \mathbb{N} \rightarrow S$.

Theorem 1.16. Let S be infinite and $S = \bigcup_{n=1}^{\infty} F_n$, where F_n are finite sets. Then S is countable.

Proof. We construct a map $f: S \rightarrow \mathbb{N}$ as follows. Define $S_n = F_1 \cup \dots \cup F_n$. Then $S_1 \subseteq S_2 \subseteq \dots$ and $\bigcup_{n=1}^{\infty} S_n = S$. We shall inductively construct bijections $f_n: S_n \rightarrow \{1, \dots, |S_n|\}$ such that f_{n+1} extends f_n . For $n = 1$, since S_1 is finite, fix a bijection $f_1: S_1 \rightarrow \{1, \dots, |S_1|\}$. Now suppose $f_n: S_n \rightarrow \{1, \dots, |S_n|\}$ has been defined. Consider $S_{n+1} = S_n \cup (S_{n+1} \setminus S_n)$, where $S_{n+1} \setminus S_n$ is finite. Fix a bijection $h_n: S_{n+1} \setminus S_n \rightarrow \{1, \dots, |S_{n+1} \setminus S_n|\}$. Then define $f_{n+1}: S_{n+1} \rightarrow \{1, \dots, |S_{n+1}|\}$ as follows:

$$f_{n+1}(s) = \begin{cases} f_n(s) & \text{if } s \in S_n, \\ h_n(s) + |S_n| & \text{if } s \in S_{n+1} \setminus S_n. \end{cases}$$

It is straightforward to check that f_{n+1} is a bijection and extends f_n .

Now define $f: S \rightarrow \mathbb{N}$ as follows: for $s \in S_n$,

$$f(s) = f_n(s).$$

This is well-defined because the maps are compatible (each f_{n+1} extends f_n).

First we show that f is injective. Suppose $f(s_1) = f(s_2)$. Choose n such that $s_1, s_2 \in S_n$. Then

$$f_n(s_1) = f(s_1) = f(s_2) = f_n(s_2).$$

Since f_n is injective, we have $s_1 = s_2$.

Next, we show that f is surjective. Let $m \in \mathbb{N}$. Since S is infinite and $S = \bigcup S_n$ with S_n increasing, the sizes $|S_n|$ are unbounded. Hence, there exists n such that $|S_n| \geq m$. Since f_n is a bijection onto $1, \dots, |S_n|$, there exists $s \in S_n$ such that $f_n(s) = m$. Thus, $f(s) = m$.

This proves that f is bijective. □

Corollary 1.17. For $n \in \mathbb{N}$, suppose $S_1, \dots, S_n$ are countable sets. Then $S = \bigcup_{i=1}^n S_i$ is countable.

Proof. Since each S_i is countable, we can enumerate its elements as

$$S_i = \{a_{i1}, a_{i2}, \dots, a_{in}, \dots\}.$$

For each i , define

$$F_i = \{a_{jk} : 1 \leq j \leq n, 1 \leq k \leq i\},$$

where we are going along the diagonals of the matrix (a_{ij}) . Each F_i is finite, and the sequence $\{F_i\}$ is increasing with $\bigcup_{i=1}^{\infty} F_i = S$. Thus S is a countable union of finite sets, and hence countable. □

Corollary 1.18. For any $j \in \mathbb{N}$, suppose S_j is countable. Then $S = \bigcup_{j=1}^{\infty} S_j$ is countable.

Proof. Since each set S_j is countable, enumerate S_j as

$$S_j = \{a_{j1}, a_{j2}, \dots, a_{jn}, \dots\}.$$

For each $i \in \mathbb{N}$, define

$$F_i = \{a_{mn} : m + n \leq i + 1\},$$

which goes along the diagonals. Then each F_i is finite, and $\{F_i\}$ is an increasing sequence with $\bigcup_{i=1}^{\infty} F_i = S$. Thus S is a countable union of finite sets, and therefore countable. $\square$

Example.

- Every infinite subset $S \subseteq \mathbb{N}$ is countable.

To see this, let $F_i = S \cap \{1, \dots, i\}$, which is finite. Then $S = \bigcup_{i=1}^{\infty} F_i$.

- $\mathbb{Z}$ is countable.

To see this, let $F_i = \{0, \pm i\}$. Then $\mathbb{Z} = \bigcup_{i=1}^{\infty} F_i$.

- $\mathbb{Q}$ is countable.

To see this, let $F_i = \left\{ \frac{p}{q} : |p| + |q| \leq i \right\}$, which is finite. Then $\mathbb{Q} = \bigcup_{i=1}^{\infty} F_i$.

- Let $\mathcal{F}$ be the set of finite subsets of $\mathbb{N}$. Then $\mathcal{F}$ is countable.

To see this, let $F_i = \{S \subseteq \mathbb{N} : S \subseteq \{1, \dots, i\}\}$, which is finite. For each finite $S \subseteq \mathbb{N}$, there exists a maximal element $s_m \in S$, so $S \subseteq \{1, \dots, s_m\}$. Thus, $S \in F_{s_m}$.

Theorem 1.19 (Schröder-Bernstein). *Let A and B be two sets. If there exist $f: A \rightarrow B$ and $g: B \rightarrow A$ such that both f and g are injective, then there exists a bijective map $h: A \rightarrow B$.*

Proof. Assume without loss of generality that A and B are disjoint. For any $a \in A$ or $b \in B$, we can form a unique two-sided sequence of elements that are alternately in A and B , by repeatedly applying f and g^{-1} to go from A to B , and g and f^{-1} to go from B to A (where defined; the inverses f^{-1} and g^{-1} are understood as partial functions.)

$$\dots \rightarrow f^{-1}(g^{-1}(a)) \rightarrow g^{-1}(a) \rightarrow a \rightarrow f(a) \rightarrow g(f(a)) \rightarrow \dots$$

For any particular a , this sequence may terminate to the left or not, at a point where f^{-1} or g^{-1} is not defined. By the fact that f and g are injective functions, each $a \in A$ and $b \in B$ is in exactly one such sequence to within identity: if an element occurs in two sequences, all elements to the left and to the right must be the same in both, by the definition of the sequences. Therefore, the sequences form a partition of the (disjoint) union of A and B . Hence, it suffices to produce a bijection between the elements of A and B in each of the sequences separately, as follows:

Call a sequence an *A-stopper* if it stops at an element of A , or a *B-stopper* if it stops at an element of B . Otherwise, call it *doubly infinite* if all the elements are distinct or cyclic if it repeats.

- For an *A-stopper*, the function f is a bijection between its elements in A and its elements in B .
- For a *B-stopper*, the function g is a bijection between its elements in B and its elements in A .
- For a doubly infinite sequence or a cyclic sequence, either f or g will do.

$\square$

Proposition 1.20. $\mathcal{P}(\mathbb{N})$ is uncountable.

Proof. Identify each subset $S \subseteq \mathbb{N}$ with its characteristic sequence $\{a_n\}$, where for each $n \in \mathbb{N}$,

$$a_n = \begin{cases} 1 & \text{if } n \in S, \\ 0 & \text{if } n \notin S. \end{cases}$$

For a contradiction, suppose that $\mathcal{P}(\mathbb{N})$ is countable. Then we can enumerate all subsets as

$$\mathcal{P}(\mathbb{N}) = \{S_1, S_2, \dots, S_n, \dots\}.$$

Let $\{a_{kn}\}$ denote the characteristic sequence of S_k , so that

$$a_{kn} = \begin{cases} 1 & \text{if } n \in S_k, \\ 0 & \text{if } n \notin S_k. \end{cases}$$

Thus each S_k is encoded by its characteristic sequence $(a_{k1}, a_{k2}, \dots)$. Define a new subset $S \subseteq \mathbb{N}$ by diagonalization:

$$S = \{n \in \mathbb{N} : a_{nn} = 0\}.$$

By construction, S differs from each S_n at least in the n -th element, namely, $n \in S \iff n \notin S_n$. Then $S \neq S_k$ for every k . This contradicts the assumption that $\{S_k\}$ lists all subsets of $\mathbb{N}$. Hence, $\mathcal{P}(\mathbb{N})$ is uncountable. $\square$

2 Real and Complex Number Systems

2.1 Fields

Definition 2.1. A *field* $(F, +, \times)$ is a non-empty set F with two operations $+$ and $\times$ satisfying the following properties:

- (A1) $x + y \in F$ for all $x, y \in F$.
- (A2) $x + y = y + x$ for all $x, y \in F$. (commutative law of addition)
- (A3) $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$ (associative law of addition)
- (A4) There exists $0 \in F$ such that $0 + x = x$ for all $x \in F$. (existence of additive identity)
- (A5) For each $x \in F$, there exists $-a \in F$ such that $a + (-a) = 0$. (existence of additive inverse)
- (M1) $xy \in F$ for all $x, y \in F$.
- (M2) $xy = yx$ for all $x, y \in F$. (commutative law of multiplication)
- (M3) $(xy)z = x(yz)$ for all $x, y, z \in F$. (associative law of multiplication)
- (M4) There exists $1 \in F$ such that $1x = x$ for all $x \in F$. (existence of multiplicative identity)
- (M5) For each $x \in F \setminus \{0\}$, there exists $\frac{1}{x} \in F$ such that $x\frac{1}{x} = 1$. (existence of multiplicative inverse)
- (D) $x(y + z) = xy + xz$ for all $x, y, z \in F$. (distributive law)

It is easy to see that the additive identity of any field is unique. Suppose a field has two additive identities 0 and $0'$. Then $0 + 0' = 0'$ and $0 + 0' = 0$, so $0 = 0'$.

Example. $\mathbb{Q}$ is a field; $\mathbb{N}$ and $\mathbb{Z}$ are not fields.

Proposition 2.2. $x^2 = 2$ does not have a rational solution, i.e., $\sqrt{2}$ is not rational.

Proof. For a contradiction, we assume that there exists $\frac{p}{q} \in \mathbb{Q}$, where $p, q \in \mathbb{N}$, $(p, q) = 1$, such that $\left(\frac{p}{q}\right)^2 = 2$. Then $p^2 = 2q^2$, so p^2 is even, so p is even. Then $p = 2m$ for some $m \in \mathbb{N}$. Thus $(2m)^2 = 2q^2$, or $2m^2 = q^2$, so q^2 is even, so q is even. Then $q = 2k$ for some $k \in \mathbb{N}$. This contradicts the assumption that $(p, q) = 1$. $\square$

2.2 Supremum property

Definition 2.3. Given an ordered set S , a subset $E \subseteq S$ is called *bounded from above* if there exists $\beta \in S$ such that

$$e \leq \beta \quad \text{for all } e \in E.$$

The number β is called an *upper bound* of E .

Definition 2.4. Given an ordered set S and a subset $E \subseteq S$ which is bounded from above, if $\beta \in S$ satisfies that:

- (1) β is an upper bound of E ;
- (2) for any $\beta' < \beta$, β' is not an upper bound of E ;

then β is called the *supremum* (or *least upper bound*) of E , denoted by $\beta = \sup E$.

Exercise. If the supremum of E exists, then it is unique.

Definition 2.5. Given an ordered set S , a subset $E \subseteq S$ is called *bounded from below* if there exists $\beta \in S$ such that

$$e \geq \beta \quad \text{for all } e \in E.$$

The number β is called a *lower bound* of E .

Definition 2.6. Given an ordered set S and a subset $E \subseteq S$ which is bounded from below, if $\beta \in S$ satisfies that:

- (1) β is a lower bound of E ;
- (2) for any $\beta' < \beta$, β' is not a lower bound of E ;

then β is called the *infimum* (or *greatest lower bound*) of E , denoted by $\beta = \inf E$.

Exercise. If the infimum of E exists, then it is unique.

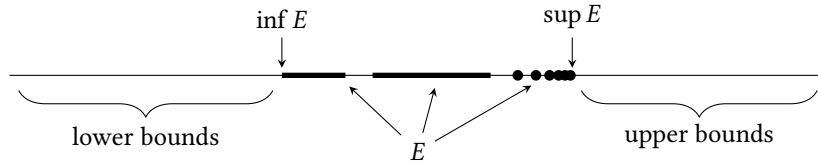


Figure 2.1: Definition of $\sup E$ and $\inf E$.

Definition 2.7. An ordered set S has the *supremum property* if for any subset $E \subseteq S$, if E is bounded from above, then $\sup E$ exists.

Example. $\mathbb{Q}$ does not have the supremum property; consider the set $\{x \in \mathbb{Q} : x > 0, x^2 < 2\}$.

2.3 Dedekind cuts

Definition 2.8. (A, A') is called a *Dedekind cut* if it satisfies

- (1) $A, A' \neq \emptyset$;
- (2) $A \cap A' = \emptyset, \mathbb{Q} = A \cup A'$;
- (3) For any $a \in A, a' \in A', a < a'$;
- (4) A does not admit a maximal element, namely, for any $a \in A$, there exists $b \in A$ such that $b > a$.

Remark. For each $r \in \mathbb{Q}$, define $A_r := \{x \in \mathbb{Q} : x < r\}$. Then the injection $r \mapsto A_r$ gives an embedding of $\mathbb{Q}$ into $\mathbb{R}$; we usually write $\mathbb{Q} \subseteq \mathbb{R}$.

Definition 2.9. The set of real numbers $\mathbb{R}$ is defined as

$$\mathbb{R} := \{(A, A') : (A, A') \text{ is a Dedekind cut}\}.$$

Definition 2.10. The order in $\mathbb{R}$ is defined as follows:

- $A < B$ if $A \subset B$;
- $A > B$ if $B \subset A$.

Proposition 2.11. The order on $\mathbb{R}$ defined as above is a total order, namely, for any $A, B \in \mathbb{R}$, exactly one of the following holds:

$$A = B, \quad A < B, \quad A > B.$$

Proof. Let $A, B \in \mathbb{R}$. Suppose $A \neq B$ and $A \not\subseteq B$. We want to show that $B \subset A$.

Since $A \not\subseteq B$, there exists $a \in A$ such that $a \notin B$. Thus, $a \in B'$.

Since B is downward closed, for every $b \in B$, we have $b < a \in A$, so $b \in A$. Hence, $B \subset A$. □

Theorem 2.12. $\mathbb{R}$ has the supremum property.

Proof. Let $C \subseteq \mathbb{R}$ be a subset bounded from above, i.e., there exists β such that $c \leq \beta$ for every $c \in C$.

For each $c \in C$, let $c = (A_c, A'_c)$ be the corresponding Dedekind cut. Take

$$A = \bigcup_{c \in C} A_c.$$

Claim: $A = \sup C$.

We first show that A is a Dedekind cut. It suffices to show (3). Let $a \in A$. Then $a \in A_c$ for some $c \in C$. Thus $a' \in A'_c \subseteq A$. Then (A, A') gives a real number denoted by $a \in \mathbb{R}$.

First, since $A_c \subseteq A$, $c \leq a$ for all $c \in C$. Thus a is an upper bound of C .

Now for any $\tilde{a} < a$, we want to show $\tilde{a}$ is not an upper bound. Write $\tilde{a} = (\tilde{A}, \tilde{A}')$, where $\tilde{A} \subset A$. If $\tilde{A}$ is an upper bound, then $\tilde{A} \supseteq A_c$ for all $c \in C$, so

$$\tilde{A} \supseteq \bigcup_{c \in C} A_c = A$$

which contradicts $\tilde{A} \subset A$. □

Definition 2.13. For $m_1 = (M_1, M'_1)$ and $m_2 = (M_2, M'_2)$ in $\mathbb{R}$, define addition as

$$m_1 + m_2 = (M_3, M'_3)$$

where $M_3 = M_1 + M_2 := \{a_1 + a_2 : a_1 \in M_1, a_2 \in M_2\}$.

Exercise. Prove that (M_3, M'_3) is a Dedekind cut.

For $m = (M, M')$, the inverse of m is defined as follows: If $m \in \mathbb{Q}$, then $-m$ is defined as in $\mathbb{Q}$. If $m \notin \mathbb{Q}$, then

$$-m = (-M', -M)$$

where $-M' := \{-m' : m' \in M'\}$.

Exercise. Prove that $(-M', -M)$ is a Dedekind cut.

Lemma 2.14. $m + (-m) = 0$.

Definition 2.15. For $m_1 = (M_1, M'_1)$ and $m_2 = (M_2, M'_2)$ in $\mathbb{R}$, define multiplication

$$m_1 \cdot m_2 = m_3 = (M_3, M'_3)$$

as follows:

- If $m_1 = 0$ or $m_2 = 0$, then $m_3 = 0$.
- If $m_1 > 0$ and $m_2 > 0$, then

$$M_3 = \{xy : x, y \geq 0, x \in M_1, y \in M_2\} \cup \{x : x < 0\}.$$

- If $m_1 > 0$ and $m_2 < 0$, then $m_3 = -(m_1 \cdot (-m_2))$.
- If $m_1 < 0$ and $m_2 > 0$, then $m_3 = -((-m_1) \cdot m_2)$.
- If $m_1 < 0$ and $m_2 < 0$, then $m_3 = (-m_1) \cdot (-m_2)$.

For $m = (M, M') \neq 0$, define the reciprocal

$$\frac{1}{m} = (B, B')$$

as follows:

- If $m > 0$, then $B' = \{\frac{1}{x} : x > 0, x \in M\}$.
- If $m < 0$, then $\frac{1}{m} = -\frac{1}{-m}$.

Lemma 2.16. $m \cdot \frac{1}{m} = 1$.

Lemma 2.17. For $n \in \mathbb{N}$, $x \in \mathbb{R}$, $n \cdot x = x + x + \cdots + x$ (sum of n copies of x).

Proposition 2.18. $(\mathbb{R}, +, \cdot)$ is a field.

Proposition 2.19. The following inequalities hold for $\mathbb{R}$:

- If $a > (\geq)b$ and $b > (\geq)c$, then $a > (\geq)c$.
- If $a \geq b$ and $c \geq d$, then $a + c \geq b + d$, where equality holds if and only if $a = b$ and $c = d$.
- If both a, b are positive (or negative), then $ab > 0$ and $a/b > 0$.
- If one of a, b is positive and the other one is negative, then $ab < 0$ and $a/b < 0$.
- If $a > (\geq)b$ and $c > 0$, then $ac > (\geq)bc$.
- If $a > (\geq)b$ and $c < 0$, then $ac < (\leq)bc$.
- If $a \geq b > 0$ and $c \geq d > 0$, then $ac \geq bd > 0$.
- If $a > b > 0$, then $\frac{1}{b} > \frac{1}{a} > 0$.

Corollary 2.20 (Infimum property). For any $S \subseteq \mathbb{R}$ bounded from below, $\inf S$ exists.

2.4 Archimedean property

The set $\mathbb{Q}$ is an extension of $\mathbb{N}$, and $\mathbb{R}$ in turn is an extension of $\mathbb{Q}$. The next few results indicate how $\mathbb{N}$ and $\mathbb{Q}$ sit inside of $\mathbb{R}$.

Proposition 2.21 (Archimedean property). For any $x, y > 0$, there exists $n \in \mathbb{N}$ such that $nx > y$.

Proof. Note that $nx > y \iff n > \frac{y}{x}$, which means that $\mathbb{N}$ is bounded above in $\mathbb{R}$. For a contradiction, assume that $\mathbb{N}$ is bounded above in $\mathbb{R}$. By the supremum property, $\beta = \sup \mathbb{N}$ exists. Then $\beta - 1$ is not an upper bound of $\mathbb{N}$. This means that there exists $n \in \mathbb{N}$ such that $\beta - 1 < n$, or $n + 1 > \beta$. This contradicts the fact that β is an upper bound of $\mathbb{N}$. $\square$

Corollary 2.22. For any $c \in \mathbb{R}$, there exists a unique integer, denoted by $\lfloor c \rfloor \in \mathbb{Z}$, such that

$$\lfloor c \rfloor \leq c \leq \lfloor c \rfloor + 1.$$

Proof. The case where $c = 0$ is trivial; take $\lfloor c \rfloor = 0$. Thus, assume $c \neq 0$. Consider the following cases:

Case 1: $c > 0$. By Archimedean property, there exists $n \in \mathbb{N}$ such that $c < n$. Define $S = \{n \in \mathbb{N} : n > c\}$. By well-ordering, S has a minimal element $m \in S$. By minimality, $m - 1 \leq c$. Then $\lfloor c \rfloor = m - 1$.

Case 2: $c < 0$. Then $-c > 0$. By Case 1, there exists $m \in \mathbb{N}$ such that $m - 1 \leq -c < m$, so

$$-m < c \leq -(m - 1).$$

If $c = -(m - 1)$, take $\lfloor c \rfloor = -(m - 1)$.

If $c < -(m - 1)$, take $\lfloor c \rfloor = -m$.

$\square$

Corollary 2.23. For any $x > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < x$.

Proof. Note that $\frac{1}{n} < x \iff n > \frac{1}{x}$. Then apply the Archimedean property. $\square$

Corollary 2.24. $\mathbb{Q}$ is dense in $\mathbb{R}$, namely, for any $a < b$, there exists $r \in \mathbb{Q}$ such that $a < r < b$.

Proof. We consider three cases based on the signs of a and b : (1) $0 \leq a < b$, (2) $a < 0 < b$, and (3) $a < b \leq 0$. We first show that Case 1 holds. Our goal is to find $m, n \in \mathbb{N}$ such that

$$a < \frac{m}{n} < b. \quad (1)$$

The first step is to choose the denominator n large enough so that consecutive increments of size $1/n$ are too close together to “step over” the interval (a, b) . By the Archimedean property, choose $n \in \mathbb{N}$ large enough such that

$$\frac{1}{n} < b - a. \quad (2)$$

Inequality (1) (which we are trying to prove) is equivalent to $na < m < nb$. With n already chosen, the idea now is to choose m to be the smallest integer greater than na , i.e.,

$$m - 1 \stackrel{(3)}{\leq} na \stackrel{(4)}{<} m.$$

Evidently $a < \frac{m}{n}$, which is half of the battle. Keeping in mind that inequality (2) is equivalent to $a < b - \frac{1}{n}$, we can use (3) to write

$$m \leq na + 1 < n\left(b - \frac{1}{n}\right) + 1 = nb.$$

Because $m < nb$ implies $\frac{m}{n} < b$, we have $a < \frac{m}{n} < b$, as desired.

For Case 2, take $r = 0$.

For Case 3, then $0 \leq -b < a$. By Case 1, there exists $r \in \mathbb{Q}$ such that $-b < r < -a$, so $a < -r < b$. $\square$

Corollary 2.25. The irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$.

Proof. By density of $\mathbb{Q}$ in $\mathbb{R}$, choose $r \in \mathbb{Q}$ such that $\frac{a}{\sqrt{2}} < r < \frac{b}{\sqrt{2}}$. Then $a < r\sqrt{2} < b$, and $r\sqrt{2}$ is irrational. $\square$

The next result shows the existence of square roots.

Theorem 2.26. There exists $x \in \mathbb{R}$ such that $x^2 = 2$, i.e., $\sqrt{2} \in \mathbb{R}$.

Proof. Consider the set

$$M = \{x \in \mathbb{R} : x > 0, x^2 \leq 2\}.$$

Since M is non-empty ($1 \in M$) and bounded above by 2, by the supremum property, $\beta = \sup M$ exists.

Claim: $\beta^2 = 2$.

We first show $\beta^2 \geq 2$. Suppose, towards a contradiction, that $\beta^2 < 2$. To obtain a contradiction, we want to find some $\varepsilon > 0$ such that $\beta^2 < (\beta + \varepsilon)^2 < 2$.

Let $c = 2 - \beta^2 > 0$. Choose $\varepsilon = \min \left\{ \frac{c}{8\beta}, \beta \right\}$. Then

$$\begin{aligned} (\beta + \varepsilon)^2 &= \beta^2 + 2\beta\varepsilon + \varepsilon^2 \\ &\leq \beta^2 + 2\beta\varepsilon + \beta\varepsilon \\ &= \beta^2 + 3\beta\varepsilon \\ &< \beta^2 + 3\beta \cdot \frac{c}{8\beta} \\ &= \beta^2 + \frac{3}{8}c < \beta^2 + c < 2. \end{aligned}$$

Thus, $\beta + \varepsilon \in M$. This contradicts the fact that β is an upper bound of M . Hence, $\beta^2 \geq 2$.

We then prove $\beta^2 \leq 2$. For a contradiction, suppose that $\beta^2 > 2$. Let $c = \beta^2 - 2$. Choose $\varepsilon = \frac{c}{4\beta}$. Then

$$\begin{aligned} (\beta - \varepsilon)^2 &= \beta^2 - 2\beta\varepsilon + \varepsilon^2 \\ &> \beta^2 - 2\beta\varepsilon \\ &> \beta^2 - 2\beta \cdot \frac{c}{4\beta} \\ &= \beta^2 - \frac{c}{2} > \beta^2 - c = 2. \end{aligned}$$

This contradicts the fact that β is the least upper bound of M . □

A small modification of this proof can be made to show that $\sqrt{x}$ exists for any $x \geq 0$. Using the binomial expansion, one can show that $\sqrt[m]{x}$ exists for arbitrary values of $m \in \mathbb{N}$.

2.5 Nested intervals

Theorem 2.27 (Nested interval theorem). *Let $\{I_n = [a_n, b_n] : n \in \mathbb{N}\}$ be a nested decreasing sequence of closed intervals. Then there exists $x \in \mathbb{R}$ such that*

$$x \in \bigcap_{n=1}^{\infty} I_n.$$



Figure 2.2: Nested interval theorem.

Proof. First, we consider $S = \{a_n : n \in \mathbb{N}\}$, the set of left endpoints. Note that S is bounded from above. In fact, b_m is an upper bound of S for all $m \in \mathbb{N}$; to see this, note that for all $m, n \in \mathbb{N}$,

$$a_m \leq a_{m+n} \leq b_{m+n} \leq b_n.$$

By the supremum property, $\sup S = \alpha$ exists.

Claim: $\alpha \in \bigcap_{n=1}^{\infty} I_n$.

Let $n \in \mathbb{N}$. We will show $\alpha \in I_n$, i.e., $a_n \leq \alpha \leq b_n$.

Since α is an upper bound for S , we have $a_n \leq \alpha$.

Since each b_n is an upper bound of S , and α is the *least* upper bound, we have $\alpha \leq b_n$. □

Remark.

- Here closed intervals cannot be replaced with open intervals, e.g., $I_n = (0, 1/n)$.
- The intersection can be a singleton, e.g. $I_n = [0, 1/n]$.
- The intersection can be a closed interval, e.g. $I_n = [-1, 1/n]$.
- Denote $T = \{b_n : n \in \mathbb{N}\}$, the set of right endpoints, and $\beta = \inf T$. By a similar argument, one can show that $\beta \geq \alpha$ and

$$\bigcap_{n=1}^{\infty} I_n = [\alpha, \beta].$$

Here if $\alpha = \beta$, then $[\alpha, \beta] = \{\alpha\}$.

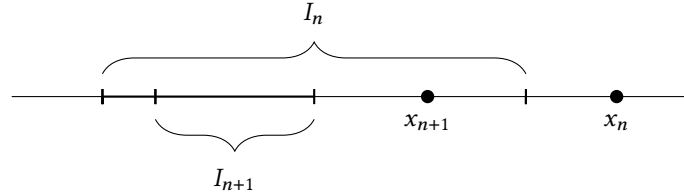
Theorem 2.28. $\mathbb{R}$ is uncountable.

Proof. For a contradiction, assume that $\mathbb{R}$ is countable. Then we can list out $\mathbb{R}$ as

$$\mathbb{R} = \{x_1, x_2, \dots, x_n, \dots\}.$$

The idea is to construct a sequence of nested intervals such that each interval avoids one more point than the previous one. Then use the nested interval theorem (2.27) to produce a real number not in the list.

Let I_1 be a closed interval such that $x_1 \notin I_1$. Next, let $I_2 \subseteq I_1$ be a closed interval such that $x_2 \notin I_2$. In general, suppose we have constructed I_n such that $x_1, \dots, x_n \notin I_n$. Then let $I_{n+1} \subseteq I_n$ be a closed interval such that $x_{n+1} \notin I_{n+1}$. Thus, we have constructed a sequence of nested intervals $\{I_n\}$.



By the nested interval property, there exists $x \in \mathbb{R}$ such that $x \in \bigcap_{n=1}^{\infty} I_n$.

We claim that $x \neq x_n$ for all $n \in \mathbb{N}$. For a contradiction, suppose $x = x_n$ for some $n \in \mathbb{N}$. But $x_n \notin I_n$ and $x \in I_n$, a contradiction. $\square$

2.6 Important inequalities

Lemma 2.29 (Bernoulli's inequality). For $x > -1$, $n \in \mathbb{N}$, we have

$$(1+x)^n \geq 1+nx. \quad (1)$$

Lemma 2.30 (AM-GM inequality). For $a_1, a_2, \dots, a_n \geq 0$,

$$\frac{a_1 + \dots + a_n}{n} \geq \sqrt[n]{a_1 \dots a_n}, \quad (2)$$

where equality holds if and only if $a_1 = a_2 = \dots = a_n$.

Corollary 2.31 (GM-HM inequality). For $a_1, a_2, \dots, a_n > 0$,

$$\sqrt[n]{a_1 \dots a_n} \geq \frac{n}{\frac{1}{a_1} + \dots + \frac{1}{a_n}}, \quad (3)$$

where equality holds if and only if $a_1 = a_2 = \dots = a_n$.

Definition 2.32. Let $x \in \mathbb{R}$. The *absolute value* of x is

$$|x| = \begin{cases} x & x \geq 0, \\ -x & x < 0. \end{cases}$$

Lemma 2.33 (Triangle inequality). For $a, b \in \mathbb{R}$,

$$||a| - |b|| \leq |a+b| \leq |a| + |b|. \quad (4)$$

Corollary 2.34. For $a_1, \dots, a_n \in \mathbb{R}$,

$$|a_1 + \dots + a_n| \leq |a_1| + \dots + |a_n|.$$

Lemma 2.35 (Cauchy-Schwarz inequality). For $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n \in \mathbb{R}$,

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right), \quad (5)$$

where equality holds if and only if there exists $k \in \mathbb{R}$ such that $a_i = kb_i$ for $i = 1, \dots, n$.

Example. For $x, y, z > 0$, prove that

$$(x + y)(y + z)(z + x) \geq 8xyz.$$

Solution. By AM-GM, $x + y \geq 2\sqrt{xy}$, $y + z \geq 2\sqrt{yz}$, and $z + x \geq 2\sqrt{zx}$. Taking the product of the three inequalities, we get $(x + y)(y + z)(z + x) \geq 8xyz$. $\square$

Example. For $x, y \in \mathbb{R}$, prove that

$$x^2 + y^2 + 1 \geq xy + x + y.$$

Solution. Similar to the previous example. More generally, we can show that

$$x^2 + y^2 + z^2 \geq xy + yz + zx.$$

$\square$

Example. For $a, b, c > 0$, prove that

$$\frac{a^3}{b^2} + \frac{b^3}{c^2} + \frac{c^3}{a^2} \geq a + b + c.$$

Solution. The idea is to cancel out the denominator by adding two b 's to cancel out the squared term b^2 :

$$\begin{aligned} \frac{a^3}{b^2} + b + b &\geq 3a, \\ \frac{b^3}{c^2} + c + c &\geq 3b, \\ \frac{c^3}{a^2} + a + a &\geq 3c. \end{aligned}$$

Then take the product of the inequalities. $\square$

Example. For $a, b, c > 0$, prove that

$$\frac{1}{a^3 + b^3 + abc} + \frac{1}{b^3 + c^3 + abc} + \frac{1}{c^3 + a^3 + abc} \leq \frac{1}{abc}.$$

Solution. By AM-GM,

$$\begin{aligned} a^3 + b^3 + b^3 &\geq 3ab^2 \\ b^3 + a^3 + a^3 &\geq 3ab^2 \end{aligned}$$

so

$$a^3 + b^3 \geq ab^2 + a^2b.$$

Hence,

$$\begin{aligned} a^3 + b^3 + abc &\geq ab^2 + a^2b + abc \\ &= ab(a + b + c) \end{aligned}$$

so

$$\frac{1}{a^3 + b^3 + abc} \leq \frac{1}{ab(a + b + c)}.$$

Similarly,

$$\frac{1}{b^3 + c^3 + abc} \leq \frac{1}{bc(a + b + c)}$$

$$\frac{1}{c^3 + a^3 + abc} \leq \frac{1}{ca(a + b + c)}$$

Then add up all three inequalities. □

2.7 Complex numbers

Definition 2.36. Let $i := \sqrt{-1}$, i.e., $i^2 = -1$. Define

$$\mathbb{C} := \{a + bi : a, b \in \mathbb{R}\}.$$

For a complex number $z = a + bi$, we call a the *real part*, b the *imaginary part*.

Definition 2.37. Let $z_1 = a_1 + b_1i$, $z_2 = a_2 + b_2i$.

- $z_1 + z_2 = (a_1 + a_2) + (b_1 + b_2)i$
- $z_1 z_2 = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i$
- $\bar{z}_1 = a_1 - b_1 i$
- $\|z_1\| = \sqrt{z_1 \bar{z}_1} = \sqrt{a_1^2 + b_1^2}$.

Proposition 2.38. $\mathbb{C}$ is a field.

Facts: For $z = a + bi$,

- $-z = -a - bi$
- if $z \neq 0$, $\frac{1}{z} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i$

3 Sequences and Series

3.1 Limits of sequences

Definition 3.1. A *sequence* is a function $a: \mathbb{N} \rightarrow \mathbb{R}$.

Notation. We usually denote a sequence by a countable collection of ordered numbers

$$\{a_n : n \in \mathbb{N}\}$$

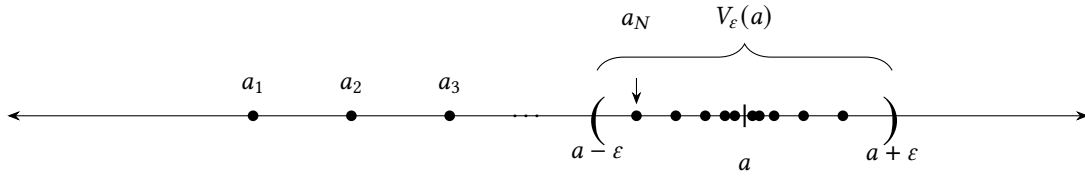
where a_n denotes the image of n .

Definition 3.2. We say a sequence $\{a_n\}_{n \in \mathbb{N}}$ *converges* to $A \in \mathbb{R}$ as $n \rightarrow \infty$, or A is the *limit* of the sequence $\{a_n\}$, if the following holds:

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad |a_n - A| < \varepsilon.$$

If a sequence has a limit, we say the sequence is *convergent*. Otherwise, we say it is *divergent*.

That is, all but finitely terms of the sequence lie in an ε -neighbourhood centered at A .



Proposition 3.3. If a sequence converges, then its limit is unique.

Proof. Let $\{a_n\}$ be a sequence such that $a_n \rightarrow A_1$ and $a_n \rightarrow A_2$. Let $\varepsilon > 0$. There exist $N_1, N_2 \in \mathbb{N}$ such that

$$\begin{aligned} |a_n - A_1| &< \frac{\varepsilon}{2} \quad \text{for all } n \geq N_1, \\ |a_n - A_2| &< \frac{\varepsilon}{2} \quad \text{for all } n \geq N_2. \end{aligned}$$

Take $N = \max\{N_1, N_2\}$. Then for any $n \geq N$, by triangle inequality,

$$|A_1 - A_2| \leq |A_1 - a_n| + |a_n - A_2| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since this inequality holds for all $\varepsilon > 0$, we must have $|A_1 - A_2| = 0$, so $A_1 = A_2$. □

Notation. If A is the limit of $\{a_n\}$, we will denote

$$\lim_{n \rightarrow \infty} a_n = A, \quad \text{or} \quad a_n \rightarrow A \text{ as } n \rightarrow \infty.$$

Example. $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Proof. Let $\varepsilon > 0$ be given. Note that $\left| \frac{1}{n} - 0 \right| = \frac{1}{n}$, so

$$\left| \frac{1}{n} - 0 \right| < \varepsilon \iff \frac{1}{n} < \varepsilon \iff n > \frac{1}{\varepsilon}.$$

By the Archimedean property, there exists $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon}$. Then for all $n \geq N$, $n > \frac{1}{\varepsilon}$. □

Example. $a_n = (-1)^n$, a_n is divergent.

Proof. For a contradiction, suppose $\{a_n\}$ is convergent; let $\lim_{n \rightarrow \infty} a_n = A$. Then

$$\forall \varepsilon > 0, \quad N \in \mathbb{N}, \quad \forall n \geq N, \quad |a_n - A| < \varepsilon.$$

In particular, for $\varepsilon = \frac{1}{2}$, take $n = 2N$ (some even number); then $a_n = 1$, so $|1 - A| < \frac{1}{2}$. Take $n = 2N + 1$ (some odd number), then $a_n = -1$, so $|-1 - A| < \frac{1}{2}$. By triangle inequality,

$$2 = |(1 - A) + (-1 - A)| \leq |1 - A| + |-1 - A| < \frac{1}{2} + \frac{1}{2} = 1.$$

This yields the desired contradiction. $\square$

Example. $\lim_{n \rightarrow \infty} \sqrt[n]{2} = 1$.

Proof. Let $\varepsilon > 0$ be given. Note that $\sqrt[n]{2} > 1$, so

$$\left| \sqrt[n]{2} - 1 \right| < \varepsilon \iff \sqrt[n]{2} - 1 < \varepsilon \iff \sqrt[n]{2} < 1 + \varepsilon \iff 2 < (1 + \varepsilon)^n.$$

By Bernoulli's inequality, $(1 + \varepsilon)^n > 1 + n\varepsilon$. Thus, it suffices to show $2 < 1 + n\varepsilon \iff n\varepsilon > 1 \iff n > \frac{1}{\varepsilon}$.

By the Archimedean property, there exists $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon}$. Then for $n \geq N$, $n > \frac{1}{\varepsilon}$. $\square$

Proposition 3.4. For a sequence $\{a_n\}$, the following statements are equivalent:

- (1) $\lim_{n \rightarrow \infty} a_n = A$;
- (2) $\lim_{n \rightarrow \infty} (a_n - A) = 0$;
- (3) $\lim_{n \rightarrow \infty} |a_n - A| = 0$.

3.2 Properties of limits

Definition 3.5. A sequence $\{a_n\}$ is *bounded* if there exists $C > 0$ such that

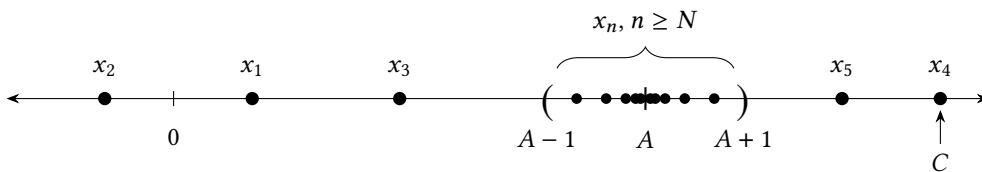
$$|a_n| \leq C \quad \text{for all } n \in \mathbb{N}.$$

Proposition 3.6. Every convergent sequence is bounded.

Proof. Let A be the limit of $\{a_n\}$. This means that given a particular value of ε , say $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that $|a_n - A| < 1$ for all $n \geq N$. By triangle inequality,

$$\begin{aligned} |a_n| &= |(a_n - A) + A| \\ &\leq |a_n - A| + |A| < 1 + |A| \end{aligned}$$

for all $n \geq N$.



We still need to worry (slightly) about the terms in the sequence that come before the N -th term. Since there are only a finitely many of them, let

$$C = \max \{|a_1|, \dots, |a_{N-1}|, |A| + 1\}.$$

Then $|a_n| \leq C$ for all $n \in \mathbb{N}$, as desired. $\square$

Remark. Boundedness is a necessary condition for convergence, and every unbounded sequence diverges. However, the converse is false; consider $a_n = (-1)^n$.

The next result gives us a lower bound for sequences that do not converge to 0.

Proposition 3.7. *Suppose $\{a_n\}$ is convergent, $a_n \neq 0$, $\lim_{n \rightarrow \infty} a_n \neq 0$. Then there exists $C > 0$ such that*

$$|a_n| \geq C \quad \text{for all } n \in \mathbb{N}.$$

Proof. Suppose $\lim_{n \rightarrow \infty} a_n = A \neq 0$. Take $\varepsilon = \frac{|A|}{2}$. There exists $N \in \mathbb{N}$ such that

$$|a_n - A| < \frac{|A|}{2} \quad \text{for all } n \geq N.$$

By triangle inequality,

$$\begin{aligned} |a_n| &= |(a_n - A) + A| \\ &\geq ||a_n - A| - |A|| \\ &= |A| - |a_n - A| \\ &> |A| - \frac{|A|}{2} = \frac{|A|}{2}. \end{aligned}$$

Take $C = \min \left\{ |a_1|, \dots, |a_{N-1}|, \frac{|A|}{2} \right\}$. Then $|a_n| \geq C$ for all $n \in \mathbb{N}$. □

The next result states that limits of convergent sequences preserve (non-strict) inequalities.

Theorem 3.8 (Order preserving). *Suppose $\{a_n\}$ and $\{b_n\}$ are convergent, and $a_n \geq b_n$. Then*

$$\lim_{n \rightarrow \infty} a_n \geq \lim_{n \rightarrow \infty} b_n.$$

Proof. Let $\lim_{n \rightarrow \infty} a_n = A$, $\lim_{n \rightarrow \infty} b_n = B$. Suppose, for a contradiction, that $A < B$.

Take $\varepsilon = \frac{B-A}{4}$. Then there exists $N_1 \in \mathbb{N}$ such that

$$|a_n - A| < \frac{B-A}{4} \quad \text{for all } n \geq N_1. \tag{1}$$

There exists $N_2 \in \mathbb{N}$ such that

$$|b_n - B| < \frac{B-A}{4} \quad \text{for all } n \geq N_2. \tag{2}$$

Put $N = \max\{N_1, N_2\}$. For all $n \geq N$, both (1) and (2) hold. Then

$$b_n > B - \frac{B-A}{4} \quad \text{and} \quad a_n < A + \frac{B-A}{4}.$$

Thus, $a_n \geq b_n$ implies

$$A + \frac{B-A}{4} > a_n \geq b_n > B - \frac{B-A}{4},$$

so $B - A < \frac{B-A}{4} + \frac{B-A}{4} = \frac{B-A}{2}$. This gives a contradiction. □

Remark. This result, of course, remains valid if the inequality $a_n \geq b_n$ holds only for all sufficiently large n . Limits need not preserve strict inequalities; for example, $1/n > 0$ for all $n \in \mathbb{N}$ but $\lim_{n \rightarrow \infty} 1/n = 0$.

Theorem 3.9 (Limit laws). *Suppose $\{a_n\}$ and $\{b_n\}$ are convergent; let $\lim_{n \rightarrow \infty} a_n = A$, $\lim_{n \rightarrow \infty} b_n = B$. Then*

$$(1) \quad \lim_{n \rightarrow \infty} \lambda a_n = \lambda A \text{ for any } \lambda \in \mathbb{R}.$$

$$(2) \quad \lim_{n \rightarrow \infty} (a_n \pm b_n) = A \pm B;$$

$$(3) \quad \lim_{n \rightarrow \infty} a_n b_n = AB;$$

$$(4) \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}, \text{ provided } b_n \neq 0, B \neq 0.$$

Proof.

(1) The case $\lambda = 0$ is easy. Thus, assume $\lambda \neq 0$. Let $\varepsilon > 0$ be given. Then

$$|\lambda a_n - \lambda A| = |\lambda| |a_n - A|.$$

We are given $a_n \rightarrow A$, so we know we can make $|a_n - A|$ as small as we like. In particular, choose $N \in \mathbb{N}$ such that

$$|a_n - A| < \frac{\varepsilon}{|\lambda|}$$

for all $n \geq N$. To see that this N indeed works, observe that, for all $n \geq N$,

$$|\lambda a_n - \lambda A| = |\lambda| |a_n - A| < |\lambda| \frac{\varepsilon}{|\lambda|} = \varepsilon.$$

(2) Let $\varepsilon > 0$ be given. By triangle inequality,

$$|(a_n + b_n) - (A + B)| \leq |a_n - A| + |b_n - B|.$$

The technique is to divide the ε between the two expressions on the RHS in the preceding inequality. Since $a_n \rightarrow A$, there exists $N_1 \in \mathbb{N}$ such that

$$|a_n - A| < \frac{\varepsilon}{2} \quad \text{for all } n \geq N_1. \quad (1)$$

Likewise, since $b_n \rightarrow B$, there exists $N_2 \in \mathbb{N}$ such that

$$|b_n - B| < \frac{\varepsilon}{2} \quad \text{for all } n \geq N_2. \quad (2)$$

By choosing $N = \max\{N_1, N_2\}$, we ensure that if $n \geq N$, then $n \geq N_1$ and $n \geq N_2$, so that (1) and (2) hold. This allows us to conclude that

$$\begin{aligned} |(a_n + b_n) - (A + B)| &\leq |a_n - A| + |b_n - B| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

for all $n \geq N$, as desired.

(3) The trick is to insert a term, so that we can use the triangle inequality.

$$\begin{aligned} |a_n b_n - AB| &= |a_n b_n - a_n B + a_n B - AB| \\ &\leq |a_n b_n - a_n B| + |a_n B - AB| \\ &= |a_n| |b_n - B| + |B| |a_n - A|. \end{aligned}$$

We try to make the above term a constant. Since $\{a_n\}$ is bounded, there exists $C > 0$ such that $|a_n| \leq C$ for all $n \in \mathbb{N}$. Then

$$|a_n b_n - AB| \leq C |b_n - B| + |B| |a_n - A|.$$

Thus, choose $N_1, N_2 \in \mathbb{N}$ such that

$$\begin{aligned} |a_n - A| &< \frac{\varepsilon}{2|B|} \quad \text{for all } n \geq N_1, \\ |b_n - B| &< \frac{\varepsilon}{2C} \quad \text{for all } n \geq N_2. \end{aligned}$$

Then choose $N = \max\{N_1, N_2\}$.

(4) Again, we employ the trick of inserting a new term.

$$\begin{aligned}
 \left| \frac{a_n}{b_n} - \frac{A}{B} \right| &= \left| \frac{a_n B - A b_n}{b_n B} \right| \\
 &= \left| \frac{a_n B - a_n b_n + a_n b_n - A b_n}{b_n B} \right| \\
 &\leq \left| \frac{a_n B - a_n b_n}{b_n B} \right| + \left| \frac{a_n b_n - A b_n}{b_n B} \right| \\
 &= \frac{|a_n| |b_n - B|}{|b_n| |B|} + \frac{|a_n - A|}{|B|}.
 \end{aligned}$$

Since $\{a_n\}$ is bounded, there exists $C > 0$ such that $|a_n| \leq C$ for all $n \in \mathbb{N}$.

Since $b_n \neq 0$, $b \neq 0$, by the lemma, there exists $\tilde{C} > 0$ such that $|b_n| \geq \tilde{C}$ for all $n \in \mathbb{N}$. Hence,

$$\left| \frac{a_n}{b_n} - \frac{A}{B} \right| \leq \frac{C}{\tilde{C}|B|} |b_n - B| + \frac{1}{|B|} |a_n - A|.$$

Choose $N_1, N_2 \in \mathbb{N}$ such that

$$\begin{aligned}
 |a_n - A| &< \frac{\varepsilon |B|}{2} \quad \text{for all } n \geq N_1, \\
 |b_n - B| &< \frac{\varepsilon |B| \tilde{C}}{2C} \quad \text{for all } n \geq N_2.
 \end{aligned}$$

Take $N = \max\{N_1, N_2\}$.

□

Remark. The convergence of $\{a_n + b_n\}$ does not imply the convergence of $\{a_n\}$ and $\{b_n\}$ separately; for example, take $a_n = n$ and $b_n = -n$. If, however, $\{a_n\}$ converges, then $\{b_n\}$ converges if and only if $\{a_n + b_n\}$ converges.

Example. To evaluate $\lim_{n \rightarrow \infty} \frac{2n^2 + 2n + 5}{n^2 + 4}$, divide by the highest power:

$$\lim_{n \rightarrow \infty} \frac{2n^2 + 2n + 5}{n^2 + 4} = \lim_{n \rightarrow \infty} \frac{2 + 2/n + 5/n^2}{1 + 4/n^2} = \frac{2 + 0 + 0}{1 + 0} = 2.$$

In general, to evaluate the limit of rational functions $\lim_{n \rightarrow \infty} \frac{P(n)}{Q(n)}$ where P, Q are polynomials, divide numerator and denominator by the highest power.

Case 1: $\deg P < \deg Q$. Then the limit equals 0.

Case 2: $\deg P > \deg Q$. Then the limit does not exist.

Case 3: $\deg P = \deg Q$. Then the limit is the quotient of leading coefficients.

Theorem 3.10 (Squeeze theorem). *Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be sequences such that $a_n \leq b_n \leq c_n$. If $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = A$, then $\{b_n\}$ converges, and*

$$\lim_{n \rightarrow \infty} b_n = A.$$

Proof. Let $\varepsilon > 0$. There exist $N_1, N_2 \in \mathbb{N}$ such that

$$\begin{aligned}
 |a_n - A| &< \varepsilon \quad \text{for all } n \geq N_1, \\
 |c_n - A| &< \varepsilon \quad \text{for all } n \geq N_2.
 \end{aligned}$$

Take $N = \max\{N_1, N_2\}$. Then for all $n \geq N$,

$$-\varepsilon < a_n - A \leq b_n - A \leq c_n - A < \varepsilon,$$

which implies that $|b_n - A| < \varepsilon$. This proves the result. $\square$

Example. To evaluate $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n}$, note that $-\frac{1}{n} \leq \frac{(-1)^n}{n} \leq \frac{1}{n}$, and $\frac{1}{n}, -\frac{1}{n} \rightarrow 0$. By squeeze theorem, $\frac{(-1)^n}{n} \rightarrow 0$.

Example. Let $a_n = \frac{1}{n^2+1} + \frac{2}{n^2+1} + \cdots + \frac{n}{n^2+1}$. To evaluate $\lim_{n \rightarrow \infty} a_n$, note that

$$\begin{aligned} a_n &\leq \frac{1+2+\cdots+n}{n^2+1} = \frac{\frac{1}{2}n^2 + \frac{1}{2}n}{n^2+1} \rightarrow \frac{1}{2}, \\ a_n &\geq \frac{1+2+\cdots+n}{n^2+n} = \frac{\frac{1}{2}(n^2+n)}{n^2+n} = \frac{1}{2}. \end{aligned}$$

By squeeze theorem, $\lim_{n \rightarrow \infty} a_n = \frac{1}{2}$.

As once consequence, we show that we can take absolute values inside limits.

Corollary 3.11. If $a_n \rightarrow A$, then $\lim_{n \rightarrow \infty} |a_n| = |A|$.

Proof. By the reverse triangle inequality,

$$0 \leq |a_n| - |A| \leq |a_n - A|,$$

and the result follows from squeeze theorem. $\square$

Corollary 3.12. If $a_n \rightarrow A$ and $a_n \geq 0$, then $\lim_{n \rightarrow \infty} \sqrt{a_n} = \sqrt{A}$.

Proof. We handle the square root by multiplying the conjugate:

$$\left| \sqrt{a_n} - \sqrt{A} \right| = \frac{|a_n - A|}{\sqrt{a_n} + \sqrt{A}} \leq \frac{|a_n - A|}{\sqrt{A}}.$$

Thus, choose $N \in \mathbb{N}$ such that

$$|a_n - A| < \sqrt{A}\varepsilon \quad \text{for all } n \geq N.$$

$\square$

Corollary 3.13. $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.

Proof. Note that

$$\left| \sqrt[n]{n} - 1 \right| < \varepsilon \iff \sqrt[n]{n} < 1 + \varepsilon \iff n < (1 + \varepsilon)^n.$$

By Bernoulli's inequality, $(1 + \varepsilon)^n > 1 + n\varepsilon$, so it suffices to show $1 + n\varepsilon > n$, but this may not hold for $n \geq N$ where N is large! Instead, consider the binomial expansion

$$\begin{aligned} (1 + \varepsilon)^n &= 1 + n\varepsilon + \frac{n(n-1)}{2}\varepsilon^2 + \cdots + n\varepsilon^{n-1} + \varepsilon^n \\ &> \frac{n(n-1)}{2}\varepsilon^2 \end{aligned}$$

We claim that there exists $N \in \mathbb{N}$ such that $\frac{n(n-1)}{2}\varepsilon^2 > n$ for all $n \geq N$. Note that

$$\begin{aligned} \frac{n(n-1)}{2}\varepsilon^2 > n &\iff \frac{n-1}{2}\varepsilon^2 > 1 \\ &\iff n-1 > \frac{2}{\varepsilon^2} \\ &\iff n > 1 + \frac{2}{\varepsilon^2}. \end{aligned}$$

Thus choose $N > 1 + \frac{2}{\varepsilon^2}$. $\square$

Example. To evaluate $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$, use difference of squares to get rid of square roots:

$$\sqrt{n^2 + n} - n = \frac{(\sqrt{n^2 + n} - n)(\sqrt{n^2 + n} + n)}{\sqrt{n^2 + n} + n} = \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \rightarrow \frac{1}{2}.$$

3.3 Monotone convergence theorem

Definition 3.14. A sequence $\{a_n\}$ is *increasing* (or *decreasing*) if for any $n \in \mathbb{N}$,

$$a_n \leq a_{n+1} \quad (\text{or } a_n \geq a_{n+1}).$$

A sequence is *monotone* if it is either increasing or decreasing.

Theorem 3.15 (Monotone convergence theorem). *If a sequence is monotone and bounded, then it converges.*

Proof. We only prove the case where the sequence is increasing (the decreasing case is handled similarly). Let $\{a_n\}$ be an increasing and bounded sequence.

To prove $\{a_n\}$ converges using the definition of convergence, we need a candidate for the limit. A natural guess is the supremum. Consider the set $S = \{a_n : n \in \mathbb{N}\}$. By assumption, S is bounded (from above), so we can let

$$A = \sup\{a_n : n \in \mathbb{N}\}.$$

Claim: $\lim_{n \rightarrow \infty} a_n = A$.

Let $\varepsilon > 0$. Since A is the supremum for $\{a_n : n \in \mathbb{N}\}$, $A - \varepsilon$ is not an upper bound, so there exists $N \in \mathbb{N}$ such that $A - \varepsilon < a_N$.

Since $\{a_n\}$ is increasing, if $n \geq N$, then $a_N \leq a_n$, so

$$A - \varepsilon < a_N \leq a_n \leq A < A + \varepsilon,$$

which implies $|a_n - A| < \varepsilon$, as desired. $\square$

We sometimes use the notation $a_n \nearrow A$ to indicate that $\{a_n\}$ is an increasing sequence that converges to A , and $a_n \searrow A$ to indicate that $\{a_n\}$ is a decreasing sequence that converges to A .

For recursively defined sequences, we try to use induction.

Example. Let $\{a_n\}$ be defined as follows: $a_1 = 2$, $a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$ for $n \in \mathbb{N}$. Prove that $\{a_n\}$ is convergent, and find $\lim_{n \rightarrow \infty} a_n$.

Solution. Compare a_n and a_{n+1} by taking their difference:

$$a_{n+1} - a_n = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right) - a_n = \frac{1}{a_n} - \frac{1}{2}a_n = \frac{2 - a_n^2}{2a_n}.$$

Claim: $2 - a_n^2 < 0$.

To see this, by AM-GM inequality,

$$a_{n+1}^2 = \frac{1}{4}a_n^2 + \frac{1}{a_n^2} + 1 \geq 2\sqrt{\frac{1}{4}a_n^2 \cdot \frac{1}{a_n^2}} + 1 = 2.$$

Then $a_{n+1} \leq a_n$, so $\{a_n\}$ is decreasing. Note that $a_n^2 > 2$ implies that $a_n > \sqrt{2}$, so $\sqrt{2}$ is a lower bound.

By the monotone convergence theorem, $\{a_n\}$ converges; denote $\lambda = \lim_{n \rightarrow \infty} a_n$. Take limits on both sides of the

recursive formula:

$$\begin{aligned}\lambda &= \frac{1}{2}\lambda + \frac{1}{\lambda} \\ \lambda^2 &= 2 \\ \lambda &= \sqrt{2}\end{aligned}$$

where we reject the negative term since every term $\geq \sqrt{2}$, so limit $\geq \sqrt{2}$. $\square$

Example (Binary expansion). Given any sequence of $\{0, 1\}$: $\{b_n\}$ ($b_n \in \{0, 1\}$), define $\{a_n\}$ as follows:

$$a_n = \sum_{k=1}^n \frac{b_k}{2^k}.$$

Prove that $\lim_{n \rightarrow \infty} a_n$ exists.

Solution. The difference $a_{n+1} - a_n = \frac{b_{n+1}}{2^{n+1}} \geq 0$, so $\{a_n\}$ is increasing. An upper bound for a_n is

$$a_n = \sum_{k=1}^n \frac{b_k}{2^k} \leq \sum_{k=1}^n \frac{1}{2^k} = 1 - \frac{1}{2^n} < 1.$$

By the monotone convergence theorem, $\{a_n\}$ converges. $\square$

Example (Binary expansion). Given any $x \in (0, 1)$, there exists a unique $\{0, 1\}$ -sequence $\{b_n\}$ such that there are infinitely many 0's in the sequence and the corresponding $\{a_n\}$ defined as above converges to x .

Solution. Define $x_1 = x$, $b_1 = \lfloor 2x_1 \rfloor$. Then define $x_2 = 2x_1 - b_1$, $b_2 = \lfloor 2x_2 \rfloor$. Suppose x_n and b_n have been defined, then define $x_{n+1} = 2x_n - b_n$ and $b_{n+1} = \lfloor 2x_{n+1} \rfloor$.

Claim: $\lim_{n \rightarrow \infty} a_n = x$.

To see this, note that

$$\begin{aligned}x_{n+1} &= 2x_n - b_n = 2(2x_{n-1} - b_{n-1}) - b_n \\ &= \cdots = 2(2(\cdots (2x_1 - b_1) - b_2) - \cdots) - b_n \\ &= 2^n x - 2^{n-1}b_1 - 2^{n-2}b_2 - \cdots - b_n.\end{aligned}$$

Thus,

$$\frac{x_{n+1}}{2^n} = x - \frac{b_1}{2} - \cdots - \frac{b_n}{2^n} = x - \sum_{k=1}^n \frac{b_k}{2^k} = a_n.$$

This implies

$$x = \frac{x_{n+1}}{2^n} + \sum_{k=1}^n \frac{b_k}{2^k} = \frac{x_{n+1}}{2^n} + a_n.$$

Note that $0 < \frac{x_{n+1}}{2^n} < \frac{1}{2^n}$. By squeeze theorem, $\frac{x_{n+1}}{2^n} \rightarrow 0$. Therefore, $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{b_k}{2^k} = x$.

Claim: In this construction, there are infinitely many 0's in $\{b_n : n \in \mathbb{N}\}$.

If $b_1 = 0$, $b_2 = b_3 = \cdots = 1$, note that

$$a_n = \sum_{k=1}^n \frac{b_k}{2^k} = \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^n} = \frac{\frac{1}{2^2} - \frac{1}{2^{n+1}}}{1 - \frac{1}{2}} = \frac{1}{2} - \frac{1}{2^n}.$$

Hence, $\lim_{n \rightarrow \infty} a_n = \frac{1}{2}$.

Claim: Suppose $\{b_n\}$ and $\{b'_n\}$ are both $\{0, 1\}$ -sequences with infinitely many zeros. Then there exists $m \in \mathbb{N}$

such that $b_m \neq b'_m$. Thus,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{b_k}{2^k} \neq \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{b'_k}{2^k}.$$

Suppose towards a contradiction that

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{b_k}{2^k} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{b'_k}{2^k}.$$

Let $S = \{k \in \mathbb{N} : b_k \neq b'_k\}$, the set of indices in which the digits differ. By assumption, $S \neq \emptyset$, so by well-ordering, choose $m \in S$ that is the smallest. Write

$$\begin{aligned} \sum_{k=1}^n \frac{b_k}{2^k} &= \frac{b_1}{2} + \cdots + \frac{b_m}{2^m} + x_n, \\ \sum_{k=1}^n \frac{b'_k}{2^k} &= \frac{b'_1}{2} + \cdots + \frac{b'_m}{2^m} + y_n. \end{aligned}$$

WLOG assume $b_m = 1, b'_m = 0$. Then

$$y_n = \frac{b'_{m+1}}{2^{m+1}} + \cdots + \frac{b'_n}{2^n} \leq \sum_{k=m+1}^n \frac{1}{2^k} = \frac{1}{2^m} - \frac{1}{2^n}.$$

Since there are infinitely many zeros, there exists $m' \geq m+1$ such that $b'_{m'} = 0$. For $n \geq m' + 1$,

$$y_n \leq \frac{1}{2^m} - \frac{1}{2^n} - \frac{1}{2^{m'}}.$$

Taking the limit $n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} y_n \leq \frac{1}{2^m} - \frac{1}{2^{m'}} < \frac{1}{2^m} \leq \lim_{n \rightarrow \infty} \left(\frac{b_m}{2^m} + x_n \right).$$

□

Lemma 3.16. The sequence $a_n = \left(1 + \frac{1}{n}\right)^n$ is convergent.

Proof. By binomial theorem,

$$a_n = \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \binom{n}{k} \left(\frac{1}{n}\right)^k = \sum_{k=0}^n \frac{n(n-1)\cdots(n-k+1)}{k!} \left(\frac{1}{n}\right)^k.$$

The idea is to use the monotone convergence theorem. We compare a_n and a_{n+1} :

$$a_{n+1} = \left(1 + \frac{1}{n+1}\right)^{n+1} = \sum_{k=0}^{n+1} \frac{(n+1)n\cdots(n-k+2)}{k!} \left(\frac{1}{n+1}\right)^k.$$

Denote

$$x_{n,k} := \frac{n-1}{n} \frac{n-2}{n} \cdots \frac{n-k+1}{n} = \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right).$$

Then

$$x_{n+1,k} = \frac{n}{n+1} \frac{n-1}{n+1} \cdots \frac{n-k+2}{n+1} = \left(1 - \frac{1}{n+1}\right) \left(1 - \frac{2}{n+1}\right) \cdots \left(1 - \frac{k-1}{n+1}\right).$$

Note that

$$1 - \frac{1}{n} < 1 - \frac{1}{n+1}, \quad 1 - \frac{2}{n} < 1 - \frac{2}{n+1}, \quad \dots, \quad 1 - \frac{k-1}{n} < 1 - \frac{k-1}{n+1}.$$

Taking the product of the inequalities yields $x_{n,k} < x_{n+1,k}$. Hence,

$$a_n = \sum_{k=0}^n \frac{1}{k!} x_{n,k} < \sum_{k=0}^n \frac{1}{k!} x_{n+1,k} < \sum_{k=0}^{n+1} \frac{1}{k!} x_{n+1,k} = a_{n+1}.$$

This means that $\{a_n\}$ is increasing.

Next, we show the boundedness of $\{a_n\}$. Since

$$x_{n,k} = \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) < 1 \cdot 1 \cdots 1 = 1,$$

it follows that $a_n = \sum_{k=0}^n \frac{1}{k!} x_{n,k} < \sum_{k=0}^n \frac{1}{k!}$. To evaluate this summation,

$$\begin{aligned} a_n &\leq 1 + \frac{1}{1} + \sum_{k=2}^n \frac{1}{k(k-1)} \\ &= 2 + \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) \\ &= 2 + \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{n-1} - \frac{1}{n} \right) \\ &= 3 - \frac{1}{n} < 3. \end{aligned}$$

By the monotone convergence theorem, $\{a_n\}$ converges. □

Definition 3.17. Euler's number is $e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$.

Lemma 3.18. The sequence $\{b_n\}$ defined by $b_n = \left(1 + \frac{1}{n}\right)^{n+1}$ is decreasing.

Remark. Note that the convergence of $\{a_n\}$ implies the convergence of $\{b_n\}$, since $b_n = a_n \left(1 + \frac{1}{n}\right)$.

Since $b_n \rightarrow \inf\{b_n\}$, $b_n > e$. Thus $a_n < e < b_n$. For large n , this gap is small so we can get some control of e .

Example. A sequence $\{a_n\}$ is defined as $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2 + a_n}$. Prove that $\{a_n\}$ is convergent, and find the limit.

Solution. We first prove $\{a_n\}$ is increasing by induction. Suppose $a_n > a_{n-1}$. Then $\sqrt{2 + a_n} > \sqrt{2 + a_{n-1}}$, so $a_{n+1} > a_n$.

Next, we prove $a_n < 2$ for all $n \in \mathbb{N}$ by induction. The base case is clear. Suppose $a_n < 2$. Then $a_{n+1} = \sqrt{2 + a_n} < \sqrt{4} = 2$.

By monotone convergence theorem, $\{a_n\}$ converges; let $L = \lim_{n \rightarrow \infty} a_n$. Taking limits $n \rightarrow \infty$ gives $L = \sqrt{2 + L}$, so $L = 2$ (reject $L = -1$ since $a_n > 0$ for all $n \in \mathbb{N}$). □

Example. Suppose $a_n \in (0, 1)$ and $a_{n+1} = \sqrt{a_n(1 - a_n)}$. Prove that a_n is convergent and find the limit.

Solution. By AM-GM inequality,

$$a_{n+1} = \sqrt{a_n(1 - a_n)} \leq \frac{a_n + (1 - a_n)}{2} = \frac{1}{2},$$

so $a_n \leq \frac{1}{2}$ for $n \geq 2$.

Next, we show $\{a_n\}$ is increasing by showing $\frac{a_{n+1}}{a_n} > 1$:

$$\frac{a_{n+1}}{a_n} = \frac{\sqrt{a_n(1 - a_n)}}{a_n} = \sqrt{\frac{1 - a_n}{a_n}} \geq \sqrt{\frac{1/2}{1/2}} = 1,$$

where we used $1 - a_n \geq \frac{1}{2}$ and $a_n \leq \frac{1}{2}$.

By monotone convergence theorem, $\{a_n\}$ converges; let $L = \lim_{n \rightarrow \infty} a_n$. Then $L = \sqrt{L(1-L)}$; solving gives $L = \frac{1}{2}$. $\square$

3.4 Subsequences, Bolzano-Weierstrass theorem

Definition 3.19. Given a sequence $\{a_n\}$, we say that $\{b_k\}_{k \in \mathbb{N}}$ is a *subsequence* of $\{a_n\}$ if for any $k \in \mathbb{N}$,

$$b_k = a_{n_k}$$

where $n_1 < n_2 < \dots$.

Example. The following are subsequences of a sequence $\{a_n\}$: $\{b_k = a_{2k} : k \in \mathbb{N}\}$, $\{b_k = a_{k+1}\}$.

Proposition 3.20. Every subsequence of a convergent sequence converges to the limit of the sequence.

Proof. Suppose $\{a_n\}$ converges to A . Let $\{a_{n_k}\}$ be a subsequence.

Let $\varepsilon > 0$ be given. There exists $N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for all $n \geq N$.

Since $n_k \geq k$, we also have $|a_{n_k} - a| < \varepsilon$ for all $k \geq N$. Hence, $a_{n_k} \rightarrow a$. $\square$

Corollary 3.21. If a sequence has subsequences that converge to different limits, then the sequence diverges.

Corollary 3.22. If $\{a_n\}$ is convergent, then $\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} a_n$.

Example. Let $\{a_n\}$ be defined by $a_n = (-1)^n$. It has two subsequences $\{a_{2n} = 1\}$ and $\{a_{2n+1} = -1\}$. Since $\lim_{n \rightarrow \infty} a_{2n} = 1$ and $\lim_{n \rightarrow \infty} a_{2n+1} = -1$, this implies that $\{a_n\}$ is divergent.

Theorem 3.23 (Monotone subsequence theorem). Any sequence admits a monotone subsequence.

Proof. Let $\{a_n\}$ be a sequence. We call a_n a *peak* if $a_n \geq a_m$ for all $m \geq n$.

Case 1: There are infinitely many peaks, say $a_{n_1}, \dots, a_{n_k}, \dots$

Let $b_k = a_{n_k}$, i.e., the k -th peak. By definition of peaks, $b_k \geq b_{k+1}$ for all $k \in \mathbb{N}$. Thus, $\{b_k\}$ is a decreasing subsequence.

Case 2: There are only finitely many peaks. By finiteness, there exists $N \in \mathbb{N}$ such that for $n \geq N$, a_n is not a peak.

Let $b_1 = a_N$. Since a_N is not a peak, there exists $N_2 > N$ such that $a_{N_2} > a_N$. Let $b_2 = a_{N_2}$.

Suppose $b_k = a_{N_k}$ has been defined. Since a_{N_k} is not a peak, there exists $N_{k+1} > N_k$ such that $a_{N_{k+1}} > a_{N_k}$. Let $b_{k+1} = a_{N_{k+1}}$. This gives an increasing subsequence. $\square$

Theorem 3.24 (Bolzano-Weierstrass). Any bounded sequence admits a convergent subsequence.

First proof. By the monotone subsequence theorem, there exists a monotone subsequence $\{b_k = a_{n_k} : k \in \mathbb{N}\}$, which is bounded since $\{a_n\}$ is bounded. By monotone convergence theorem, $\{b_k\}$ is bounded. $\square$

Second proof. This proof uses the nested interval property. Fix $C, D, C < D$ such that $c \leq a_n \leq D$ for all $n \in \mathbb{N}$. Then $a_n \in [C, D]$ for all $n \in \mathbb{N}$.

Let $I_1 = [C, D]$. Partition I_1 into two sub-intervals $I'_1 = \left[C, \frac{C+D}{2}\right]$ and $I''_1 = \left[\frac{C+D}{2}, D\right]$. One of I'_1 and I''_1 must contain infinitely many terms from $\{a_n\}$. Let I_2 be that sub-interval.

Inductively, suppose $I_k = [C_k, D_k]$ has been constructed, such that I_k contains infinitely many terms from $\{a_n\}$, and $|I_k| = \frac{|I_1|}{2^{k-1}}$. Now let $I'_k = \left[C_k, \frac{C_k+D_k}{2}\right]$ and $I''_k = \left[\frac{C_k+D_k}{2}, D_k\right]$. One of them must contain infinitely many terms from $\{a_n\}$. Let I_{k+1} be that one.

This constructs a decreasing sequence of nested intervals $I_1 \supset I_2 \supset \dots \supset I_k \supset \dots$ such that every I_k contains infinitely many terms from $\{a_n\}$.

Now choose $a_{n_1} \in I_1$ and set $b_1 = a_{n_1}$. Suppose $b_k = a_{n_k} \in I_k$ has been chosen. Since there are infinitely many terms in I_{k+1} , there exists $n_{k+1} > n_k$ such that $a_{n_{k+1}} \in I_{k+1}$, so set $b_{k+1} = a_{n_{k+1}}$.

Claim: The subsequence $\{b_k = a_{n_k} : k \in \mathbb{N}\}$ of $\{a_n\}$ as constructed above is convergent.

Note that $|I_k| = \frac{|I_1|}{2^{k-1}}$, so $|I_k| \rightarrow 0$. Thus, Problem A.1.5, $\bigcap_{k=1}^{\infty} I_k = \{\xi\}$. Since $b_k, \xi \in I_k$,

$$|b_k - \xi| \leq |I_k| = \frac{|I_1|}{2^{k-1}} \rightarrow 0.$$

By squeeze theorem, $|b_k - \xi| \rightarrow 0$. Hence, $b_k \rightarrow \xi$. □

3.5 Cauchy sequences

Roughly speaking, a Cauchy sequence is a sequence whose terms eventually get arbitrarily close together.

Definition 3.25. A sequence $\{a_n\}$ is a *Cauchy sequence* if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \quad |a_m - a_n| < \varepsilon.$$

Lemma 3.26. Every Cauchy sequence is bounded.

Proof. Let $\{a_n\}$ be a Cauchy sequence. Given $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that

$$|a_m - a_n| < 1 \quad \text{for all } m, n \geq N.$$

In particular, we have $|a_n - a_N| < 1$ for all $n \geq N$, so

$$|a_n| \leq |a_n - a_N| + |a_N| < |a_N| + 1$$

for all $n \geq N$. Let $C = \max\{|a_1|, \dots, |a_{N-1}|, |a_N| + 1\}$. Then $|a_n| \leq C$ for all $n \in \mathbb{N}$. □

Theorem 3.27 (Cauchy convergence criterion). A sequence is convergent if and only if it is a Cauchy sequence.

Proof.

$\Rightarrow$ Suppose $\{a_n\}$ is convergent; let $A = \lim_{n \rightarrow \infty} a_n$. Let $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that

$$|a_n - A| < \frac{\varepsilon}{2} \quad \text{for all } n \geq N.$$

Then for $m, n \geq N$,

$$\begin{aligned} |a_m - a_n| &= |(a_m - A) - (a_n - A)| \\ &\leq |a_m - A| + |a_n - A| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

$\Leftarrow$ Suppose $\{a_n\}$ is a Cauchy sequence. Then $\{a_n\}$ is bounded. By Bolzano-Weierstrass, $\{a_n\}$ has a convergent subsequence $\{b_k = a_{n_k} : k \in \mathbb{N}\}$. Let $A = \lim_{k \rightarrow \infty} b_k$. This gives us a candidate for the limit of $\{a_n\}$:

Claim: $\lim_{n \rightarrow \infty} a_n = A$.

Let $\varepsilon > 0$. Since $\{a_n\}$ is a Cauchy sequence, there exists $N \in \mathbb{N}$ such that

$$|a_m - a_n| < \frac{\varepsilon}{2} \quad \text{for all } m, n \geq N.$$

Since $a_{n_k} \rightarrow A$, there exists $n_k \in \mathbb{N}$ such that $n_k \geq N$ and

$$|a_{n_k} - A| < \frac{\varepsilon}{2}.$$

Then for $n \geq N$,

$$\begin{aligned} |a_n - A| &= |(a_n - a_{n_k}) + (a_{n_k} - A)| \\ &\leq |a_n - a_{n_k}| + |a_{n_k} - A| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

This proves that $\lim_{n \rightarrow \infty} a_n = A$. □

Definition 3.28. A real sequence $\{a_n\}$ is *contractive* if there exists a constant $0 < C < 1$ such that

$$|a_{n+1} - a_n| \leq C|a_n - a_{n-1}|$$

for all $n \geq 2$.

Proposition 3.29. Every contractive sequence is Cauchy, hence convergent.

Proof. By induction, we can show that

$$|a_{n+1} - a_n| \leq C|a_n - a_{n-1}| \leq C^2|a_{n-1} - a_{n-2}| \leq \cdots \leq C^{n-1}|a_2 - a_1|.$$

Therefore, for $m \geq n + 1$,

$$\begin{aligned} |a_m - a_n| &= \left| \sum_{k=n}^{m-1} (a_{k+1} - a_k) \right| \leq \sum_{k=n}^{m-1} |a_{k+1} - a_k| \leq \sum_{k=n}^{m-1} C^{k-1} |a_2 - a_1| \\ &= |a_2 - a_1| \sum_{k=n}^{m-1} C^{k-1} \leq C^{n-2} |a_2 - a_1|. \end{aligned}$$

Since $0 < C < 1$, $C^{n-2} \rightarrow 0$. Hence, $\{a_n\}$ is a Cauchy sequence. □

Example. The alternating harmonic series $\sum \frac{(-1)^{n-1}}{n}$ is convergent.

Proof. Let $\varepsilon > 0$. For $m \geq n + 1$,

$$|a_m - a_n| = \left| \sum_{k=n+1}^m \frac{(-1)^{k-1}}{k} \right| = \left| \frac{1}{n+1} - \frac{1}{n+2} + \frac{1}{n+3} - \cdots + \frac{(-1)^{m-n+1}}{m} \right|.$$

If the last term is positive, we group terms as follows:

$$\frac{1}{n+1} - \underbrace{\left(\frac{1}{n+2} - \frac{1}{n+3} \right)}_{>0} - \cdots - \underbrace{\left(\frac{1}{m-1} - \frac{1}{m} \right)}_{>0} < \frac{1}{n+1}$$

and

$$\underbrace{\left(\frac{1}{n+1} - \frac{1}{n+2} \right)}_{>0} + \underbrace{\left(\frac{1}{n+3} - \frac{1}{n+4} \right)}_{>0} + \cdots + \underbrace{\left(\frac{1}{m-2} - \frac{1}{m-1} \right)}_{>0} + \frac{1}{m} > 0$$

If the last term is negative, then group as follows:

$$\frac{1}{n+1} - \underbrace{\left(\frac{1}{n+2} - \frac{1}{n+3} \right)}_{>0} - \cdots - \underbrace{\left(\frac{1}{m-2} - \frac{1}{m-1} \right)}_{>0} - \frac{1}{m} < \frac{1}{n+1}$$

and

$$\underbrace{\left(\frac{1}{n+1} - \frac{1}{n+2}\right)}_{>0} + \underbrace{\left(\frac{1}{n+3} - \frac{1}{n+4}\right)}_{>0} + \cdots + \underbrace{\left(\frac{1}{m-1} - \frac{1}{m}\right)}_{>0} > 0.$$

Combining both cases, $|a_m - a_n| < \frac{1}{n+1}$. Thus, choose $N \in \mathbb{N}$ such that $\frac{1}{N+1} < \varepsilon$. Then for all $m > n \geq N$,

$$|a_m - a_n| < \frac{1}{n+1} < \frac{1}{N+1} < \varepsilon.$$

This proves that $\{a_n\}$ is a Cauchy sequence, and thus convergent. $\square$

3.6 Properly divergent sequences

The following notation for sequences that “diverge to infinity” is convenient.

Definition 3.30. Let $\{a_n\}$ be a sequence.

- We write $\lim_{n \rightarrow \infty} a_n = +\infty$ if

$$\forall C \in \mathbb{R}, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad a_n > C.$$

- We write $\lim_{n \rightarrow \infty} a_n = -\infty$ if

$$\forall C \in \mathbb{R}, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad a_n < C.$$

- We write $\lim_{n \rightarrow \infty} a_n = \infty$ if $\{|a_n|\}$ tends to $+\infty$.

That is, $a_n \rightarrow +\infty$ means that the terms of the sequence $\{a_n\}$ get arbitrarily positive large for all sufficiently large n , while $a_n \rightarrow -\infty$ means that the terms get arbitrarily negative large for all sufficiently large n .

Example. Consider the sequence $\{a_n\}$ defined by $a_n = n^3$. To show $a_n \rightarrow +\infty$, let $M \in \mathbb{R}$ be given. Choose $N \in \mathbb{N}$ such that $N > M^{1/3}$. Then for all $n \geq N$, we have $n^3 > N^3 > M$, which proves the result.

Theorem 3.31. A monotone sequence is properly divergent iff it is unbounded.

Proof.

$\Rightarrow$ By monotone convergence theorem, a bounded monotone sequence converges.

$\Leftarrow$ Suppose $\{a_n\}$ is increasing and unbounded. By unboundedness, for any $C \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $a_n > C$. Since $\{a_n\}$ is increasing, $a_m \geq a_n > C$ for all $m \geq n$. Hence, $a_n \rightarrow +\infty$.

The decreasing case is similar. $\square$

Theorem 3.32 (Squeeze theorem). For $\{a_n\}$ and $\{b_n\}$,

- (1) if $a_n \geq b_n$ and $\lim_{n \rightarrow \infty} b_n = +\infty$, then $\lim_{n \rightarrow \infty} a_n = +\infty$;
- (2) if $a_n \leq b_n$ and $\lim_{n \rightarrow \infty} a_n = -\infty$, then $\lim_{n \rightarrow \infty} b_n = -\infty$.

Theorem 3.33 (Reciprocals).

- (1) Suppose $a_n \neq 0$ and $\lim_{n \rightarrow \infty} a_n = \infty$. Then $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$.
- (2) Suppose $a_n \neq 0$ and $\lim_{n \rightarrow \infty} a_n = 0$. Then $\lim_{n \rightarrow \infty} \frac{1}{a_n} = \infty$.

3.7 Upper and lower limits

The upper and lower limits allow us to reduce questions about the convergence and limits of general real sequences to ones about monotone sequences.

In order to define the upper and lower limits of a sequence $\{a_n\}$, define the sequences $\{b_n\}$ and $\{c_n\}$ obtained by taking the supremum and infimum, respectively, of the “tails” of $\{a_n\}$:

$$b_n = \sup\{a_k : k \geq n\}, \quad c_n = \inf\{a_k : k \geq n\}.$$

As n increases, the supremum and infimum are taken over smaller sets, so $\{b_n\}$ is decreasing and $\{c_n\}$ is increasing. Then the limits of these sequences are the lim sup and lim inf, respectively, of the original sequence.

Definition 3.34. Suppose $\{a_n\}$ is bounded from above. The *upper limit* of $\{a_n\}$ is defined as follows

$$\limsup_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \sup\{a_k : k \geq n\}.$$

By monotone convergence theorem, $\{b_n\}$ is either convergent or $b_n \rightarrow -\infty$.

Definition 3.35. Suppose $\{a_n\}$ is bounded from below. The *lower limit* of $\{a_n\}$ is defined as follows

$$\liminf_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \inf\{a_k : k \geq n\}.$$

By monotone convergence theorem, $\{c_n\}$ is either convergent or $c_n \rightarrow +\infty$.

Hence, limsup and liminf of a sequence always exist, whereas limit may not always exist e.g. $(-1)^n$.

Definition 3.36. A is called a *limit point* of a sequence $\{a_n\}$ if A is the limit of a subsequence of $\{a_n\}$, namely, there exists a subsequence $\{b_k = a_{n_k}\}$ such that

$$\lim_{k \rightarrow \infty} b_k = A.$$

Theorem 3.37. Let $\{a_n\}$ be a bounded sequence, and E denote the set of limit points of $\{a_n\}$. Then both upper limit and lower limit of $\{a_n\}$ are contained in E , and moreover,

$$\limsup_{n \rightarrow \infty} a_n = \max E, \quad \liminf_{n \rightarrow \infty} a_n = \min E.$$

Proof. We only prove the statement for limsup; the proof for liminf is similar. Denote $s = \limsup_{n \rightarrow \infty} a_n$. We first show that $s \in E$, by constructing a subsequence of $\{a_n\}$ that converges to s .

For each $n \in \mathbb{N}$, since $b_n := \sup\{a_k : k \geq n\}$, there exists $n_k \geq n$ such that

$$b_n - \frac{1}{n} < a_{n_k}.$$

By choosing $n_1 < n_2 < \dots$ inductively, we obtain a subsequence $\{a_{n_k}\}$ satisfying

$$b_{n_k} - \frac{1}{n_k} < a_{n_k} \leq b_{n_k}.$$

Taking limits and using $b_{n_k} \rightarrow s$, we obtain $a_{n_k} \rightarrow s$. Hence, s is a limit point, so $s \in E$.

Next, we prove that $s = \max E$, namely, $t \leq s$ for any $t \in E$. Since $t \in E$, there exists a subsequence $\{a_{m_k}\}$ such that $a_{m_k} \rightarrow t$. For each k ,

$$a_{m_k} \leq \sup\{a_l : l \geq m_k\} = b_{m_k}.$$

Taking limits $k \rightarrow \infty$ on both sides gives

$$t = \lim_{k \rightarrow \infty} a_{m_k} \leq \lim_{k \rightarrow \infty} b_{m_k} = s.$$

Hence, every limit point t satisfies $t \leq s$, so $s = \max E$. □

Corollary 3.38. For any bounded sequence $\{a_n\}$,

$$\liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n,$$

where equality holds if and only if $\{a_n\}$ is convergent.

Proposition 3.39 (Order-preserving). *Let $\{a_n\}$ and $\{b_n\}$ be bounded sequences, and $a_n \leq b_n$ for all $n \in \mathbb{N}$. Then*

$$\begin{aligned}\liminf_{n \rightarrow \infty} a_n &\leq \liminf_{n \rightarrow \infty} b_n, \\ \limsup_{n \rightarrow \infty} a_n &\leq \limsup_{n \rightarrow \infty} b_n.\end{aligned}$$

Proof. use definition, take sup on both sides, then lim on both sides. $\square$

3.8 Infinite series

Definition 3.40. Given a sequence $\{a_n\}$, we define the sequence of partial sums $\{s_n\}$ as follows:

$$s_n = \sum_{k=1}^n a_k.$$

If $\lim_{n \rightarrow \infty} s_n = A$, we say the series $\sum_{n=1}^{\infty} a_n$ converges to A , and write $\sum_{n=1}^{\infty} a_n = A$.

If $\{s_n\}$ is divergent, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

Notation. If there is no ambiguity, we simply write the series as $\sum a_n$.

Theorem 3.41 (Divergent test). *Suppose $\sum a_n$ is convergent. Then $\lim_{n \rightarrow \infty} a_n = 0$.*

Proof. Since $\sum a_n$ is convergent, let $A = \lim_{n \rightarrow \infty} s_n$, where $s_n = \sum_{k=1}^n a_k$ is the n -th partial sum. Note that $a_n = s_n - s_{n-1}$, so taking limits on both sides yields

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} s_n - \lim_{n \rightarrow \infty} s_{n-1} = A - A = 0.$$

$\square$

Example. Consider the geometric series $\sum r^n$, where $r > 0$.

- If $r \geq 1$, then $r^n \not\rightarrow 0$ as $n \rightarrow \infty$. By divergence test, $\sum r^n$ is divergent.
- If $0 < r < 1$, then the partial sums converge:

$$\sum_{k=1}^n r^k = \frac{r(1 - r^n)}{1 - r} \rightarrow \frac{r}{1 - r}.$$

Example. Consider the series $\sum \frac{1}{n(n+1)}$. Using telescoping sums, the partial sum is

$$\begin{aligned}s_n &= \sum_{k=1}^n \frac{1}{k(k+1)} = \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) \\ &= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= 1 - \frac{1}{n+1} \rightarrow 1.\end{aligned}$$

Theorem 3.42 (Cauchy criterion). *A series $\sum a_n$ is convergent if and only if*

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m > n \geq N, \quad \left| \sum_{k=n+1}^m a_k \right| < \varepsilon.$$

Proof. Observe that $|s_n - s_m| = \left| \sum_{k=n+1}^m a_k \right|$ and apply the Cauchy criterion for sequences on $\{s_n\}$. $\square$

Theorem 3.43 (Monotone convergence theorem). Suppose $\{a_n\}$ is a non-negative sequence. Then $\sum a_n$ converges if and only if the partial sums are bounded.

Example.

- Harmonic series: $\sum \frac{1}{n} = +\infty$, since the series is not bounded so divergent
- Alternating harmonic series: $\sum \frac{(-1)^{n+1}}{n}$ is convergent, since it is Cauchy (proven above)
- 2-series: $\sum \frac{1}{n^2}$ is convergent, since bounded so convergent by MCT - proven in tutorial

Definition 3.44. The series $\sum a_n$ is *absolutely convergent* if $\sum |a_n|$ is convergent. The series $\sum a_n$ is *conditionally convergent* if it is convergent but not absolutely convergent.

Proposition 3.45. If $\sum a_n$ is absolutely convergent, then it is convergent.

Example.

- $\sum \frac{(-1)^n}{n^2}$ is absolutely convergent.
- $\sum \frac{(-1)^n}{n}$ is conditionally convergent.

3.9 Various convergence tests

Theorem 3.46 (Comparison test). Suppose $\{a_n\}$ and $\{b_n\}$ are sequences such that $0 \leq a_n \leq \lambda b_n$ for all $n \geq n_0$.

- (1) If $\sum b_n$ converges, then $\sum a_n$ converges.
- (2) If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Proof. We prove (1); statement (2) follows by contrapositive. Denote the partial sums $A_n = \sum_{k=1}^n a_k$, $B_n = \sum_{k=1}^n b_k$. Since $a_n \geq 0$, the sequence $\{A_n\}$ is increasing. For $n \geq n_0$, we have

$$\begin{aligned} A_n &= \sum_{k=1}^{n_0-1} a_k + \sum_{k=n_0}^n a_k \\ &\leq \sum_{k=1}^{n_0-1} a_k + \lambda \sum_{k=n_0}^n b_k \\ &= \sum_{k=1}^{n_0-1} a_k + \lambda(B_n - B_{n_0-1}) \\ &= C + \lambda B_n, \end{aligned}$$

where $C = \sum_{k=1}^{n_0-1} a_k - \lambda B_{n_0-1}$ is constant.

Since $\sum b_n$ converges, $\{B_n\}$ is bounded. Hence $\{A_n\}$ is bounded. Being increasing and bounded, $\{A_n\}$ converges. Therefore, $\sum a_n$ converges. $\square$

Example.

- For $0 < p \leq 1$, $\frac{1}{n^p} \geq \frac{1}{n} > 0$, so $\sum \frac{1}{n^p}$ is divergent.
- For $p \geq 2$, $0 < \frac{1}{n^p} \leq \frac{1}{n^2}$, so $\sum \frac{1}{n^p}$ is convergent.

Theorem 3.47 (Limit comparison test). Suppose $\{a_n\}$ and $\{b_n\}$ are positive sequences and

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lambda.$$

- (1) If $\lambda > 0$, then $\sum a_n$ converges if and only if $\sum b_n$ converges.

(2) If $\lambda = 0$, then $\sum b_n$ convergent implies $\sum a_n$ convergent; $\sum a_n$ divergent implies $\sum b_n$ divergent.

Proof.

(1) By definition of the limit, there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$,

$$\frac{1}{2}\lambda \leq \frac{a_n}{b_n} \leq \frac{3}{2}\lambda.$$

Multiplying through by b_n gives

$$\frac{\lambda}{2}b_n \leq a_n \leq \frac{3\lambda}{2}b_n.$$

If $\sum b_n$ diverges, use first inequality. If $\sum b_n$ converges, use second inequality.

(2) If $\lambda = 0$, then for any $\varepsilon > 0$, there exists n_0 such that for all $n \geq n_0$,

$$\frac{a_n}{b_n} < \varepsilon,$$

so

$$a_n < \varepsilon b_n.$$

If $\sum b_n$ converges, then by the comparison test, $\sum a_n$ also converges. For the second statement, we argue by contrapositive: if $\sum b_n$ converges, then $\sum a_n$ converges.

□

Example. To determine the convergence of $\sum \frac{1}{n-\sqrt{n}+1}$, consider the series $\sum \frac{1}{n}$. Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n}{n - \sqrt{n} + 1} = 1.$$

Since $\sum \frac{1}{n}$ is divergent, by limit comparison test, $\sum \frac{1}{n-\sqrt{n}+1}$ is divergent.

Example. To determine the convergence of $\sum \frac{1}{\sqrt{n}-1+1}$, consider the series $\sum \frac{1}{\sqrt{n}}$. Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{n} - 1 + 1} = 1.$$

Since $\sum \frac{1}{\sqrt{n}}$ is divergent, by limit comparison test, $\sum \frac{1}{\sqrt{n}-1+1}$ is divergent.

Proposition 3.48 (*p*-series). For $p > 1$, $\sum \frac{1}{n^p}$ is convergent.

Proof. We use dyadic partition, i.e., cleverly grouping terms of the series. Consider the case $k = 2^m$:

$$\begin{aligned}
\sum_{n=1}^k \frac{1}{n^p} &= \frac{1}{1} + \frac{1}{2^p} + \left(\frac{1}{3^p} + \frac{1}{4^p} \right) + \left(\frac{1}{5^p} + \frac{1}{6^p} + \frac{1}{7^p} + \frac{1}{8^p} \right) + \cdots \\
&\quad + \sum_{n=2^{k-1}+1}^{2^k} \frac{1}{n^p} + \cdots + \sum_{n=2^{m-1}+1}^{2^m} \frac{1}{n^p} \\
&\leq 1 + \frac{1}{2^p} + \left(\frac{1}{3^p} + \frac{1}{3^p} \right) + \left(\frac{1}{5^p} + \frac{1}{5^p} + \frac{1}{5^p} + \frac{1}{5^p} \right) \\
&\quad + \cdots + \sum_{n=2^{k-1}+1}^{2^k} \frac{1}{(2^{k-1}+1)^p} + \cdots + \sum_{n=2^{m-1}+1}^{2^m} \frac{1}{(2^{m-1}+1)^p} \\
&= 1 + \frac{1}{2^p} + \frac{2}{3^p} + \frac{4}{5^p} + \cdots + \frac{2^{k-1}}{(2^{k-1}+1)^p} + \cdots + \frac{2^{m-1}}{(2^{m-1}+1)^p} \\
&< 1 + \frac{1}{2^p} + \frac{2}{2^p} + \frac{4}{4^p} + \cdots + \frac{2^{k-1}}{(2^{k-1})^p} + \cdots + \frac{2^{m-1}}{(2^{m-1})^p} \\
&= 1 + \frac{1}{2^p} + \left(\frac{1}{2} \right)^{p-1} + \left(\frac{1}{4} \right)^{p-1} + \cdots + \left(\frac{1}{2^{k-1}} \right)^{p-1} + \cdots + \left(\frac{1}{2^{m-1}} \right)^{p-1} \\
&= 1 + \frac{1}{2^p} + \sum_{l=1}^{m-1} \left(\frac{1}{2^l} \right)^{p-1} \\
&= 1 + \frac{1}{2^p} + \sum_{l=1}^{m-1} \left(\frac{1}{2^{p-1}} \right)^l \leq C
\end{aligned}$$

for some constant C . Since the sequence of partial sums is bounded and non-negative, by monotone convergence theorem, $\sum \frac{1}{n^p}$ is convergent for $p > 1$. $\square$

Theorem 3.49 (Cauchy condensation test). *Suppose $\{a_n\}$ is positive and decreasing. Then $\sum a_n$ is convergent if and only if*

$$\sum_{n=1}^{\infty} 2^n a_{2^n}$$

is convergent.

Proof. The proof is similar to the proof of convergence of p -series. It is left as an exercise. $\square$

Example. To determine the convergence of $\sum \frac{1}{n \log_2 n}$, note that

$$2^n a_{2^n} = 2^n \cdot \frac{1}{2^n \log_2 2^n} = \frac{1}{n}.$$

Since $\sum \frac{1}{n}$ is divergent, by Cauchy condensation test, $\sum \frac{1}{n \log_2 n}$ is divergent.

Example. To determine the convergence of $\sum \frac{1}{n(\log_2 n)^2}$, note that

$$2^n a_{2^n} = 2^n \cdot \frac{1}{2^n (\log_2 2^n)^2} = \frac{1}{k^n}.$$

Since $\sum \frac{1}{k^2}$ is convergent, by Cauchy condensation test, $\sum \frac{1}{n(\log_2 n)^2}$ is convergent.

The Leibniz test is used to determine the convergence of an alternating series. It can be viewed as a general case of the alternating harmonic series.

Theorem 3.50 (Leibniz test). *Suppose $\{a_n\}$ is a decreasing sequence with $a_n \geq 0$ and $a_n \rightarrow 0$. Then $\sum (-1)^{n+1} a_n$ converges.*

Proof. We use the Cauchy criterion. Let $m > n$. We want to bound the absolute value of

$$S_{n,m} = \sum_{k=n+1}^m (-1)^{k-1} a_k.$$

We claim that $|S_{n,m}| \leq a_{n+1}$.

Case 1: last term is positive. Then group the terms as follows:

$$\underbrace{(a_{n+1} - a_{n+2})}_{\geq 0} + \cdots + \underbrace{(a_{m-2} - a_{m-1})}_{\geq 0} + a_m > 0.$$

On the other hand, group as

$$a_{n+1} - \underbrace{(a_{n+2} - a_{n+3})}_{\geq 0} - \cdots - \underbrace{(a_{m-1} - a_m)}_{\geq 0} < a_{n+1}.$$

Case 2: last term is negative. Then group as

$$\underbrace{(a_{n+1} - a_{n+2})}_{\geq 0} + \cdots + \underbrace{(a_{m-1} - a_m)}_{\geq 0} > 0.$$

On the other hand,

$$a_{n+1} - \underbrace{(a_{n+2} - a_{n+3})}_{\geq 0} - \cdots - \underbrace{(a_{m-2} - a_{m-1})}_{\geq 0} - a_m < a_{n+1}.$$

In both cases, we have

$$\left| \sum_{k=n+1}^m (-1)^{k-1} a_k \right| < a_{n+1}.$$

Since $a_n \rightarrow 0$, there exists $N \in \mathbb{N}$ such that $a_{n+1} < \varepsilon$ for all $n \geq N$. Then for $m > n \geq N$,

$$\left| \sum_{k=n+1}^m (-1)^{k-1} a_k \right| < a_{n+1} < \varepsilon.$$

□

Example. To show that $\sum \frac{(-1)^n}{\sqrt{n}}$ is convergent, let $a_n = \frac{1}{\sqrt{n}}$. Then $a_n \geq 0$ is decreasing and $a_n \rightarrow 0$. By Leibniz test, $\sum \frac{(-1)^n}{\sqrt{n}}$ is convergent.

Theorem 3.51 (Root test). *Let $\{a_n\}$ be a sequence.*

- (1) *If $|a_n|^{1/n} \leq r$ for all $n \geq n_0$, where $0 \leq r < 1$, then $\sum a_n$ is absolutely convergent.*
- (2) *If $|a_n|^{1/n} \geq 1$ for all $n \geq n_0$, then $\sum a_n$ is divergent.*

Proof.

- (1) Suppose $|a_n|^{1/n} \leq r < 1$ for all $n \geq n_0$. Taking n -th power gives $|a_n| \leq r^n$. Since $\sum r^n$ converges $0 < r < 1$, by comparison test, $\sum |a_n|$ converges. Hence, $\sum a_n$ absolutely converges.
- (2) Suppose $|a_n|^{1/n} \geq 1$ for all $n \geq n_0$. Then $|a_n| \geq 1$, so $a_n \not\rightarrow 0$. By divergence test, $\sum a_n$ diverges.

□

Corollary 3.52. *Suppose $\limsup_{n \rightarrow \infty} |a_n|^{1/n} = r \geq 0$.*

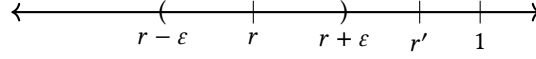
- (1) *If $r < 1$, then $\sum a_n$ absolutely converges.*

(2) If $r > 1$, then $\sum a_n$ diverges.

Proof.

(1) Suppose $r = \limsup_{n \rightarrow \infty} |a_n|^{1/n} < 1$. Let $b_n = \sup\{|a_k|^{1/k} : k \geq n\}$. By definition, $\lim_{n \rightarrow \infty} b_n = r$, so for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|b_n - r| < \varepsilon \quad \text{for all } n \geq N.$$



Choose $\varepsilon = \frac{1-r}{2}$. Then there exists r' such that $b_n < r' < 1$ for all $n \geq N$, namely,

$$|a_n|^{1/n} \leq b_n < r' < 1 \quad \text{for all } n \geq N.$$

By root test, $\sum a_n$ is absolutely convergent.

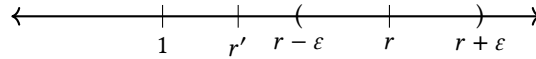
(2) Suppose $r = \limsup_{n \rightarrow \infty} |a_n|^{1/n} > 1$. Since r is a limit point of $\{|a_n|^{1/n}\}$, there exists a subsequence $\{b_k = |a_{n_k}|^{1/n_k} : k \in \mathbb{N}\}$ such that $\lim_{k \rightarrow \infty} b_k = r > 1$. Thus, there exists $K \in \mathbb{N}$ where

$$|b_k - r| < \varepsilon \quad \text{for all } k \geq K.$$

In particular, take $\varepsilon = \frac{r-1}{2}$. Then there exists r' such that

$$b_k > r' > 1 \quad \text{for all } k \geq K.$$

That is, $|a_{n_k}|^{1/n_k} \geq 1$, so $|a_{n_k}| \geq 1$. Since the subsequence $a_{n_k} \not\rightarrow 0$, this implies that $a_n \not\rightarrow 0$. By divergence test, $\sum a_n$ is divergent.



□

Theorem 3.53 (Ratio test). *Let $\{a_n\}$ be a sequence with non-zero terms.*

(1) If $\left| \frac{a_{n+1}}{a_n} \right| \leq r$ for all $n \geq n_0$, where $0 < r < 1$, then $\sum a_n$ absolutely converges.

(2) If $\left| \frac{a_{n+1}}{a_n} \right| \geq 1$ for all $n \geq n_0$, then $\sum a_n$ diverges.

Proof.

(1) The idea is to compare $\sum a_n$ with a geometric series. Suppose $\left| \frac{a_{n+1}}{a_n} \right| \leq r < 1$ for $n \geq n_0$. Then for all $n \geq n_0 + 1$,

$$\left| \frac{a_n}{a_{n-1}} \right| \leq r, \quad \left| \frac{a_{n-1}}{a_{n-2}} \right| \leq r, \quad \dots, \quad \left| \frac{a_{n_0+1}}{a_{n_0}} \right| \leq r.$$

Multiplying these inequalities gives

$$\left| \frac{a_n}{a_{n_0}} \right| \leq r^{n-n_0},$$

so $|a_n| \leq |a_{n_0}| \cdot r^{n-n_0}$ for all $n \geq n_0 + 1$. Since $\sum |a_{n_0}| r^{n-n_0}$ is a convergent geometric series, by comparison test, $\sum |a_n|$ converges.

(2) Using the same argument, we have $\left| \frac{a_n}{a_{n_0}} \right| \geq 1$, so $|a_n| \leq |a_{n_0}| > 0$ for all $n \geq n_0 + 1$.

Since $a_n \not\rightarrow 0$, by divergence test, $\sum a_n$ diverges.

□

Corollary 3.54. Suppose $a_n \neq 0$ and $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = r \geq 0$.

- (1) If $r < 1$, then $\sum a_n$ absolutely converges.
- (2) If $r > 1$, then $\sum a_n$ diverges.
- (3) If $r = 1$, then the test is inconclusive, i.e. $\sum a_n$ may converge or diverge.

For (3), $\sum \frac{1}{n}$ is divergent but $\sum \frac{1}{n^2}$ is convergent.

Example. To determine the convergence of $\sum \frac{(-1)^n n^2}{2^n}$, let $a_n = \frac{(-1)^n n^2}{2^n}$.

- To use the root test, evaluate

$$|a_n|^{1/n} = \left| \frac{(-1)^n n^2}{2^n} \right|^{1/n} = \frac{(n^2)^{1/n}}{2} \rightarrow \frac{1}{2}$$

since $n^{1/n} \rightarrow 1$. By root test, $\sum a_n$ absolutely converges and thus converges.

- To use the ratio test, evaluate

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{2} \left(\frac{n+1}{n} \right)^2 \rightarrow \frac{1}{2}.$$

By ratio test, $\sum a_n$ converges.

Example. To determine the convergence of $\sum \frac{2^n}{n!}$, let $a_n = \frac{2^n}{n!}$.

- To use the root test, evaluate

$$|a_n|^{1/n} = \left| \frac{2^n}{n!} \right|^{1/n} = \frac{2}{(n!)^{1/n}}.$$

To bound this term from above, we need to find a lower bound for $(n!)^{1/n}$. Note that

$$n! = n!(n-1) \cdots 2 \cdot 1 > n(n-1) \cdots \frac{n}{2} > \left(\frac{n}{2} \right)^{n/2},$$

so taking roots gives $(n!)^{1/n} > \left(\frac{n}{2} \right)^{1/2}$. Thus,

$$\frac{2}{(n!)^{1/n}} \leq \frac{2}{\left(\frac{n}{2} \right)^{1/2}}.$$

By squeeze theorem, $|a_n|^{1/n} \rightarrow 0$. By root test, $\sum a_n$ is absolutely convergent.

- To use the ratio test, evaluate $\left| \frac{a_{n+1}}{a_n} \right| = \frac{2}{n+1} \rightarrow 0$, so that $\sum a_n$ converges.

Proposition 3.55. $\sum_{n=0}^{\infty} \frac{1}{n!} = e$.

Proof. One can show that the series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges using the root test or ratio test. Let $a_n = \left(1 + \frac{1}{n} \right)^n$. By

binomial theorem,

$$\begin{aligned} a_n &= \sum_{k=0}^n \binom{n}{k} \left(\frac{1}{n}\right)^k \\ &= \sum_{k=0}^n \frac{n(n-1)\cdots(n-k+1)}{k!} \cdot \frac{1}{n^k} \\ &= \sum_{k=0}^n \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right). \end{aligned}$$

Denote $x_{n,k} = \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right)$. Then $x_{n,k} < 1$, so $a_n < \sum_{k=0}^n \frac{1}{k!}$.

Let $\lambda = \sum_{k=0}^{\infty} \frac{1}{k!}$. Take limits $n \rightarrow \infty$ on both sides. By order-preserving property of limits,

$$e = \lim_{n \rightarrow \infty} a_n \leq \lambda.$$

Then we prove $\lambda \leq e$. It suffices to prove that every partial sum is no larger than e , i.e., for any $N \in \mathbb{N}$, $\sum_{k=0}^N \frac{1}{k!} \leq e = \lim_{n \rightarrow \infty} a_n$. Note that

$$a_n = \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \frac{1}{k!} x_{n,k} > \sum_{k=0}^N \frac{1}{k!} x_{n,k}. \quad (*)$$

Also note that for $k = 0, 1, \dots, N$, one has $x_{n,k} \rightarrow 1$. Taking limit on both sides of (*),

$$e = \lim_{n \rightarrow \infty} a_n \geq \lim_{n \rightarrow \infty} \sum_{k=0}^N \frac{1}{k!} x_{n,k} = \sum_{k=0}^N \frac{1}{k!}.$$

□

Corollary 3.56. e is irrational.

Proof. Let $s_n = \sum_{k=0}^n \frac{1}{k!}$ denote the partial sum. Then $e - s_n = \sum_{k=n+1}^{\infty} \frac{1}{k!}$. For $k \geq n+1$,

$$\frac{1}{k!} = \frac{1}{(n+1)!} \cdot \frac{1}{k(k-1)\cdots(n+2)} \leq \frac{1}{(n+1)!} \cdot \left(\frac{1}{n+2}\right)^{k-n-1}.$$

Thus,

$$e - s_n \leq \frac{1}{(n+1)!} \sum_{k=n+1}^{\infty} \left(\frac{1}{n+2}\right)^{k-n-1} = \frac{1}{(n+1)!} \cdot \frac{n+2}{n+1}.$$

For a contradiction, assume e is rational. Then $e = \frac{p}{q}$ for some $p, q \in \mathbb{N}$. Thus,

$$\frac{p}{q} - \sum_{k=0}^n \frac{1}{k!} \leq \frac{1}{(n+1)!} \cdot \frac{n+2}{n+1}.$$

Take $n = q$. Then

$$\frac{p}{q} - \sum_{k=0}^q \frac{1}{k!} \leq \frac{1}{(q+1)!} \cdot \frac{q+2}{q+1}.$$

Notice that $q! \left(\frac{p}{q} - \sum_{k=0}^q \frac{1}{k!}\right)$ is a natural number, so $q!(e - s_q) \geq 1$. On the other hand,

$$q!(e - s_q) \leq \frac{q!}{(q+1)!} \cdot \frac{q+2}{q+1} = \frac{q+2}{(q+1)^2}.$$

Since $\frac{q+2}{(q+1)^2} \rightarrow 0$ as $q \rightarrow \infty$, we can pick q large enough such that $\frac{q+2}{(q+1)^2} < 1$. This leads to a contradiction. □

3.10 Abel's summation formula

Theorem 3.57 (Abel's summation formula). *Let $A_n = \sum_{k=1}^n a_k$. For any $q > p$,*

$$\sum_{n=p}^q a_n b_n = A_q b_q - A_{p-1} b_p + \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}). \quad (6)$$

Proof. Note that $a_n = A_n - A_{n-1}$, so we can rewrite LHS as

$$\begin{aligned} \sum_{n=p}^q (A_n - A_{n-1}) b_n &= \sum_{n=p}^q A_n b_n - \sum_{n=p}^q A_{n-1} b_n \\ &= \sum_{n=p}^q A_n b_n - \sum_{n=p-1}^{q-1} A_n b_{n+1} \\ &= A_q b_q + \sum_{n=p}^{q-1} A_n b_n - \left(A_{p-1} b_p + \sum_{n=p}^{q-1} A_n b_{n+1} \right) \\ &= A_q b_q - A_{p-1} b_p + \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}). \end{aligned}$$

□

Remark. Abel's summation formula is useful in proofs to show that $\sum a_n b_n$ converges using Cauchy criterion.

Theorem 3.58 (Dirichlet's test). *Let $\{a_n\}$ satisfy that its partial sums $\{A_n = \sum_{k=1}^n a_k\}$ are bounded. Let $\{b_n\}$ be a positive decreasing sequence approaching zero. Then $\sum a_n b_n$ is convergent.*

Proof. Let $\varepsilon > 0$. We will prove $\sum a_n b_n$ is Cauchy, namely, there exists $N \in \mathbb{N}$ such that for all $q > p \geq N$,

$$\left| \sum_{n=p}^q a_n b_n \right| < \varepsilon.$$

By Abel's summation formula,

$$\begin{aligned} \left| \sum_{n=p}^q a_n b_n \right| &= \left| A_q b_q - A_{p-1} b_p + \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) \right| \\ &\leq |A_q| |b_q| + |A_{p-1}| |b_p| + \sum_{n=p}^{q-1} |A_n| |b_n - b_{n+1}|. \end{aligned}$$

Since $|A_q| |b_q| + |A_{p-1}| |b_p|$ is constant, it remains to bound $\sum_{n=p}^{q-1} |A_n| |b_n - b_{n+1}|$.

Since the partial sums $\{A_n\}$ are bounded, there exists $C > 0$ such that $|A_n| \leq C$ for all $n \in \mathbb{N}$. Since $\{b_n\}$ is decreasing, we have $b_n \geq b_{n+1}$, so $|b_n - b_{n+1}| = b_n - b_{n+1}$. Hence,

$$\sum_{n=p}^{q-1} |A_n| |b_n - b_{n+1}| \leq C \sum_{n=p}^{q-1} (b_n - b_{n+1}) = C(b_p - b_q) < C b_p.$$

Therefore,

$$\left| \sum_{n=p}^q a_n b_n \right| \leq C b_p + C b_p + C b_p \leq 3C b_N.$$

□

Example. To determine the convergence of $\sum (-1)^n \frac{1}{n}$, let $a_n = (-1)^n$ and $b_n = \frac{1}{n}$. We check that the conditions of Dirichlet's test are satisfied.

- (1) The partial sums of $\sum a_n$ are bounded, since

$$A_n = (-1) + 1 + (-1) + 1 + \cdots + (-1)^n \in \{-1, 0\}.$$

- (2) It is obvious that $\{b_n\}$ is a positive decreasing sequence approaching zero.

By Dirichlet's test, $\sum (-1)^n \frac{1}{n}$ converges.

Example. To determine the convergence of $\sum \frac{\sin n}{n}$, let $a_n = \sin n$ and $b_n = \frac{1}{n}$.

- (1) We check that the partial sums of $\sum a_n$ are bounded. Let $A_n = \sum_{k=1}^n \sin k$. To evaluate this summation, we need the trigonometric formula $\sin \alpha \sin \beta = \frac{1}{2}(\cos(\alpha - \beta) - \cos(\alpha + \beta))$. Plug $\beta = k$ to get

$$\sin \alpha \sin k = \frac{1}{2}(\cos(\alpha - k) - \cos(\alpha + k)).$$

We want to find some α such that we can use telescoping sum to cancel out terms. Note that the preceding term is

$$\sin \alpha \sin(k-1) = \frac{1}{2}(\cos(k-1-\alpha) - \cos(k-1+\alpha)).$$

Equating $\cos(k-\alpha) = \cos(k-1+\alpha)$ and solving yields $\alpha = \frac{1}{2}$. Thus,

$$\begin{aligned} \sum_{k=1}^n \sin \frac{1}{2} \sin k &= \frac{1}{2} \left(\sum_{k=1}^n \cos \left(k - \frac{1}{2} \right) - \cos \left(k + \frac{1}{2} \right) \right) \\ &= \frac{1}{2} \left(\cos \frac{1}{2} - \cos \frac{3}{2} + \cos \frac{3}{2} - \cos \frac{5}{2} + \cdots + \cos \left(n - \frac{1}{2} \right) - \cos \left(n + \frac{1}{2} \right) \right) \\ &= \frac{1}{2} \left(\cos \frac{1}{2} - \cos \left(n + \frac{1}{2} \right) \right). \end{aligned}$$

Hence,

$$A_n = \sum_{k=1}^n \sin k = \frac{\cos \frac{1}{2} - \cos \left(n + \frac{1}{2} \right)}{2 \sin \frac{1}{2}},$$

so that

$$|A_n| = \frac{\left| \cos \frac{1}{2} - \cos \left(n + \frac{1}{2} \right) \right|}{\left| 2 \sin \frac{1}{2} \right|} \leq \frac{\left| \cos \frac{1}{2} \right| + \left| \cos \left(n + \frac{1}{2} \right) \right|}{2 \sin \frac{1}{2}} \leq \frac{1+1}{2 \sin \frac{1}{2}} = \frac{1}{\sin \frac{1}{2}}$$

which is a constant independent of n . Hence, $\{A_n\}$ is bounded.

- (2) It is obvious that $\{b_n\}$ is a positive decreasing sequence approaching zero.

Theorem 3.59 (Abel's test). *Let $\{a_n\}$ satisfy that the partial sums of $\sum a_n$ are convergent. Let $\{b_n\}$ be a monotone and bounded sequence. Then $\sum a_n b_n$ is convergent.*

Proof. Let the partial sums be $A_n = \sum_{k=1}^n a_k$. Since $\{A_n\}$ converges, let $A_n \rightarrow A$ as $n \rightarrow \infty$.

Let $R_n = A_n - A$. Then $R_n \rightarrow 0$. Write

$$\begin{aligned} a_n &= A_n - A_{n-1} \\ &= (A_n - A) - (A_{n-1} - A) \\ &= R_n - R_{n-1}. \end{aligned}$$

By Abel's summation formula,

$$\left| \sum_{n=p}^q a_n b_n \right| = \left| R_q b_q - R_{p-1} b_p + \sum_{n=p}^{q-1} R_n (b_n - b_{n+1}) \right|.$$

Case 1: $\{b_n\}$ is decreasing. Then $|b_n - b_{n+1}| = b_n - b_{n+1}$, so

$$\left| \sum_{n=p}^q a_n b_n \right| \leq |R_q| |b_q| - |R_{p-1}| |b_p| + \sum_{n=p}^{q-1} |R_n| (b_n - b_{n+1}).$$

Since $R_n \rightarrow 0$, for any $\varepsilon_1 > 0$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|R_n| < \varepsilon_1$. Then for $p \geq N_1 + 1$,

$$\begin{aligned} \left| \sum_{n=p}^q a_n b_n \right| &\leq \varepsilon_1 |b_q| + \varepsilon_1 |b_p| + \sum_{n=p}^{q-1} \varepsilon_1 (b_n - b_{n+1}) \\ &= \varepsilon_1 |b_q| + \varepsilon_1 |b_p| + \varepsilon_1 (b_p - b_q) \\ &= \varepsilon_1 (|b_q| + |b_p| + (b_p - b_q)) \\ &\leq 4C \cdot \varepsilon_1, \end{aligned}$$

where the last inequality follows from the boundedness of $\{b_n\}$. Choose $\varepsilon_1 = \frac{\varepsilon}{4C}$. Hence, $\sum a_n b_n$ satisfies the Cauchy criterion, so it is convergent.

Case 2: $\{b_n\}$ is increasing. Then $|b_n - b_{n+1}| = b_{n+1} - b_n$. The proof is similar. □

Example. To determine the convergence of $\sum \frac{(-1)^n}{n} \left(1 + \frac{1}{n}\right)^n$, let $a_n = \frac{(-1)^n}{n}$ and $b_n = \left(1 + \frac{1}{n}\right)^n$.

(1) From above, $\sum a_n$ is convergent, so its partial sums are convergent.

(2) Since $\{b_n\}$ is increasing and converges to e , it is monotone and bounded.

By Abel's test, $\sum \frac{(-1)^n}{n} \left(1 + \frac{1}{n}\right)^n$ is convergent.

3.11 Rearrangements of series

Definition 3.60. $\{b_n\}$ is called a *rearrangement* of $\{a_n\}$ if

$$b_n = a_{f(n)}$$

where $f: \mathbb{N} \rightarrow \mathbb{N}$ is a bijective map.

The next result states that for any rearrangement, the series is absolutely convergent, and converges to the same limit as the original series.

Theorem 3.61. Let $\{a_n\}$ be a sequence such that $\sum a_n$ is absolutely convergent. Let $\{b_n\}$ be a rearrangement of $\{a_n\}$. Then $\sum b_n$ is absolutely convergent, and

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n.$$

Proof. Suppose $\sum a_n$ converges absolutely to A , and let $\sum b_k$ be a rearrangement of $\sum a_n$. Denote the partial sums

$$s_n = \sum_{k=1}^n a_k, \quad t_m = \sum_{k=1}^m b_k.$$

We want to show that $t_m \rightarrow A$. Let $\varepsilon > 0$. Since $s_n \rightarrow A$, choose $N_1 \in \mathbb{N}$ such that

$$|s_n - A| < \frac{\varepsilon}{2} \quad \text{for all } n \geq N_1.$$

Since the convergence is absolute, we can choose $N_2 \in \mathbb{N}$ such that

$$\sum_{k=m+1}^n |a_k| < \frac{\varepsilon}{2} \quad \text{for all } n > m \geq N_2.$$

Take $N = \max\{N_1, N_2\}$. We know that the finite set of terms $\{a_1, a_2, \dots, a_N\}$ must all appear in the rearranged series, and we want to move far enough out in the series $\sum b_n$ so that we have included all of these terms. Thus, choose

$$M = \max\{f(k) : 1 \leq k \leq N\}.$$

It should now be evident that if $m \geq M$, then $\{t_m - s_N\}$ consists of a finite set of terms, the absolute values of which appear in the tail $\sum_{k=N+1}^{\infty} |a_k|$. Our choice of N_2 earlier then guarantees $|t_m - s_N| < \frac{\varepsilon}{2}$, and so

$$\begin{aligned} |t_m - A| &= |(t_m - s_N) + (s_N - A)| \\ &\leq |t_m - s_N| + |s_N - A| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

whenever $m \geq M$. □

If $\sum a_n$ is conditionally convergent, then $\sum a_n$ can be re-arranged to sum to any limit we want:

Theorem 3.62 (Riemann rearrangement theorem). *Let $\{a_n\}$ be a sequence such that $\sum a_n$ converges conditionally. Then for any $A \in \mathbb{R}$, there exists a rearrangement $\{b_n\}$ of $\{a_n\}$ such that*

$$\sum_{n=1}^{\infty} b_n = A.$$

4 Continuity

4.1 Limits of functions

Notation. For $x \in \mathbb{R}$ and $\varepsilon > 0$, we denote the ε -neighbourhood centered at x by

$$V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon).$$

Definition 4.1. Let $A \subseteq \mathbb{R}$ be a non-empty. We call $x \in \mathbb{R}$ a *limit point* of A if for any $\varepsilon > 0$,

$$A \cap (V_\varepsilon(x) \setminus \{x\}) \neq \emptyset.$$

Theorem 4.2. $x \in \mathbb{R}$ is a limit point of A if and only if there exists a sequence $\{a_n\}$ such that $a_n \in A$, $a_n \neq x$, and

$$\lim_{n \rightarrow \infty} a_n = x.$$

Proof. use the definition □

Definition 4.3. Let $f: A \rightarrow \mathbb{R}$ be a function on A , and c be a limit point of A . We say that L is the *limit* of f at c , and write $\lim_{x \rightarrow c} f(x) = L$, if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (V_\delta(c) \setminus \{c\}), \quad |f(x) - L| < \varepsilon.$$

Remark. This definition means that the values of $f(x)$ can get arbitrarily close to L by choosing x closer and closer to c . The issue of what happens when $x = c$ is irrelevant from the point of view of limits of functions. In fact, c need not even be in the domain of f .

Definition 4.4. If $f: A \rightarrow \mathbb{R}$ does not have a limit at a limit point c of A , we say that f is *divergent* at c .

Exercise (Uniqueness of limit). Let $f: A \rightarrow \mathbb{R}$ be a function on A , and c be a limit point of A . If the limit of f at c exists, then it is unique.

Example. To show $\lim_{x \rightarrow 2} x^2 = 4$, factorize

$$|x^2 - 4| = |x - 2||x + 2|.$$

We can make $|x - 2|$ as small as we like, but we need an upper bound on $|x + 2|$ in order to know how small to choose δ .

If we agree that our δ -neighborhood around $c = 2$ must have radius no bigger than $\delta = 1$, then we get the upper bound $|x + 2| \leq |3 + 2| = 5$ for all $x \in V_\delta(c)$.

Now, choose $\delta = \min\{\varepsilon/5, 1\}$. If $0 < |x - 2| < \delta$, then

$$|x^2 - 4| = |x - 2||x + 2| < 5 \cdot \frac{\varepsilon}{5} = \varepsilon,$$

and the limit is proved.

Example. To show $\lim_{x \rightarrow 2} \sqrt{x} = \sqrt{2}$, factorize $|x - 2| = |\sqrt{x} - \sqrt{2}| \cdot |\sqrt{x} + \sqrt{2}|$, so that

$$|\sqrt{x} - \sqrt{2}| = \frac{|x - 2|}{|\sqrt{x} + \sqrt{2}|}.$$

Take $\delta < 1$. Then $|\sqrt{x} + \sqrt{2}| > 1 + \sqrt{2}$, so

$$|\sqrt{x} - \sqrt{2}| \leq \frac{|x - 2|}{1 + \sqrt{2}} < \frac{\delta}{1 + \sqrt{2}}.$$

Take $\delta = \min\{1, \varepsilon/(1 + \sqrt{2})\}$.

Definition 4.5. Let $f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be a function on A , and c be a limit point of A . We say that f is *bounded in a neighbourhood* of c if there exist $\delta > 0$ and $C > 0$ such that

$$|f(x)| \leq C \quad \text{for all } x \in A \cap V_\delta(c).$$

Theorem 4.6. If $f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$ has a limit at a limit point c of A , then f is bounded in a neighbourhood of c .

Proof. Take $\varepsilon = 1$. There exists $\delta > 0$ such that

$$|f(x) - L| < 1 \quad \text{for all } x \in A \cap (V_\delta(c) \setminus \{c\}).$$

By triangle inequality, $|f(x)| \leq |L| + 1$. Take $C = \max\{|L| + 1, |f(c)|\}$. □

We now look at a very important theorem, which roughly states that if $\lim_{x \rightarrow c} f(x) = L$, then no matter how we approach c (using a sequence), the function values still tend to L .

Theorem 4.7 (Sequential criterion). Let $f: A \rightarrow \mathbb{R}$ be a function, and c be a limit point of A . Then the following are equivalent:

- (1) $\lim_{x \rightarrow c} f(x) = L$;
- (2) For any sequence $\{a_n\} \subseteq A \setminus \{c\}$ such that $\lim_{n \rightarrow \infty} a_n = c$,

$$\lim_{n \rightarrow \infty} f(a_n) = L.$$

Proof.

$\Rightarrow$ Suppose $\lim_{x \rightarrow c} f(x) = L$. To prove (ii), take any sequence $\{a_n\} \subseteq A \setminus \{c\}$ such that $a_n \rightarrow c$. Our goal is to show that $f(a_n) \rightarrow L$.

Let $\varepsilon > 0$. Then there exists $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{for all } x \in (V_\delta(c) \setminus \{c\}) \cap A. \quad (1)$$

All we need to do then is argue that our particular sequence $\{a_n\}$ is eventually in $V_\delta(c)$. Since $a_n \rightarrow c$, there exists $N \in \mathbb{N}$ such that for $n \geq N$,

$$|a_n - c| < \delta,$$

so that $a_n \in V_\delta(c)$. By (1), we have $|f(a_n) - L| < \varepsilon$ for all $n \geq N$, as desired.

$\Leftarrow$ Suppose for any sequence $\{a_n\} \subseteq A \setminus \{c\}$ such that $a_n \rightarrow c$, we have $f(a_n) \rightarrow L$. For a contradiction, assume that $\lim_{x \rightarrow c} f(x) \neq L$. Then there exists $\varepsilon > 0$ such that

$$\forall \delta > 0, \quad \exists x \in (V_\delta(c) \setminus \{c\}) \cap A, \quad |f(x) - L| \geq \varepsilon. \quad (2)$$

To get a contradiction, we will construct a sequence $\{a_n\}$ such that $a_n \rightarrow c$ but $f(a_n) \not\rightarrow L$.

For each $n \in \mathbb{N}$, define $\delta_n = \frac{1}{n}$. By (2), choose $a_n \in V_{\delta_n}(c)$, $a_n \neq c$ such that $|f(a_n) - L| \geq \varepsilon$. Then $\{f(a_n)\}$ does not converge to L . This gives the desired contradiction. □

Example. $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist. To show this using the sequential criterion, consider the sequence $\{a_n\}$ defined by $a_n = \frac{1}{n}$, where $a_n \neq 0$ and $a_n \rightarrow 0$. However,

$$f(a_n) = \frac{1}{1/n} = n$$

is unbounded and thus divergent. By the sequential criterion, $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist.

Corollary 4.8 (Divergence criterion). *Let $f: A \rightarrow \mathbb{R}$ be a function, and c be a limit point of A . If there exist two sequences $\{a_n\}, \{b_n\} \subseteq A \setminus \{c\}$ such that*

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = c \quad \text{but} \quad \lim_{n \rightarrow \infty} f(a_n) \neq \lim_{n \rightarrow \infty} f(b_n),$$

then $\lim_{x \rightarrow c} f(x)$ does not exist.

Example. To show that $\lim_{x \rightarrow 1} \lfloor x \rfloor$ does not exist, consider sequences $\{a_n\}$ and $\{b_n\}$ both converging to 1:

$$\begin{aligned} a_n &= 1 - \frac{0.01}{n} \rightarrow 1 \implies f(a_n) = \lfloor a_n \rfloor = 0 \rightarrow 0 \\ b_n &= 1 + \frac{0.01}{n} \rightarrow 1 \implies f(b_n) = \lfloor b_n \rfloor = 1 \rightarrow 1 \end{aligned}$$

Since $\{f(a_n)\}$ and $\{f(b_n)\}$ converge to different limits, by the sequential criterion, $\lim_{x \rightarrow 1} \lfloor x \rfloor$ does not exist.

Corollary 4.9 (Algebraic operators). *Let $f, g: A \rightarrow \mathbb{R}$, c be a limit point of A , and $\lim_{x \rightarrow c} f(x) = L$, $\lim_{x \rightarrow c} g(x) = M$. Then*

- (1) $\lim_{x \rightarrow c} \lambda f(x) = \lambda L$ for all $\lambda \in \mathbb{R}$;
- (2) $\lim_{x \rightarrow c} (f \pm g)(x) = L \pm M$;
- (3) $\lim_{x \rightarrow c} fg(x) = LM$;
- (4) $\lim_{x \rightarrow c} \left(\frac{f}{g} \right)(x) = \frac{L}{M}$, provided $g(x) \neq 0$, $M \neq 0$.

Theorem 4.10 (Order preserving). *Let $f, g: A \rightarrow \mathbb{R}$, c be a limit point of A . Suppose $\lim_{x \rightarrow c} f(x) = L$, $\lim_{x \rightarrow c} g(x) = M$, and there exists $\delta > 0$ such that*

$$f(x) \geq g(x) \quad \text{for all } x \in A \cap (V_\delta(c) \setminus \{c\}).$$

Then $L \geq M$.

Theorem 4.11 (Squeeze theorem). *Let $f, g, h: A \rightarrow \mathbb{R}$, c a limit point of A . Suppose $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = L$, and there exists $\delta > 0$ such that*

$$f(x) \leq h(x) \leq g(x) \quad \text{for all } x \in A \cap (V_\delta(c) \setminus \{c\}).$$

Then $\lim_{x \rightarrow c} h(x) = L$.

Theorem 4.12 (Sign preserving). *Let $f: A \rightarrow \mathbb{R}$ be a function, and c be a limit point of A .*

- (1) *Suppose $\lim_{x \rightarrow c} f(x) = L > 0$. Then there exists $\delta > 0$ and $C > 0$ such that*

$$f(x) \geq C \quad \text{for all } x \in A \cap (V_\delta(c) \setminus \{c\}).$$

- (2) *Suppose $\lim_{x \rightarrow c} f(x) = L < 0$. Then there exists $\delta > 0$ and $C < 0$ such that*

$$f(x) \leq C \quad \text{for all } x \in A \cap (V_\delta(c) \setminus \{c\}).$$

Proof. We only prove (1). The proof of (2) is similar. Suppose $\lim_{x \rightarrow c} f(x) = L > 0$. This means that for all $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{for all } x \in (V_\delta(c) \setminus \{c\}) \cap A.$$

In particular, for $\varepsilon = \frac{L}{2}$, there exists $\delta > 0$ such that for all $x \in (V_\delta(c) \setminus \{c\}) \cap A$,

$$|f(x) - L| < \frac{L}{2} \implies f(x) - L > -\frac{L}{2} \implies f(x) > L - \frac{L}{2} = \frac{L}{2} > 0.$$

□

4.2 One-sided limits

Definition 4.13. Let $f: A \rightarrow \mathbb{R}$ be a function, c be a limit point of $A \cap (-\infty, c)$. We say that L is the *left limit* of f at c , and write $\lim_{x \rightarrow c^-} f(x) = L$, if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (c - \delta, c), \quad |f(x) - L| < \varepsilon.$$

Definition 4.14. Let $f: A \rightarrow \mathbb{R}$ be a function, c be a limit point of $A \cap (c, +\infty)$. We say that L is the *right limit* of f at c , and write $\lim_{x \rightarrow c^+} f(x) = L$, if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (c, c + \delta), \quad |f(x) - L| < \varepsilon.$$

Using a similar argument, one can show that one-sided limits satisfy the properties of limits:

- Boundedness
- Sequential criterion
- Order preserving
- Algebraic operators
- Squeeze theorem
- Sign preserving

Proposition 4.15. *The following are equivalent:*

- (1) $\lim_{x \rightarrow c} f(x) = L$;
- (2) $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L$.

Example. Consider $\lim_{x \rightarrow 2} \lfloor x \rfloor$. Since $\lim_{x \rightarrow 2^-} \lfloor x \rfloor = 1$ and $\lim_{x \rightarrow 2^+} \lfloor x \rfloor = 2$, the one-sided limits are not equal, so the limit does not exist.

4.3 Infinite limits

Definition 4.16. Let $f: A \rightarrow \mathbb{R}$ be a function, and c be a limit point of A .

- We say that $\lim_{x \rightarrow c} f(x) = +\infty$ if

$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (V_\delta(c) \setminus \{c\}), \quad f(x) > L.$$

- We say that $\lim_{x \rightarrow c} f(x) = -\infty$ if

$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (V_\delta(c) \setminus \{c\}), \quad f(x) < -L.$$

- We say that $\lim_{x \rightarrow c} f(x) = \infty$ if

$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (V_\delta(c) \setminus \{c\}), \quad |f(x)| > L.$$

We can also define one-sided infinite limits: Let $f: A \rightarrow \mathbb{R}$ be a function, and c be a limit point of $A \cap (-\infty, c)$.

- We say that $\lim_{x \rightarrow c^-} f(x) = +\infty$ if

$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (c - \delta, c), \quad f(x) > L.$$

- We say that $\lim_{x \rightarrow c^-} f(x) = -\infty$ if

$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (c - \delta, c), \quad f(x) < -L.$$

- We say that $\lim_{x \rightarrow c^-} f(x) = \infty$ if

$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (c - \delta, c), \quad |f(x)| > L.$$

Let $f: A \rightarrow \mathbb{R}$ be a function, and c be a limit point of $A \cap (c, +\infty)$.

- We say that $\lim_{x \rightarrow c^+} f(x) = +\infty$ if

$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (c, c + \delta), \quad f(x) > L.$$

- We say that $\lim_{x \rightarrow c^+} f(x) = -\infty$ if

$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (c, c + \delta), \quad f(x) < -L.$$

- We say that $\lim_{x \rightarrow c^+} f(x) = \infty$ if

$$\forall L > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap (c, c + \delta), \quad |f(x)| > L.$$

4.4 Limits at infinity

We define the limit of a function $f(x)$ as x becomes arbitrary positively large or arbitrarily negatively large.

Definition 4.17. Let $f: A \rightarrow \mathbb{R}$ be a function. Suppose there exists $c \in \mathbb{R}$ such that $(c, +\infty) \subset A$.

- We say that $\lim_{x \rightarrow +\infty} f(x) = L$ if

$$\forall \varepsilon > 0, \quad \exists M > c, \quad \forall x > M, \quad |f(x) - L| < \varepsilon.$$

- We say that $\lim_{x \rightarrow +\infty} f(x) = +\infty$ if

$$\forall L > 0, \quad \exists M > c, \quad \forall x > M, \quad f(x) > L.$$

- We say that $\lim_{x \rightarrow +\infty} f(x) = -\infty$ if

$$\forall L > 0, \quad \exists M > c, \quad \forall x > M, \quad f(x) < -L.$$

Definition 4.18. Let $f: A \rightarrow \mathbb{R}$ be a function. Suppose there exists $c \in \mathbb{R}$ such that $(-\infty, c) \subset A$.

- We say that $\lim_{x \rightarrow -\infty} f(x) = L$ if

$$\forall \varepsilon > 0, \quad \exists M < c, \quad \forall x < M, \quad |f(x) - L| < \varepsilon.$$

- We say that $\lim_{x \rightarrow -\infty} f(x) = +\infty$ if

$$\forall L > 0, \quad \exists M < c, \quad \forall x < M, \quad f(x) > L.$$

- We say that $\lim_{x \rightarrow -\infty} f(x) = -\infty$ if

$$\forall L > 0, \quad \exists M < c, \quad \forall x < M, \quad f(x) < -L.$$

Definition 4.19.

- We say that $\lim_{x \rightarrow +\infty} f(x) = \infty$ if $\lim_{x \rightarrow +\infty} |f(x)| = +\infty$.
- We say that $\lim_{x \rightarrow -\infty} f(x) = \infty$ if $\lim_{x \rightarrow -\infty} |f(x)| = +\infty$.
- We say that $\lim_{x \rightarrow \infty} f(x) = L$ if

$$\lim_{x \rightarrow +\infty} f(x) = L = \lim_{x \rightarrow -\infty} f(x).$$

Definition 4.20.

- We say that $\lim_{x \rightarrow \infty} f(x) = +\infty$ if

$$\lim_{x \rightarrow +\infty} f(x) = +\infty = \lim_{x \rightarrow -\infty} f(x).$$

- We say that $\lim_{x \rightarrow \infty} f(x) = -\infty$ if

$$\lim_{x \rightarrow +\infty} f(x) = -\infty = \lim_{x \rightarrow -\infty} f(x).$$

- We say that $\lim_{x \rightarrow \infty} f(x) = \infty$ if

$$\lim_{x \rightarrow +\infty} |f(x)| = +\infty = \lim_{x \rightarrow -\infty} |f(x)|.$$

Remark. $\lim_{x \rightarrow c, \pm\infty} f(x) = \pm\infty$ is still considered as divergent.

Example.

- $\lim_{x \rightarrow +\infty} 2^x = +\infty$, $\lim_{x \rightarrow -\infty} 2^x = 0$.
- $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$, $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$.
- $\lim_{x \rightarrow 0^+} \log x = -\infty$, $\lim_{x \rightarrow +\infty} \log x = +\infty$.
- $\lim_{x \rightarrow \infty} x^2 = +\infty$.
- $\lim_{x \rightarrow +\infty} x = +\infty$, $\lim_{x \rightarrow -\infty} x = -\infty$.

Theorem 4.21 (Sequential criterion).

- (1) $\lim_{x \rightarrow +\infty} f(x) = L$ if and only if for any sequence $\{a_n\}$ such that $\lim_{n \rightarrow \infty} a_n = +\infty$,

$$\lim_{n \rightarrow \infty} f(a_n) = L.$$

- (2) $\lim_{x \rightarrow -\infty} f(x) = L$ if and only if for any sequence $\{a_n\}$ such that $\lim_{n \rightarrow \infty} a_n = -\infty$,

$$\lim_{n \rightarrow \infty} f(a_n) = L.$$

Similar statements hold with L replaced by $\pm\infty, \infty$.

Theorem 4.22 (Boundedness).

- (1) If $\lim_{x \rightarrow +\infty} f(x) = L$, then $f(x)$ is bounded in $(C, +\infty)$ for some $C \in \mathbb{R}$.
- (2) If $\lim_{x \rightarrow -\infty} f(x) = L$, then $f(x)$ is bounded in $(-\infty, C)$ for some $C \in \mathbb{R}$.

Theorem 4.23 (Order preserving).

(1) Suppose $\lim_{x \rightarrow +\infty} f(x) = L$, $\lim_{x \rightarrow +\infty} g(x) = M$. If there exists $C \in \mathbb{R}$ such that

$$f(x) \geq g(x) \quad \text{for all } x > C.$$

Then $L \geq M$.

(2) Suppose $\lim_{x \rightarrow -\infty} f(x) = L$, $\lim_{x \rightarrow -\infty} g(x) = M$. If there exists $C \in \mathbb{R}$ such that

$$f(x) \geq g(x) \quad \text{for all } x < C.$$

Then $L \geq M$.

Theorem 4.24 (Algebraic operators for limits at infinity). Let $f, g: A \rightarrow \mathbb{R}$. Suppose there exists c such that $(c, +\infty) \subset A$, and $\lim_{x \rightarrow +\infty} f(x) = L$, $\lim_{x \rightarrow +\infty} g(x) = M$.

$$(1) \lim_{x \rightarrow +\infty} (f \pm g)(x) = L \pm M;$$

$$(2) \lim_{x \rightarrow +\infty} fg(x) = LM;$$

$$(3) \lim_{x \rightarrow +\infty} \left(\frac{f}{g}\right)(x) = \frac{L}{M}, \text{ provided } g(x) \neq 0, M \neq 0.$$

Theorem 4.25 (Algebraic operators for limits at infinity). Let $f, g: A \rightarrow \mathbb{R}$. Suppose there exists c such that $(-\infty, c) \subset A$, and $\lim_{x \rightarrow -\infty} f(x) = L$, $\lim_{x \rightarrow -\infty} g(x) = M$.

$$(1) \lim_{x \rightarrow -\infty} (f \pm g)(x) = L \pm M;$$

$$(2) \lim_{x \rightarrow -\infty} fg(x) = LM;$$

$$(3) \lim_{x \rightarrow -\infty} \left(\frac{f}{g}\right)(x) = \frac{L}{M}, \text{ provided } g(x) \neq 0, M \neq 0.$$

Theorem 4.26 (Squeeze theorem for limits at infinity).

(1) Suppose $\lim_{x \rightarrow +\infty} f(x) = L = \lim_{x \rightarrow +\infty} g(x)$. If $h(x)$ satisfies that there exists $C \in \mathbb{R}$ such that

$$f(x) \leq h(x) \leq g(x) \quad \text{for all } x > C.$$

Then $\lim_{x \rightarrow +\infty} h(x) = L$.

(2) Suppose $\lim_{x \rightarrow -\infty} f(x) = L = \lim_{x \rightarrow -\infty} g(x)$. If $h(x)$ satisfies that there exists $C \in \mathbb{R}$ such that

$$f(x) \leq h(x) \leq g(x) \quad \text{for all } x < C.$$

Then $\lim_{x \rightarrow -\infty} h(x) = L$.

Theorem 4.27 (Squeeze theorem for infinite limits). Let $f, g: A \rightarrow \mathbb{R}$ be functions, and c be a limit point of A .

(1) Suppose $\lim_{x \rightarrow c} f(x) = +\infty$ and there exists $\delta > 0$ such that

$$g(x) \geq f(x) \quad \text{for all } x \in A \cap (V_\delta(c) \setminus \{c\}).$$

Then $\lim_{x \rightarrow c} g(x) = +\infty$.

(2) Suppose $\lim_{x \rightarrow c} f(x) = -\infty$ and there exists $\delta > 0$ such that

$$g(x) \leq f(x) \quad \text{for all } x \in A \cap (V_\delta(c) \setminus \{c\}).$$

Then $\lim_{x \rightarrow c} g(x) = -\infty$.

Here the same statement holds with c replaced by $c^\pm, \pm\infty$.

Theorem 4.28 (Sign preserving for limits at infinity). Let $f: A \rightarrow \mathbb{R}$ be a function, and c satisfy that $(c, +\infty) \subset A$.

(1) Suppose $\lim_{x \rightarrow +\infty} f(x) = L > 0$. Then there exists $M > c$ and $C > 0$ such that

$$f(x) \geq C \quad \text{for all } x > M.$$

(2) Suppose $\lim_{x \rightarrow +\infty} f(x) = L < 0$. Then there exists $M > c$ and $C < 0$ such that

$$f(x) \leq C \quad \text{for all } x > M.$$

Theorem 4.29 (Sign preserving for limits at infinity). Let $f: A \rightarrow \mathbb{R}$ be a function, and c satisfy that $(-\infty, c) \subset A$.

(1) Suppose $\lim_{x \rightarrow -\infty} f(x) = L > 0$. Then there exists $M < c$ and $C > 0$ such that

$$f(x) \geq C \quad \text{for all } x < M.$$

(2) Suppose $\lim_{x \rightarrow -\infty} f(x) = L < 0$. Then there exists $M < c$ and $C < 0$ such that

$$f(x) \leq C \quad \text{for all } x < M.$$

Proposition 4.30. $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$.

Proof. By definition, we need to prove $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x = e$ and $\lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x}\right)^x = e$.

We first prove $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x = e$. We use a squeezing argument. Let $m \in \mathbb{N}$ be such that $m \leq x < m + 1$. This allows us to bound $\left(1 + \frac{1}{x}\right)^x$ from above and below by sequences related to e :

$$\begin{aligned} \left(1 + \frac{1}{x}\right)^x &\geq \left(1 + \frac{1}{x}\right)^m \geq \left(1 + \frac{1}{m+1}\right)^m \\ \left(1 + \frac{1}{x}\right)^x &\leq \left(1 + \frac{1}{x}\right)^{m+1} \leq \left(1 + \frac{1}{m}\right)^{m+1}. \end{aligned}$$

Hence,

$$\left(1 + \frac{1}{m+1}\right)^m \leq \left(1 + \frac{1}{x}\right)^x \leq \left(1 + \frac{1}{m}\right)^{m+1}.$$

Recall the limits $\lim_{m \rightarrow \infty} \left(1 + \frac{1}{m+1}\right)^m = e$ and $\lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^{m+1} = e$. Let $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that for $m \geq N$,

$$\left(1 + \frac{1}{m+1}\right)^m > e - \varepsilon \quad \text{and} \quad \left(1 + \frac{1}{m}\right)^{m+1} < e + \varepsilon.$$

If $x \geq N + 1$, then $m \geq N$, so combining the above inequalities yields

$$e - \varepsilon < \left(1 + \frac{1}{m+1}\right)^m \leq \left(1 + \frac{1}{x}\right)^x \leq \left(1 + \frac{1}{m}\right)^{m+1} < e + \varepsilon,$$

so that

$$\left| \left(1 + \frac{1}{x}\right)^x - e \right| < \varepsilon.$$

This proves that $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x = e$.

Next, we prove $\lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x}\right)^x = e$. Consider the change of variables $x = -t$. Then

$$\begin{aligned} \left(1 + \frac{1}{x}\right)^x &= \left(1 + \frac{1}{-t}\right)^{-t} = \left(1 - \frac{1}{t}\right)^{-t} = \frac{1}{\left(1 - \frac{1}{t}\right)^t} = \frac{1}{\left(\frac{t-1}{t}\right)^t} \\ &= \left(\frac{t}{t-1}\right)^t = \left(1 + \frac{1}{t-1}\right)^t = \left(1 + \frac{1}{t-1}\right)^{t-1} \left(1 + \frac{1}{t-1}\right). \end{aligned}$$

Since $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x = e$, for any $\varepsilon > 0$, there exists $M > 0$ such that

$$\left| \left(1 + \frac{1}{x}\right)^x - e \right| < \varepsilon \quad \text{for all } x > M.$$

Then for all $t > M + 1$,

$$\left| \left(1 + \frac{1}{t-1}\right)^{t-1} - e \right| < \varepsilon,$$

which shows that $\lim_{t \rightarrow +\infty} \left(1 + \frac{1}{t-1}\right)^{t-1} = e$. Also note that $\lim_{t \rightarrow +\infty} \left(1 + \frac{1}{t-1}\right) = 1$. Hence, taking the product, $\lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x}\right)^x = e$. $\square$

4.5 Continuous functions

Definition 4.31. A function $f: A \rightarrow \mathbb{R}$ is *continuous* at $c \in A$ if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in A \cap V_\delta(c), \quad |f(x) - f(c)| < \varepsilon.$$

Otherwise, f is *discontinuous* at c . If f is continuous at every point of A , then f is *continuous on A* .

Remark. If c is not a limit point of A , i.e., an isolated point, then f is always continuous at c .

Remark. If c is a limit point of A , then f is continuous at c if and only if

$$\lim_{x \rightarrow c} f(x) = f(c).$$

Theorem 4.32 (Sequential criterion). *Let $f: A \rightarrow \mathbb{R}$ be a function, and $c \in A$. Then f is continuous at c if and only if for any sequence $\{a_n\} \subseteq A$ with $\lim_{n \rightarrow \infty} a_n = c$,*

$$\lim_{n \rightarrow \infty} f(a_n) = f(c).$$

Corollary 4.33. *Let $f: A \rightarrow \mathbb{R}$ be a function, and $c \in A$. If there exists a sequence $\{a_n\} \subseteq A$ such that $\lim_{n \rightarrow \infty} a_n = c$ but $\lim_{n \rightarrow \infty} f(a_n) \neq f(c)$, then f is discontinuous at c .*

Theorem 4.34 (Algebraic operators). *Let $f, g: A \rightarrow \mathbb{R}$ be functions, and $c \in A$. If f and g are continuous at c , then $f \pm g, fg$ are continuous at c . If $g(x) \neq 0$ in a neighborhood of c , then $\frac{f}{g}$ is also continuous at c .*

Example. The following functions are continuous:

- Constant functions $f(x) = C$
- $f(x) = x$
- Polynomials $f(x) = \sum_{k=1}^n a_k x^k$
- Rational functions $f(x) = \frac{p(x)}{q(x)}$, $q(c) \neq 0$
- $f(x) = |x|$, $c = 0$

Example. The Dirichlet's function

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

is discontinuous.

Proof. Let $c \in \mathbb{R}$. We want to show that f is discontinuous at $c \in \mathbb{R}$. We will use the sequential criterion.

For any $c \in \mathbb{R}$, we can exploit the density of the rationals and irrationals:

- Choose a sequence $\{a_n\} \subseteq \mathbb{Q}$ such that $a_n \rightarrow c$. Since each $a_n \in \mathbb{Q}$, this implies $f(a_n) = 1$, so that $\lim_{n \rightarrow \infty} f(a_n) = 1$.
- Choose a sequence $\{b_n\} \subseteq \mathbb{R} \setminus \mathbb{Q}$ such that $b_n \rightarrow c$. Since each $b_n \notin \mathbb{Q}$, this implies $f(b_n) = 0$, so that $\lim_{n \rightarrow \infty} f(b_n) = 0$.

By the sequential criterion, $\lim_{x \rightarrow c} f(x)$ does not exist, so f is discontinuous at c . $\square$

Example. The functions $\sin x$ and $\cos x$ are continuous.

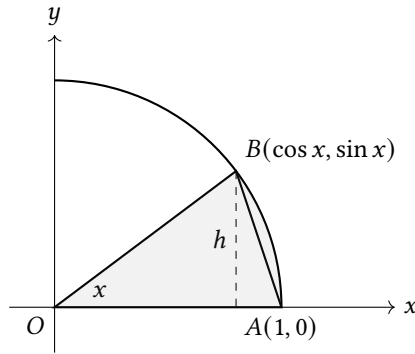
Proof. Let $c \in \mathbb{R}$ be arbitrary. We first prove that $\sin x$ is continuous at c , i.e., $\lim_{x \rightarrow c} \sin x = \sin c$, namely

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad |x - c| < \delta \implies |\sin x - \sin c| < \varepsilon.$$

We first need an inequality:

Lemma. $|\sin x| \leq |x|$ for $|x| \leq 1$.

Proof. We shall prove this geometrically. Since $|\sin x| = \sin |x|$, it suffices to prove it for the case $x \geq 0$. Consider a unit circle and an angle x in radians.



The area of $\triangle OAB$ is $\frac{1}{2} \sin x$. The area of sector OAB is $\frac{1}{2}x$. Since $\triangle OAB$ is contained in the sector OAB , we have $\frac{1}{2} \sin x \leq \frac{1}{2}x$, or $\sin x \leq x$. $\square$

Using the sum-to-product identity,

$$|\sin x - \sin c| = \left| 2 \cos \frac{x+c}{2} \sin \frac{x-c}{2} \right| = 2 \left| \cos \frac{x+c}{2} \right| \left| \sin \frac{x-c}{2} \right|.$$

Since $|\cos \theta| \leq 1$ for all θ , and using our inequality $|\sin \theta| \leq |\theta|$,

$$|\sin x - \sin c| \leq 2 \cdot 1 \cdot \left| \frac{x-c}{2} \right| = |x-c|.$$

To ensure $|\sin x - \sin c| < \varepsilon$, choose $\delta = \varepsilon$. Then, $|x - c| < \delta \implies |\sin x - \sin c| \leq |x - c| < \delta = \varepsilon$.

Thus, $\sin x$ is continuous for all $c \in \mathbb{R}$. $\square$

Similarly, for $\cos x$, we use

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}.$$

By limit laws, it follows that the following trigonometric functions are also continuous:

- $\tan x$ at c , where $\cos c \neq 0$
- $\cot x$ at c , where $\sin c \neq 0$
- $\sec x$ at c , where $\cos c \neq 0$
- $\csc x$ at c , where $\sin c \neq 0$

Example. The exponential function $f(x) = e^x$ is continuous.

Proof. Let $c \in \mathbb{R}$. We want to show that $\lim_{x \rightarrow c} e^x = e^c$, i.e.,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad |x - c| < \delta \implies |e^x - e^c| < \varepsilon.$$

First factorize the expression we wish to bound:

$$|e^x - e^c| = |e^c \cdot e^{x-c} - e^c| = e^c |e^{x-c} - 1|.$$

Since e^c is constant, it remains to show that $|e^{x-c} - 1|$ can be made arbitrarily small by choosing x sufficiently close to c .

Lemma. For any $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that

$$e^{1/n} < 1 + \varepsilon \quad \text{and} \quad e^{-1/n} > 1 - \varepsilon.$$

Proof. This is equivalent to showing $e < (1 + \varepsilon)^n$, which follows from the fact that $\lim_{n \rightarrow \infty} a^{1/n} = 1$ for any $a > 0$. $\square$

By the lemma, there exists $n \in \mathbb{N}$ such that

$$1 - \frac{\varepsilon}{e^c} < e^{-\frac{1}{n}} < e^{\frac{1}{n}} < 1 + \frac{\varepsilon}{e^c}. \quad (1)$$

Choose $\delta = \frac{1}{n}$. If $|x - c| < \frac{1}{n}$, then $-\frac{1}{n} < x - c < \frac{1}{n}$. Since e^x is strictly increasing, applying the function to the inequality gives

$$e^{-\frac{1}{n}} < e^{x-c} < e^{\frac{1}{n}}.$$

Using (1), we have

$$1 - \frac{\varepsilon}{e^c} < e^{x-c} < 1 + \frac{\varepsilon}{e^c},$$

so that

$$|e^{x-c} - 1| < \frac{\varepsilon}{e^c}.$$

$\square$

Example. The logarithmic function $f(x) = \log x$ is continuous (on $\mathbb{R}^+$).

Proof. The proof is similar to the proof of continuity of e^x . $\square$

Theorem 4.35 (Composition). Let $f: A \rightarrow \mathbb{R}$, $g: B \rightarrow \mathbb{R}$ be functions such that $f(A) \subseteq B$. If f is continuous at $c \in A$, and g is continuous at $f(c)$, then $g \circ f$ is continuous at c .

Proof. We want to show that $\lim_{x \rightarrow c} g(f(x)) = g(f(c))$, namely,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad |x - c| < \delta \implies |g(f(x)) - g(f(c))| < \varepsilon.$$

Since f is continuous at c , for any $\varepsilon_1 > 0$, there exists $\delta_1 > 0$ such that

$$|x - c| < \delta_1 \implies |f(x) - f(c)| < \varepsilon_1. \quad (1)$$

Since g is continuous at $f(c)$, for any $\varepsilon_2 > 0$, there exists $\delta_2 > 0$ such that

$$|y - f(c)| < \delta_2 \implies |g(y) - g(f(c))| < \varepsilon_2. \quad (2)$$

Let $\varepsilon > 0$ be given. In (2), set $\varepsilon_2 = \varepsilon$. This gives us $\delta_2 > 0$ such that

$$|y - f(c)| < \delta_2 \implies |g(y) - g(f(c))| < \varepsilon.$$

In (1), set $\varepsilon_1 = \delta_2$. This gives us $\delta_1 > 0$ such that

$$|x - c| < \delta_1 \implies |f(x) - f(c)| < \delta_2.$$

Choose $\delta = \delta_1$. If $|x - c| < \delta$, then $|f(x) - f(c)| < \delta_2$. By letting $y = f(x)$, we satisfy the condition for the continuity of g , which implies

$$|g(f(x)) - g(f(c))| < \varepsilon.$$

Therefore, $g \circ f$ is continuous at c . □

Example. More examples of continuous functions:

- $f(x) = x^a = e^{a \log x}$, where $a > 0$.
- x^a is continuous at $c > 0$.
- $f(x) = \sqrt{x^2 + 1}$ is continuous at any $c \in \mathbb{R}$.
- $f(x) = \log(x^2 + 1)$ is continuous at any $c \in \mathbb{R}$.
- $f(x) = |2x^2 - 5x + 6|$ is continuous at any $c \in \mathbb{R}$.

Theorem 4.36 (Limit of composite functions). *Let $f: A \rightarrow \mathbb{R}$ and $g: B \rightarrow \mathbb{R}$ be functions such that $f(A) \subseteq B$. Let c be a limit point of A and*

$$\lim_{x \rightarrow c} f(x) = L.$$

Suppose $L \in B$ and g is continuous at L . Then

$$\lim_{x \rightarrow c} g(f(x)) = g(L).$$

Theorem 4.37 (Limit of composite functions). *Let $f: A \rightarrow \mathbb{R}$ and $g: B \rightarrow \mathbb{R}$ be functions such that $f(A) \subseteq B$. Let c be a limit point of A and*

$$\lim_{x \rightarrow c} f(x) = L.$$

Suppose further that $f(x) \neq L$ for $x \in A \cap (V_\delta(c) \setminus \{c\})$. If L is a limit point of B and $\lim_{y \rightarrow L} g(y) = M$. Then

$$\lim_{x \rightarrow c} g(f(x)) = M.$$

Theorem 4.38 (Composition of functions at infinity). *Let $f: A \rightarrow \mathbb{R}$ and $g: B \rightarrow \mathbb{R}$ be functions such that $f(A) \subseteq B$, and there exists c such that $(c, +\infty) \subseteq A$. Suppose $\lim_{x \rightarrow +\infty} f(x) = L$, g is continuous at L , then*

$$\lim_{x \rightarrow +\infty} g(f(x)) = g(L).$$

Theorem 4.39 (Composition of functions at infinity). *Let $f: A \rightarrow \mathbb{R}$ and $g: B \rightarrow \mathbb{R}$ be functions such that $f(A) \subseteq B$, and there exists c such that $(-\infty, c) \subseteq A$. Suppose $\lim_{x \rightarrow -\infty} f(x) = L$, g is continuous at L , then*

$$\lim_{x \rightarrow -\infty} g(f(x)) = g(L).$$

Remark. If $\lim_{x \rightarrow +\infty} f(x) = L$ and $f(x) \neq L$ for $x > C$ and $\lim_{y \rightarrow L} g(y) = M$. Then $\lim_{x \rightarrow +\infty} g(f(x)) = M$.

Theorem 4.40. Let $f : A \rightarrow \mathbb{R}$ and $g : B \rightarrow \mathbb{R}$ be functions such that $f(A) \subseteq B$, and there exists c such that $(c, +\infty) \subseteq B$. Suppose $\lim_{x \rightarrow x_0} f(x) = +\infty$ and $\lim_{y \rightarrow +\infty} g(y) = L$, then

$$\lim_{x \rightarrow x_0} g(f(x)) = L.$$

Theorem 4.41. Let $f : A \rightarrow \mathbb{R}$ and $g : B \rightarrow \mathbb{R}$ be functions such that $f(A) \subseteq B$, and there exists c such that $(-\infty, c) \subseteq B$. Suppose $\lim_{x \rightarrow x_0} f(x) = -\infty$ and $\lim_{y \rightarrow -\infty} g(y) = L$, then

$$\lim_{x \rightarrow x_0} g(f(x)) = L.$$

Example. More examples:

- $\lim_{x \rightarrow +\infty} x(\log(x+1) - \log x) = 1.$

To see this, note that by continuity of logarithm function,

$$\lim_{x \rightarrow +\infty} x(\log(x+1) - \log x) = \lim_{x \rightarrow +\infty} \log\left(1 + \frac{1}{x}\right)^x = \log \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x = \log e = 1.$$

- $\lim_{x \rightarrow 0} (1+x)^{1/x} = e.$

Using the substitution $x = \frac{1}{u}$,

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = \lim_{u \rightarrow \infty} \left(1 + \frac{1}{u}\right)^u = e.$$

- $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1.$

We apply the previous result:

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = \lim_{x \rightarrow 0} \log(1+x)^{\frac{1}{x}} = \log \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = \log e = 1.$$

- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1.$

Using the substitution $u = e^x - 1$,

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{u \rightarrow 0} \frac{u}{\log(u+1)} = \lim_{u \rightarrow 0} \frac{1}{\frac{\log(u+1)}{u}} = \frac{1}{1} = 1.$$

- $\lim_{x \rightarrow 0} \frac{(1+x)^a - 1}{x} = a.$

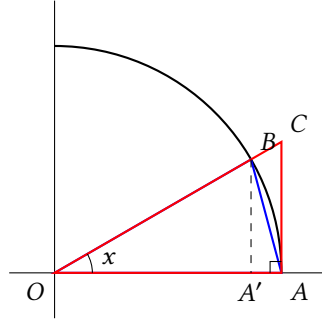
Rewrite the exponential term, and use the fact that $a \log(1+x) \rightarrow 0$ as $x \rightarrow 0$:

$$\frac{(1+x)^a - 1}{x} = \frac{e^{a \log(1+x)} - 1}{x} = \frac{e^{a \log(1+x)} - 1}{a \log(1+x)} \cdot \frac{a \log(1+x)}{x} \rightarrow a.$$

Example. Important limit: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$

Proof. To prove this, we first consider $x \rightarrow 0^+$. Based on the geometric construction of a unit circle, we observe the following relationship between areas:

$$\text{Area of } \triangle OAB \leq \text{Area of Sector } OAB \leq \text{Area of } \triangle OAC$$



Calculate each area:

- Area of $\triangle OAB = \frac{1}{2}(1)(\sin x) = \frac{1}{2} \sin x$
- Area of Sector $OAB = \frac{1}{2}(1)^2 x = \frac{1}{2} x$
- Area of $\triangle OAC = \frac{1}{2}(1)(\tan x) = \frac{1}{2} \tan x$

Substituting these into our inequality,

$$\frac{1}{2} \sin x \leq \frac{1}{2} x \leq \frac{1}{2} \frac{\sin x}{\cos x},$$

so

$$1 \leq \frac{x}{\sin x} \leq \frac{1}{\cos x}.$$

Taking the limit as $x \rightarrow 0$, by squeeze theorem, we conclude that $\lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$, so $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. $\square$

Example. We consider several related limits:

- $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$.

To see this, note that $\frac{\tan x}{x} = \frac{\sin x}{x} \cdot \frac{1}{\cos x}$. Since $\frac{\sin x}{x} \rightarrow 1$ and $\frac{1}{\cos x} \rightarrow 1$, we have $\frac{\tan x}{x} \rightarrow 1$.

- $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$.

To see this, note that

$$\frac{1 - \cos x}{x^2} = \frac{2 \sin^2\left(\frac{x}{2}\right)}{x^2} = 2 \left(\frac{\sin \frac{x}{2}}{x} \right)^2$$

By change of variables, let $u = \frac{x}{2}$. Then $\frac{\sin u}{2u} = \frac{1}{2} \frac{\sin u}{u} \rightarrow \frac{1}{2}$.

- $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = \frac{1}{2}$.

To see this, note that $\frac{\tan x - \sin x}{x^3} = \frac{\tan x}{x} \cdot \frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$.

Definition 4.42. A function $f: A \rightarrow \mathbb{R}$ is called *bounded* if there exists $C > 0$ such that

$$|f(x)| \leq C \quad \text{for all } x \in A.$$

Theorem 4.43 (Boundedness). Suppose $f: [a, b] \rightarrow \mathbb{R}$ is continuous. Then f is bounded.

Proof. For a contradiction, suppose that f is not bounded. Then for every $C > 0$, there exists $x \in [a, b]$ such that $|f(x)| > C$. In particular, for every $n \in \mathbb{N}$, there exists $x_n \in [a, b]$ such that $|f(x_n)| > n$. This gives a sequence $\{x_n\} \subseteq [a, b]$ such that $\lim_{n \rightarrow \infty} |f(x_n)| = +\infty$.

By Bolzano-Weierstrass, $\{x_n\}$ admits a convergent subsequence $\{y_k = x_{n_k}\}$; let $\lim_{k \rightarrow \infty} y_k = x_0 \in [a, b]$.

Since f is continuous at x_0 , by the sequential criterion,

$$\lim_{k \rightarrow \infty} f(y_k) = f(x_0).$$

This contradicts the fact that $\lim_{k \rightarrow \infty} |f(y_k)| = +\infty$. Therefore, f is bounded. $\square$

Remark. The statement is not true for open intervals; consider $f(x) = \frac{1}{x}$ on $(0, 1)$.

Definition 4.44. Let $f: A \rightarrow \mathbb{R}$.

- We say that f has an *absolute maximum* on A if there exists $x^* \in A$ such that $f(x) \leq f(x^*)$ for all $x \in A$.
- We say that f has an *absolute minimum* on A if there exists $x_* \in A$ such that $f(x) \geq f(x_*)$ for all $x \in A$.

Theorem 4.45 (Extreme value theorem). *Suppose $f: [a, b] \rightarrow \mathbb{R}$ is continuous. Then f has attains an absolute maximum and absolute minimum.*

Proof. We only prove the existence of maximum; the proof of the existence of minimum is similar.

Consider the set $\{f(x) : x \in [a, b]\}$, which is non-empty. Let $M = \sup_{x \in [a, b]} f(x)$.

We prove that there exists $x^* \in [a, b]$ such that $f(x^*) = M$.

For each $n \in \mathbb{N}$, since M is the supremum, $M - \frac{1}{n}$ is not an upper bound of the set, so there exists $x_n \in [a, b]$ such that

$$M \geq f(x_n) > M - \frac{1}{n}.$$

Taking the limit as $n \rightarrow \infty$, we obtain $\lim_{n \rightarrow \infty} f(x_n) = M$.

By Bolzano-Weierstrass, there exists a convergent subsequence $\{y_k = x_{n_k} : k \in \mathbb{N}\}$ such that $\lim_{k \rightarrow \infty} y_k = x^* \in [a, b]$.

We now show that $f(x^*) = M$. Since f is continuous at x^* , we have

$$f(x^*) = f\left(\lim_{k \rightarrow \infty} y_k\right) = \lim_{k \rightarrow \infty} f(y_k) = M,$$

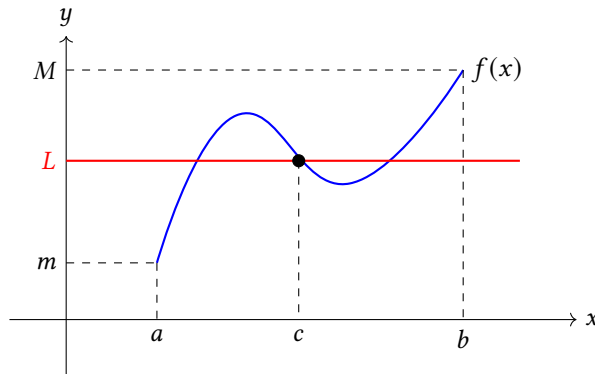
where the last line follows since $\{f(y_k)\}$ is a subsequence of $\{f(x_n)\}$, so it converges to the same limit M . Hence, f achieves its absolute maximum at x^* . $\square$

4.6 Intermediate value theorem

The Intermediate Value Theorem (IVT) is the name given to the very intuitive observation that a continuous function f on a closed interval $[a, b]$ attains every value that falls between the range values $f(a)$ and $f(b)$.

Theorem 4.46 (Bolzano's intermediate value theorem). *Suppose $f: [a, b] \rightarrow \mathbb{R}$ is continuous. Let $m = \min\{f(a), f(b)\}$ and $M = \max\{f(a), f(b)\}$. Then for any $L \in [m, M]$, there exists $c \in [a, b]$ such that*

$$f(c) = L.$$



Proof. Without loss of generality, suppose $f(a) < L < f(b)$. Let $I_0 = [a, b]$. We use the bisection method to construct a sequence of nested intervals. Consider the value of f at the midpoint.

- **Case 1:** If $f\left(\frac{a+b}{2}\right) = L$, we are done, and $c = \frac{a+b}{2}$.
- **Case 2:** If $f\left(\frac{a+b}{2}\right) > L$, choose the left sub-interval $I_1 = \left[a, \frac{a+b}{2}\right]$.
- **Case 3:** If $f\left(\frac{a+b}{2}\right) < L$, choose the right sub-interval $I_1 = \left[\frac{a+b}{2}, b\right]$.

In Cases 2 or 3, we obtain a new interval $I_1 = [a_1, b_1]$ such that $f(a_1) < L < f(b_1)$ and the length of the interval is halved. By repeating this process inductively, we construct a sequence of nested intervals $I_n = [a_n, b_n]$ such that for each $n \in \mathbb{N}$,

$$f(a_n) \leq L \leq f(b_n) \quad \text{and} \quad b_n - a_n = \frac{b-a}{2^n}.$$

By the nested interval theorem, since $|I_n| \rightarrow 0$, we have $\bigcap_{n=1}^{\infty} I_n = \{x_0\}$. It follows that $\lim_{k \rightarrow \infty} a_k = x_0$ and $\lim_{k \rightarrow \infty} b_k = x_0$.

Claim: $f(x_0) = L$.

Since f is continuous at x_0 , we have

$$\lim_{k \rightarrow \infty} f(a_k) = f(x_0).$$

Since $f(a_k) < L$ for all k , taking $k \rightarrow \infty$ gives $f(x_0) \leq L$.

Conversely, we have $\lim_{k \rightarrow \infty} f(b_k) = f(x_0)$. Since $f(b_k) > L$ for all k , taking $k \rightarrow \infty$ gives $f(x_0) \geq L$.

Hence, $f(x_0) = L$. Thus, there exists $c = x_0 \in [a, b]$ such that $f(c) = L$. □

An important corollary is in location of roots, used to prove that some equation has a solution.

Corollary 4.47 (Location of roots). *Suppose $f: [a, b] \rightarrow \mathbb{R}$ is continuous. Let $m = \min\{f(a), f(b)\}$ and $M = \max\{f(a), f(b)\}$. If $m \leq 0$ and $M \geq 0$, then there exists $c \in [a, b]$ such that*

$$f(c) = 0.$$

Any value between max and min can be obtained

Corollary 4.48. *Suppose $f: [a, b] \rightarrow \mathbb{R}$ is continuous. Let $m = \min_{x \in [a, b]} f(x)$ and $M = \max_{x \in [a, b]} f(x)$. Then for any $L \in [m, M]$, there exists $c \in [a, b]$ such that*

$$f(c) = L.$$

Proof. Restrict the domain of function to change the endpoints. If $M = f(a')$, $m = f(b')$, then apply IVT on $[a', b']$. □

Theorem 4.49. *Suppose $f: [a, b] \rightarrow \mathbb{R}$ is continuous. Then the range $f([a, b])$ is a closed interval.*

Proof. By the extreme value theorem, there exist $x_1, x_2 \in [a, b]$ such that

$$f(x_1) = \max_{x \in [a, b]} f(x), \quad f(x_2) = \min_{x \in [a, b]} f(x).$$

We will show that $f([a, b]) = [f(x_2), f(x_1)]$.

☐ This is clear, since $f(x_2)$ is lower bound, $f(x_1)$ is upper bound

☐ For any $L \in [f(x_2), f(x_1)]$, by corollary of intermediate value theorem, there exists $c \in [a, b]$ such that $f(c) = L$. □

sufficient condition to show interval

Lemma 4.50. *If $S \subseteq \mathbb{R}$ contains at least two elements and satisfies that for any $x < y \in S$, $[x, y] \subseteq S$, then S is an interval.*

The next theorem states that continuous functions preserve connectedness.

Theorem 4.51. Suppose $f: I \rightarrow \mathbb{R}$ is continuous, where I is an interval. Then the range $f(I)$ is an interval.

Proof. Denote $S = f(I)$. To show that S is an interval, we use the above lemma; that is, we want to show that for all $x, y \in S$, $x < y$, then $[x, y] \subseteq S$.

Since $x, y \in f(I)$, there exist $a, b \in I$ such that $f(a) = x$, $f(b) = y$. Without loss of generality, assume $a < b$. Then for any $L \in [x, y]$, apply IVT to f on $[a, b]$, so there exists $c \in [a, b]$ such that $f(c) = L$, so $L \in S$. Thus, $[x, y] \subseteq S$. $\square$

Example. Let $f: (a, b) \rightarrow \mathbb{R}$ be an injective continuous function on (a, b) . Prove that its range $f((a, b))$ is an open interval.

Proof. By the previous theorem, $f((a, b))$ is an interval. Thus, it suffices to show that its endpoints are not contained in $f((a, b))$.

Suppose $f((a, b)) = I$, where $I = [m, M]$ is an interval. We show that $m \notin I$; the argument for M is similar.

For a contradiction, suppose that $m \in I$. Then there exists $c \in (a, b)$ such that $f(c) = m$.

Since m is the minimum value of f , choose points c_1, c_2 such that

$$a < c_1 < c < c_2 < b.$$

Since f is continuous, apply IVT on $[c_1, c]$ and $[c, c_2]$:

- For any $L \in (f(c), f(c_1))$, there exists $c' \in (c_1, c)$ such that $f(c') = L$.
- For any $L' \in (f(c), f(c_2))$, there exists $c'' \in (c, c_2)$ such that $f(c'') = L'$.

Take $T = \min\{f(c_1), f(c_2)\}$. For any $L \in (f(c), T)$, there exist points $c' \in (c_1, c)$, $c'' \in (c, c_2)$ such that

$$f(c') = L = f(c'')$$

which contradicts the injectivity of f . $\square$

Example. Let $f: [0, 1] \rightarrow \mathbb{R}$ be continuous, such that $f(0) = f(1)$. Prove that there exists $x_0 \in [0, 1/2]$ such that $f(x_0) = f(x_0 + 1/2)$.

Proof. We want to find an appropriate function, then apply IVT.

Consider $g(x) = f(x) - f(x + 1/2)$ on $[0, 1/2]$.

At the endpoints, $g(0) = f(0) - f(1/2)$, $g(1/2) = f(1/2) - f(1)$. Since $f(0) = f(1)$, this implies $g(0) = -g(1/2)$.

If $g(0) = 0$, then we are done. If $g(0) \neq 0$, apply IVT to $g(x)$ on $[0, 1/2]$. There exists $c \in [0, 1/2]$ such that $g(c) = 0$, so $f(c) = f(c + 1/2)$. $\square$

4.7 Uniform continuity

Definition 4.52. A function $f: A \rightarrow \mathbb{R}$ is *uniformly continuous* on A if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x, y \in A, \quad |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Remark. Note the difference between continuity and uniform continuity: For continuity, δ depends on x and ε ; for uniform continuity, δ only depends on ε , it works for all $x \in A$.

Uniform continuity implies continuity, but converse is not true.

Example. Consider $f(x) = \frac{1}{x}$ on $(0, 1)$.

Take $\varepsilon = 1$. We want to show that for any $\delta > 0$, there exist $x_1, x_2 \in (0, 1)$ such that $|x_1 - x_2| < \delta$ but $|f(x_1) - f(x_2)| \geq 1$.

Choose $N \in \mathbb{N}$ such that $\frac{1}{N} < \delta$. Choose $x_1 = \frac{1}{2N}$, $x_2 = \frac{1}{4N}$, so $|x_1 - x_2| = \frac{1}{4N} < \delta$, $|f(x_1) - f(x_2)| = 2N > 1$.

If the domain is a closed interval, it turns out that continuous functions are uniformly continuous.

Theorem 4.53. *Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous. Then f is uniformly continuous on $[a, b]$.*

Proof. Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous. For a contradiction, suppose f is not uniformly continuous. Then there exists $\varepsilon > 0$ such that for all $\delta > 0$, there exist $x, y \in [a, b]$ satisfying

$$|x - y| < \delta \quad \text{and} \quad |f(x) - f(y)| \geq \varepsilon.$$

In particular, for each $n \in \mathbb{N}$, taking $\delta_n = \frac{1}{n}$, then there exist $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < \frac{1}{n} \quad \text{and} \quad |f(x_n) - f(y_n)| \geq \varepsilon.$$

Since $\{x_n\}$ is bounded, by Bolzano-Weierstrass, $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$. Let $x_{n_k} \rightarrow \xi$ for some $\xi \in [a, b]$.

Since f is continuous at ξ , there exists $\delta > 0$ such that

$$|x - \xi| < \delta \implies |f(x) - f(\xi)| < \frac{\varepsilon}{2}. \quad (1)$$

Since $x_{n_k} \rightarrow \xi$ and $|x_{n_k} - y_{n_k}| \rightarrow 0$, for sufficiently large k , both x_{n_k} and y_{n_k} are contained in the neighbourhood $V_\delta(\xi)$. By (1),

$$|f(x_{n_k}) - f(\xi)| < \frac{\varepsilon}{2}, \quad |f(y_{n_k}) - f(\xi)| < \frac{\varepsilon}{2}.$$

By triangle inequality,

$$|f(x_{n_k}) - f(y_{n_k})| \leq |f(x_{n_k}) - f(\xi)| + |f(\xi) - f(y_{n_k})| < \varepsilon,$$

which contradicts $|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon$. □

The next theorem states that uniformly continuous functions preserve Cauchy sequences.

Theorem 4.54. *Suppose $f: A \rightarrow \mathbb{R}$ is uniformly continuous on A . If $\{a_n\}$ is a Cauchy sequence in A , then $\{f(a_n)\}$ is a Cauchy sequence.*

Proof. Let $\varepsilon > 0$. Known:

$$\forall \varepsilon_1 > 0, \quad \exists N_1 \in \mathbb{N}, \quad \forall m > n \geq N_1, \quad |a_m - a_n| < \varepsilon_1 \quad (1)$$

$$\forall \varepsilon_2 > 0, \quad \exists \delta > 0, \quad \forall x, y \in A, \quad |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon_2. \quad (2)$$

The idea is to make a_m, a_n to be δ -close, so that we can apply (2).

Choose $\varepsilon_2 = \varepsilon$ in (2). Then we get $\delta > 0$ such that $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$.

Then choose $\varepsilon_1 = \delta$ in (1). We get $N_1 \in \mathbb{N}$ such that for all $m > n \geq N_1$, $|a_m - a_n| < \delta \implies |f(a_m) - f(a_n)| < \varepsilon$.

Choose $N = N_1$. □

Remark. Continuous functions might not preserve Cauchy sequences. For example, consider $f(x) = \frac{1}{x}$ on $(0, 1)$. The sequence $\{a_n = \frac{1}{n}\}$ is a Cauchy sequence, but $\{f(a_n) = n\}$ is clearly not a Cauchy sequence.

Theorem 4.55 (Continuous extension theorem). *$f: (a, b) \rightarrow \mathbb{R}$ is uniformly continuous on (a, b) if and only if f can be extended to a continuous function on $[a, b]$, i.e., both $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow b^-} f(x)$ exist.*

Proof.

$\Rightarrow$ Suppose f is uniformly continuous on (a, b) . We want to prove that $\lim_{x \rightarrow a^+} f(x)$ exists.

Take any sequence $\{a_n\} \subseteq (a, b)$ such that $a_n \rightarrow a$. By 4.54, $\{f(a_n)\}$ is Cauchy; let $f(a_n) \rightarrow A$.

Claim: $\lim_{x \rightarrow a^+} f(x) = A$.

We use the sequential criterion. For a contradiction, suppose $\lim_{x \rightarrow a^+} f(x) \neq A$. Then there exists a sequence $\{b_n\} \subseteq (a, b)$ such that

$$\lim_{n \rightarrow \infty} b_n = a \quad \text{but} \quad \lim_{n \rightarrow \infty} f(b_n) = A' \neq A.$$

Consider the sequence $\{c_n\}$ defined by

$$c_n = \begin{cases} a_{n/2} & \text{if } n \text{ is even,} \\ b_{(n+1)/2} & \text{if } n \text{ is odd.} \end{cases}$$

That is, $\{b_1, a_1, b_2, a_2, \dots, b_n, a_n\}$. Note that $\lim_{n \rightarrow \infty} c_n = a$. Therefore, $\{f(c_n)\}$ is convergent. But $\{a_n\}$ and $\{b_n\}$ are subsequences of $\{c_n\}$, so we must have

$$\lim_{n \rightarrow \infty} f(b_n) = A' = \lim_{n \rightarrow \infty} f(a_n).$$

This yields the desired contradiction.

$\square$ Suppose f can be extended to a continuous function $\tilde{f}$ on $[a, b]$. Then $\tilde{f}$ is continuous, and thus uniformly continuous, on $[a, b]$, and $\tilde{f} = f$ on (a, b) .

Let $\varepsilon > 0$. Since $\tilde{f}$ is uniformly continuous on $[a, b]$, for all $x, y \in [a, b]$,

$$|x - y| < \delta \implies |\tilde{f}(x) - \tilde{f}(y)| < \varepsilon.$$

In particular, we can replace the closed interval $[a, b]$ with the open interval (a, b) , namely, for all $x, y \in (a, b)$,

$$|x - y| < \delta \implies |\tilde{f}(x) - \tilde{f}(y)| < \varepsilon.$$

Since $\tilde{f} = f$ on (a, b) ,

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

for all $x, y \in (a, b)$. Hence, f is uniformly continuous on (a, b) . $\square$

Corollary 4.56. *If limits at endpoint do not exist, then f is not uniformly continuous.*

4.8 Lipschitz functions

Definition 4.57. We say that $f: A \rightarrow \mathbb{R}$ is a *Lipschitz function* if there exists $K > 0$ such that for all $x, y \in A$,

$$|f(x) - f(y)| \leq K|x - y|.$$

K is called the *Lipschitz constant* for f .

Theorem 4.58. *If $f: A \rightarrow \mathbb{R}$ is a Lipschitz function on A , then f is uniformly continuous on A .*

Proof. Suppose $|f(x) - f(y)| \leq K|x - y|$. Let $\varepsilon > 0$. Choose $\delta = \varepsilon/K$. If $|x - y| < \varepsilon/K$, then $|f(x) - f(y)| \leq K|x - y| < K \cdot \varepsilon/K = \varepsilon$. $\square$

Uniform continuity does not imply Lipschitz, e.g., $f(x) = \sqrt{x}$ on $[0, 1]$. We show $f(x) = \sqrt{x}$ is not Lipschitz, i.e., for all $K > 0$, there exist $x, y \in [0, 1]$ such that

$$|f(x) - f(y)| > K|x - y| \iff |\sqrt{x} - \sqrt{y}| > K|x - y| \iff 1 > K(\sqrt{x} + \sqrt{y}).$$

Simply take some small value of x, y , e.g. $x = 0, y = \frac{1}{4K^2}$.

Definition 4.59. Given $\alpha \in (0, 1)$, $f: A \rightarrow \mathbb{R}$ is called α -Hölder if there exists $K > 0$ such that for all $x, y \in A$,

$$|f(x) - f(y)| \leq K|x - y|^\alpha.$$

Example. $f(x) = \sqrt{x}$ is $\frac{1}{2}$ -Hölder.

Holder implies uniform continuity.

4.9 Monotone functions

Definition 4.60. A function $f: A \rightarrow \mathbb{R}$ is *increasing* on A if

$$f(x) \leq f(y) \quad \text{for all } x < y$$

and *decreasing* if

$$f(x) \geq f(y) \quad \text{for all } x < y.$$

A *monotone* function is one that is either increasing or decreasing.

The next theorem states that one-sided limits for monotone functions defined on intervals exist.

Theorem 4.61. Suppose $f: I \rightarrow \mathbb{R}$ is monotone on I , where I is an interval. Let $c \in I$ be an interior point. Then the one-sided limits $\lim_{x \rightarrow c^-} f(x)$ and $\lim_{x \rightarrow c^+} f(x)$ exist.

Remark. The left and right limits may not be equal.

Proof. Suppose f is increasing; the case where f is decreasing is similar. We shall show that the left limit $\lim_{x \rightarrow c^-} f(x)$ exists; the proof of the right limit is similar.

The idea of the proof is similar to the monotone convergence theorem.

Since f is increasing, for all $x < c$, $f(x) \leq f(c)$. Thus, the set $S = \{f(x) : x < c\}$ is bounded from above by $f(c)$, so its supremum exists; denote $M = \sup_{x < c} f(x)$.

Claim: $\lim_{x \rightarrow c^-} f(x) = M$.

Let $\varepsilon > 0$. Since $M - \varepsilon$ is not an upper bound of S , there exists $x_0 < c$ such that $f(x_0) > M - \varepsilon$.

Choose $\delta = c - x_0$. Then for any $x \in (c - \delta, c)$, i.e., $x_0 < x < c$,

$$M \geq f(x) \geq f(x_0) > M - \varepsilon \implies |f(x) - M| < \varepsilon.$$

Hence, $\lim_{x \rightarrow c^-} f(x) = M$. □

Corollary 4.62. Suppose $f: I \rightarrow \mathbb{R}$ is increasing on I , where I is an interval. Let $c \in I$ be an interior point. Then the following are equivalent:

- (1) f is continuous at c ;
- (2) $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x)$;
- (3) $\sup_{x < c} f(x) = f(c) = \inf_{x > c} f(x)$.

The same statements hold for decreasing functions.

Definition 4.63. Suppose $f: I \rightarrow \mathbb{R}$ is a monotone function, where I is an interval. Let $c \in I$ be an interior point. The *jump* of f at c is

$$j_f(c) := \lim_{x \rightarrow c^+} f(x) - \lim_{x \rightarrow c^-} f(x).$$

That is, the jump is the difference between the left and right limits.

Corollary 4.64. Suppose $f: I \rightarrow \mathbb{R}$ is a monotone function, where I is an interval. Let $c \in I$ be an interior point. Then f is continuous at c if and only if $j_f(c) = 0$.

Theorem 4.65. Suppose $f: I \rightarrow \mathbb{R}$ is a monotone function, where I is an interval. Then the set of discontinuous points of f on I is at most countable.

Proof. Let C be the set of discontinuous points of f . If C is finite, then we are done. Otherwise, if C is infinite, we want to show that C is countable.

For any $c \in C$, map c to an open interval $I_c = (l_c, r_c)$, where $l_c = \lim_{x \rightarrow c^-} f(x)$, $r_c = \lim_{x \rightarrow c^+} f(x)$. This gives a family of open intervals $\{I_c\}_{c \in C}$.

Claim: $\{I_c\}_{c \in C}$ is a *disjoint* family of open intervals, i.e., for any $c, c' \in C$, $I_c \cap I_{c'} = \emptyset$.

WLOG assume $c < c'$. To show that (l_c, r_c) and $(l_{c'}, r_{c'})$ are disjoint, it suffices to show that $r_c \leq l_{c'}$. Note that

$$r_c = \inf_{x > c} f(x) \leq f\left(\frac{c + c'}{2}\right) \leq \sup_{x < c'} f(x) = l_{c'}$$

since $(c + c')/2$ is larger than c , and smaller than c' .

By Tutorial 1, this family of disjoint open intervals is at most countable. $\square$

Theorem 4.66 (Continuous inverse theorem). *Suppose $f: I \rightarrow \mathbb{R}$ is strictly increasing and continuous on an interval I . Then f^{-1} is strictly increasing and continuous on the interval $J = f(I)$.*

Proof. By 4.51, $J = f(I)$ is an interval.

To show that f^{-1} is strictly increasing, let $y_1, y_2 \in J$, $y_1 < y_2$. We want to show that $f^{-1}(y_1) < f^{-1}(y_2)$. There exists $x_1, x_2 \in I$ such that $x_1 = f^{-1}(y_1)$, $x_2 = f^{-1}(y_2)$. Then $f(x_1) = y_1$, $f(x_2) = y_2$. We want to show $x_1 < x_2$. For a contradiction, suppose $x_1 \geq x_2$. Then $f(x_1) \geq f(x_2)$, i.e., $y_1 \geq y_2$, a contradiction.

To show that f^{-1} is continuous on J , for a contradiction, suppose f^{-1} is discontinuous at some point $c \in J$.

Case 1: c is an interior point of J .

Then the one-sided limits at c are not equal, namely $\lim_{y \rightarrow c^-} f^{-1}(y) \neq \lim_{y \rightarrow c^+} f^{-1}(y)$, so

$$l_c = \lim_{y \rightarrow c^-} f^{-1}(y) < \lim_{y \rightarrow c^+} f^{-1}(y) = r_c,$$

so there is a jump at c . Thus, $(l_c, r_c) \not\subseteq f^{-1}(J) = I$. Note that there exist $x_1 \leq l_c$, $x_2 \geq r_c$ such that $x_1, x_2 \in f^{-1}(J)$. Then $x_1 < x_2$ but $[x_1, x_2] \not\subseteq f^{-1}(J)$, so by the lemma, $f^{-1}(J)$ is not an interval, contradicting the assumption that I is an interval.

Case 2: c is an endpoint of J , say the left endpoint.

If f^{-1} is not continuous at c , then

$$r_c = \lim_{y \rightarrow c^+} f^{-1}(y) > f^{-1}(c).$$

Then $(f^{-1}(c), r_c) \not\subseteq f^{-1}(J) = I$. There exists $x_1 \leq f^{-1}(c)$ (take $x = f^{-1}(c)$), $x_2 \geq r_c$ such that $x_1, x_2 \in I$. Thus similarly I is not an interval, a contradiction. $\square$

5 Basic Topology

5.1 Topology on $\mathbb{R}$

A *topology* is a family of subsets, called *open sets*. Hence, to define a topology, we need to define the open sets.

Definition 5.1. A subset $U \subseteq \mathbb{R}$ is called *open* if for any $x \in U$, there exists $\delta > 0$ such that $V_\delta(x) \subseteq U$.

That is, for every point in U , we can find a neighbourhood centered at x that is totally contained in U .

Definition 5.2. A subset $C \subseteq \mathbb{R}$ is called *closed* if $\mathbb{R} \setminus C$ is open.

Proposition 5.3.

- (1) $\emptyset, \mathbb{R}$ are open.
- (2) For any collection of open sets $\{U_\alpha\}_{\alpha \in \mathcal{A}}$, the union $\bigcup_{\alpha \in \mathcal{A}} U_\alpha$ is open.
- (3) For any finite collection of open sets $\{U_i\}_{i=1}^n$, the intersection $\bigcap_{i=1}^n U_i$ is open.

Proof.

- (1) Obvious.
- (2) Let $\{U_\alpha\}_{\alpha \in \mathcal{A}}$ be a collection of open sets. Let $x \in \bigcup_{\alpha \in \mathcal{A}} U_\alpha$ be arbitrary. Then $x \in U_\alpha$ for some U_α . Since U_α is open, there exists $\delta > 0$ such that $V_\delta(x) \subseteq U_\alpha \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$.
- (3) Let $\{U_i\}_{i=1}^n$ be a finite collection of open sets. Let $x \in \bigcap_{i=1}^n U_i$ be arbitrary. Then $x \in U_i$ for all $i = 1, \dots, n$. Since each U_i is open, there exists $\delta_i > 0$ such that $V_{\delta_i}(x) \subseteq U_i$. Set $\delta = \min_{1 \leq i \leq n} \delta_i$. Then $V_\delta(x) \subseteq V_{\delta_i}(x) \subseteq U_i$ for each i , so $V_\delta(x) \subseteq \bigcap_{i=1}^n U_i$.

□

Remark. The finiteness condition in (3) is required for us to take minimum.

Corollary 5.4.

- (1) $\emptyset, \mathbb{R}$ are closed.
- (2) For any collection of closed sets $\{U_\alpha\}_{\alpha \in \mathcal{A}}$, the intersection $\bigcap_{\alpha \in \mathcal{A}} U_\alpha$ is closed.
- (3) For any finite collection of closed sets $\{U_i\}_{i=1}^n$, the union $\bigcup_{i=1}^n U_i$ is closed.

Proof. Take complements, and use de Morgan's law.

□

Definition 5.5. A point $x \in \mathbb{R}$ is called a *limit point* of $E \subseteq \mathbb{R}$ if for any $\delta > 0$,

$$(V_\delta(x) \setminus \{x\}) \cap E \neq \emptyset.$$

Let E' denote the set of limit points of E .

That is, if every neighborhood $V_\varepsilon(x)$ of x intersects the set E at some point other than x .

Theorem 5.6. A point x is a limit point of E if and only if $x = \lim_{n \rightarrow \infty} a_n$ for some sequence $\{a_n\} \subseteq E$ such that $a_n \neq x$.

Proof.

⇒ Suppose x is a limit point of E . To produce a sequence $\{a_n\}$ converging to x , consider the ε -neighbourhoods obtained using $\varepsilon = \frac{1}{n}$.

Since x is a limit point, every neighbourhood intersects E in some point other than x . For each $n \in \mathbb{N}$, fix

$$a_n \in V_{1/n}(x) \cap E$$

where $a_n \neq x$. We then show that $a_n \rightarrow x$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $\frac{1}{N} < \varepsilon$. For all $n \geq N$, we have $|a_n - x| < \frac{1}{n} \leq \frac{1}{N} < \varepsilon$, as desired.

$\Rightarrow$ Suppose $x = \lim_{n \rightarrow \infty} a_n$ for some sequence $\{a_n\} \subseteq E$ such that $a_n \neq x$. Let $V_\varepsilon(x)$ be an arbitrary ε -neighbourhood. By definition of convergence, there exists a term a_N in the sequence satisfying $a_N \in V_\varepsilon(x)$, and the proof is complete. $\square$

The next result characterizes closed sets.

Theorem 5.7. *E is closed if and only if $E' \subseteq E$, i.e., E contains all its limit points.*

Proof.

$\Rightarrow$ Suppose E is closed. Let $x \in E'$. We will prove $x \in E$.

For a contradiction, suppose $x \notin E$. Then $x \in E^c$. Since E^c is open, there exists $\delta > 0$ such that $V_\delta(x) \subseteq E^c$, so $V_\delta(x) \cap E = \emptyset$. This contradicts the assumption that $x \in E$.

$\Leftarrow$ Suppose $E' \subseteq E$. We want to show E is closed, i.e., $U = E^c$ is open.

Let $x \in U$. Then $x \notin E \supseteq E'$, so x is not a limit point of E . Thus, there exists $\delta > 0$ such that $(V_\delta(x) \setminus \{x\}) \cap E = \emptyset$. Since $x \notin E$, add it in: $V_\delta(x) \cap E = \emptyset$, so $V_\delta(x) \subseteq E^c = U$. This shows that U is open, so E is closed. $\square$

Definition 5.8. Let $E \subseteq \mathbb{R}$. The *closure* of E is $\bar{E} = E \cup E'$.

Corollary 5.9. *For any $E \subseteq \mathbb{R}$, its closure $\bar{E}$ is closed.*

Proof. We want to prove $U = \bar{E}^c$ is open. Let $x \in U$ be arbitrary. Then $x \notin E$ and $x \notin E'$.

Since x is not a limit point of E , there exists $\delta > 0$ such that $(V_\delta(x) \setminus \{x\}) \cap E = \emptyset$. Since $x \notin E$, add it in: $V_\delta(x) \cap E = \emptyset$.

We also need to show $V_{\delta/2}(x) \cap E' = \emptyset$. Suppose not. Fix $c' \in E' \cap V_{\delta/2}(x)$. Then there exists a sequence $\{a_n\} \subseteq E$ such that $\lim_{n \rightarrow \infty} a_n = c'$. By definition, there exists $N \in \mathbb{N}$ such that $|a_N - c'| < \frac{\delta}{4}$. Also, $|c' - x| < \frac{\delta}{2}$. By triangle inequality, $|a_N - x| \leq |a_N - c'| + |c' - x| < \delta$ which is a contradiction. $\square$

Corollary 5.10. *E is closed if and only if $E = \bar{E}$.*

The next result is a characterization of open sets in $\mathbb{R}$.

Theorem 5.11. *A set $U \subseteq \mathbb{R}$ is open if and only if it is a disjoint union of finite or countably many open intervals.*

Proof. We only prove the forward direction; the other direction is trivial.

Let $U \subseteq \mathbb{R}$ be open. For each $x \in U$, we will find the “maximal” open interval I_x such that $x \in I_x \subseteq U$. Here “maximal” means that for any open interval J such that $x \in J \subseteq U$, we have $J \subseteq I_x$.

Let $x \in U$. Define $I_x = (a_x, b_x)$, where $a_x = \inf \{a \in \mathbb{R} : (a, x) \subseteq U\}$ and $b_x = \sup \{b \in \mathbb{R} : (x, b) \subseteq U\}$. Either could be infinite. They are well-defined since there exists an open interval I such that $x \in I \subseteq U$, because U is open.

Clearly, $x \in I_x \subseteq U$. Suppose $J = (p, q)$ is such that $x \in J \subseteq U$. Then $(p, x) \subseteq U$, so $p \geq a_x$. Likewise $q \leq b_x$. Thus, $J \subseteq I_x$ and so I_x is indeed maximal.

Suppose $a_x \in U$. (In particular, we are supposing a_x is finite.) Then there exists $\varepsilon > 0$ such that $(a_x - \varepsilon, a_x + \varepsilon) \subseteq U$. The $(a_x - \varepsilon, b_x)$ is larger than I_x contradicting maximality. Thus, $a_x \notin U$ and likewise $b_x \notin U$.

Claim: For $x, y \in U$, the intervals I_x and I_y are either disjoint or identical.

Proof. If $I_x \cap I_y \neq \emptyset$, then $I_x \cup I_y$ is an open interval. Since $x \in I_x \cup I_y \subseteq U$, we have, by maximality, $I_x \cup I_y \subseteq I_x$. Likewise, $I_x \cup I_y \subseteq I_y$. Thus, $I_x = I_y$. $\square$

We now have that U is a disjoint union of maximal open intervals. Call this collection $\mathcal{I} = \{I_x\}_{x \in U}$. Then $\mathcal{I}$ is at most countable, i.e., finite or countable. $\square$

Remark. Closed sets are more complicated.

5.2 Metric spaces

Definition 5.12. Given a set X , a function $d: X \times X \rightarrow \mathbb{R}$ is a *metric* on X if for all $x, y \in X$,

- (1) $d(x, y) \geq 0$, where equality holds if and only if $x = y$; (non-negativity)
- (2) $d(x, y) = d(y, x)$; (symmetry)
- (3) for all $z \in X$, $d(x, y) \leq d(x, z) + d(z, y)$. (triangle inequality)

A *metric space* is a set X together with a metric d .

Example.

- The real line $(\mathbb{R}, |\cdot|)$, where $d(x, y) := |x - y|$ is the absolute value distance.
- The n -dimensional space $(\mathbb{R}^n, \|\cdot\|)$, where $d(x, y) := \|x - y\|$. There is more than one definition of the norm; the *Euclidean norm* is

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Note that the triangle inequality for this norm is equivalent to the Cauchy-Schwarz inequality.

The *supremum norm* is defined as

$$\|x\| = \max_{1 \leq i \leq n} |x_i|.$$

- $(\mathbb{Q}, \|\cdot\|_p)$, where $d(x, y) := \|x - y\|_p$ is the p -adic norm (p is prime) defined as follows: if $x = p^k \frac{r}{s}$, $(p, rs) = 1$, i.e., extract all powers of p , then $\|p\| = p^{-k}$. This norm is from number theory. Intuitively, two points are close if they are congruent to some large power of p . The p -adic norm satisfies a stronger triangle inequality:

$$\|x - z\|_p \leq \max\{\|x - y\|_p, \|y - z\|_p\}.$$

- The function space $C[0, 1]$ is the set of continuous functions defined on $[0, 1]$, with metric

$$d(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|.$$

Definition 5.13. The ε -neighbourhood at $x \in X$ is defined as

$$V_\varepsilon(x) := \{y \in X : d(y, x) < \varepsilon\}.$$

A metric d on X induces a topology on X as follows: a set $U \subseteq X$ is open if for any $x \in U$, there exists $\varepsilon > 0$ such that $V_\varepsilon(x) \subseteq U$.

This gives an important family of topological spaces. There are nice properties with metric on a space.

Proposition 5.14. Let (X, d) be a metric space.

- (1) $\emptyset, X$ are open;
- (2) For any collection of open sets $\{U_\alpha\}_{\alpha \in \mathcal{A}}$, the union $\bigcup_{\alpha \in \mathcal{A}} U_\alpha$ is open.
- (3) For any finite collection of open sets $\{U_i\}_{i=1}^n$, the intersection $\bigcap_{i=1}^n U_i$ is open.

Definition 5.15. Let (X, d) be a metric space. A sequence $\{x_n\} \subseteq X$ converges to $x \in X$ if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad d(x_n, x) < \varepsilon.$$

Then x is the *limit* of $\{x_n\}$, denoted by $\lim_{n \rightarrow \infty} x_n = x$.

Exercise. If a sequence converges, then its limit is unique.

Definition 5.16. Given a metric space (X, d) and a subset $E \subseteq X$, a point $x \in X$ is called a *limit point* of E if for any $\delta > 0$,

$$(V_\delta(x) \setminus \{x\}) \cap E \neq \emptyset.$$

Let E' denote the set of limit points of E .

Theorem 5.17. Let (X, d) be a metric space, and $E \subseteq X$.

- (1) E is closed if and only if $E' \subseteq E$;
- (2) Let $\bar{E} := E \cup E'$. Then $\bar{E}$ is closed.

5.3 General topology

Definition 5.18. A *topological space* is a non-empty set X with a collection of subsets $\mathcal{O}$, called *open sets*, with the following properties:

- (1) $\emptyset$ and X are open.
- (2) For any subset $\mathcal{O}' \subseteq \mathcal{O}$, the union $\bigcup_{O \in \mathcal{O}'} O \in \mathcal{O}$.
- (3) For any finite subset $\mathcal{O}' \subseteq \mathcal{O}$, the intersection $\bigcap_{O \in \mathcal{O}'} O \in \mathcal{O}$.

Definition 5.19. Let $(X, \mathcal{O})$ be a topological space. A subset $E \subseteq X$ is called *closed* if $X \setminus E$ is open.

It is easy to see that closed subsets satisfy the following:

- (1) $\emptyset, X$ are closed.
- (2) For any collection $\mathcal{E}$ of closed sets, $\bigcap_{E \in \mathcal{E}} E$ is closed.
- (3) For any finite collection $\mathcal{E}$ of closed sets, $\bigcup_{E \in \mathcal{E}} E$ is closed.

Example. Let X be a set. The *finite complement topology* (or *cofinite topology*) on X defined as follows: $U \subseteq X$ is open if $X \setminus U$ is finite or $U = \emptyset$. Show that this is indeed a topology.

Proof.

- (1) $\emptyset$ is open by definition. Also, $X \setminus X = \emptyset$ is finite, so X is open.
- (2) Suppose $\{U_\alpha\}_{\alpha \in \mathcal{A}}$ are open. Note that $U_\alpha \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$, so $X \setminus U_\alpha \supseteq X \setminus (\bigcup_{\alpha \in \mathcal{A}} U_\alpha)$. Since $X \setminus U_\alpha$ is finite for each $\alpha \in \mathcal{A}$, it follows that $X \setminus (\bigcup_{\alpha \in \mathcal{A}} U_\alpha)$ is finite. Hence, $\bigcup_{\alpha \in \mathcal{A}} U_\alpha$ is open.
- (3) Suppose $\{U_i\}_{i=1}^n$ are open. Then each $X \setminus U_i$ is finite. Since $X \setminus (\bigcap_{i=1}^n U_i) = \bigcup_{i=1}^n X \setminus U_i$, it follows that $X \setminus (\bigcap_{i=1}^n U_i)$ is a finite union of finite sets, so it is finite. Hence, $\bigcap_{i=1}^n U_i$ is open.

□

Definition 5.20. A topological space has the *Hausdorff property* if for any $x, y \in X$, $x \neq y$, there exist disjoint U_x and U_y such that $x \in U_x$, $y \in U_y$.

That is, any two distinct points can be separated by disjoint open sets.

Example. The finite complement topology, as defined above, is not Hausdorff.

Given a subset Y of a topological space X , we can use the topology on X to define the topology on Y :

Definition 5.21. Given a topological space X and a subset $Y \subseteq X$, we define the *induced topology* on Y as follows: $U \subseteq Y$ is called *open relative to Y* if there exists an open set $O \subseteq X$ such that

$$U = O \cap Y.$$

5.4 Compactness

Definition 5.22. Let X be a topological space and $K \subseteq X$. If $K \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$ and U_α is open for any $\alpha \in \mathcal{A}$, we call $\bigcup_{\alpha \in \mathcal{A}} U_\alpha$ an *open cover* of K .

K is *compact* if every open cover of K has a finite subcover, namely, if $K \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$, then there exists a finite subcover $K \subseteq \bigcup_{i=1}^n U_{\alpha_i}$.

Theorem 5.23. Let $K \subseteq Y \subseteq X$. Then K is compact in Y (with respect to the induced topology) if and only if K is compact in X .

This result tells us that compactness does not change if we change to induced topology; different from relative openness. In some sense, compactness is a “universal property”.

Proof.

$\Rightarrow$ Suppose K is compact in Y . We will prove K is compact in X . Take any open cover of K :

$$K \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha,$$

where U_α is open in X . Then, since $K \subseteq Y$,

$$K = K \cap Y = \bigcup_{\alpha \in \mathcal{A}} (U_\alpha \cap Y),$$

where each $U_\alpha \cap Y$ is open in Y . Since K is compact in Y , there exists a finite cover

$$K \subseteq \bigcup_{i=1}^n (U_{\alpha_i} \cap Y).$$

Thus, $K \subseteq \bigcup_{i=1}^n U_{\alpha_i}$. Hence, K is compact in X .

$\Leftarrow$ Suppose K is compact in X . We will prove K is compact in Y . Take any open cover of K :

$$K \subseteq \bigcup_{\alpha \in \mathcal{A}} (U_\alpha \cap Y),$$

where each $U_\alpha \cap Y$ is open in Y . Then $K \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$, which is an open cover in X . Since K is compact in X , there exists a finite subcover $K \subseteq \bigcup_{i=1}^n U_{\alpha_i}$. Since $K \subseteq Y$, $K = K \cap Y \subseteq \bigcup_{i=1}^n (U_{\alpha_i} \cap Y)$. Thus, $K \subseteq \bigcup_{i=1}^n (U_{\alpha_i} \cap Y)$. Hence, K is compact in Y . $\square$

Theorem 5.24. Let $K \subseteq X$ be compact and $Y \subseteq X$ be a closed subset. Then $Y \cap K$ is compact.

Proof. Take any open cover of $Y \cap K$:

$$Y \cap K = \bigcup_{\alpha \in \mathcal{A}} U_\alpha,$$

where U_α are open. Since Y is closed, $U := Y^c$ is open. We claim that $K \subseteq U \cup \bigcup_{\alpha \in \mathcal{A}} U_\alpha$.

Since K is compact, there exists a finite subcover

$$K \subseteq U \cup \bigcup_{i=1}^n U_{\alpha_i}.$$

Then $K \cap Y \subseteq U \cup \bigcup_{i=1}^n U_{\alpha_i}$. Since $(K \cap Y) \cap U = \emptyset$, we don't need this U , so $K \cap Y \subseteq \bigcup_{i=1}^n U_{\alpha_i}$. $\square$

Theorem 5.25. Let (X, d) be a metric space. Then compact subsets of X are closed.

Proof. Let $K \subseteq X$ be compact. To prove that K is closed, we need to prove that K^c is open, i.e., for all $y \notin K$, there exists $\delta > 0$ such that $V_\delta(y) \cap K = \emptyset$.

For each $x \in K$, let $\delta_x = \frac{1}{4}d(x, y)$. Then $V_{\delta_x}(y) \cap V_{\delta_x}(x) = \emptyset$. Note that $K \subseteq \bigcup_{x \in K} V_{\delta_x}(x)$ is an open cover. By compactness of K , there exist $x_1, \dots, x_n \in K$ such that $K \subseteq \bigcup_{i=1}^n V_{\delta_{x_i}}(x_i)$. Take $\delta = \min_{1 \leq i \leq n} \delta_{x_i}$.

We claim that $V_\delta(y) \cap K = \emptyset$. Note that $V_\delta(y) \cap V_{\delta_{x_i}}(x_i) = \emptyset$ for $i = 1, \dots, n$. This implies

$$V_\delta(y) \cap \left(\bigcup_{i=1}^n V_{\delta_{x_i}}(x_i) \right) = \emptyset.$$

Since $K \subseteq \bigcup_{i=1}^n V_{\delta_{x_i}}(x_i) = \emptyset$ covers K , this implies that $V_\delta(y) \cap K = \emptyset$. $\square$

Theorem 5.26 (Heine-Borel). *A set $K \subseteq \mathbb{R}$ is compact if and only if K is closed and bounded.*

Proof.

$\Rightarrow$ Suppose $K \subseteq \mathbb{R}$ is compact. By the previous theorem, K is closed. Thus, it remains to show that K is bounded. Take an open cover $\{V_1(x) : x \in \mathbb{R}\}$ of K , i.e., the set of neighbourhoods of radius 1. Then

$$K \subseteq \bigcup_{x \in \mathbb{R}} V_1(x).$$

By compactness, there exist $x_1, \dots, x_n \in \mathbb{R}$ such that

$$K \subseteq \bigcup_{i=1}^n V_1(x_i).$$

Take $C = \max_{1 \leq i \leq n} |x_i| + 1$. For each $x \in K \subseteq \bigcup_{i=1}^n V_1(x_i)$, there exists i such that $|x - x_i| < 1$. By triangle inequality,

$$\begin{aligned} |x| &\leq |x - x_i| + |x_i| \\ &< 1 + |x_i| \leq C. \end{aligned}$$

$\Leftarrow$ Suppose K is closed and bounded.

Since K is bounded, there exists an initial closed interval $I_1 = [a_1, b_1]$ such that $K \subseteq I_1$. To prove K is compact, let $\{U_\alpha : \alpha \in \mathcal{A}\}$ be any open cover of K , such that $K \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$.

Suppose, for sake of contradiction, that K does not have a finite subcover.

Divide I_1 into two sub-intervals, I'_1 and I''_1 , by bisecting it at the midpoint. Let $K'_1 = K \cap I'_1$ and $K''_1 = K \cap I''_1$. At least one of these sets, denote it K_2 , must not have a finite subcover from $\{U_\alpha\}$. Let I_2 be the corresponding sub-interval such that $K_2 = K \cap I_2$.

By repeating this process inductively, we construct a sequence of nested intervals $\{I_\ell\}$ and a sequence of subsets $\{K_\ell\}$ such that for each $\ell \in \mathbb{N}$:

- (1) $I_{\ell+1} \subseteq I_\ell$ and $K_{\ell+1} \subseteq K_\ell$.
- (2) The length of I_ℓ is $|I_\ell| = \frac{b_1 - a_1}{2^{\ell-1}}$, which tends to 0 as $\ell \rightarrow \infty$.
- (3) K_ℓ does not have a finite subcover.

By the Nested Interval Property, there exists a unique point ξ such that $\bigcap_{\ell=1}^\infty I_\ell = \{\xi\}$. Since $K_\ell \subseteq I_\ell$ is non-empty for all ℓ , we can choose a sequence $x_\ell \in K_\ell$ such that $\lim_{\ell \rightarrow \infty} x_\ell = \xi$. Because K is closed and each $x_\ell \in K$, it follows that $\xi \in K$.

Since $\{U_\alpha\}$ covers K and $\xi \in K$, there exists some $\alpha_0 \in \mathcal{A}$ such that $\xi \in U_{\alpha_0}$. Because U_{α_0} is open, there exists a neighborhood $V_\delta(\xi) \subseteq U_{\alpha_0}$ for some $\delta > 0$.

For sufficiently large ℓ , the interval I_ℓ is contained within $V_\delta(\xi)$ because the length of I_ℓ tends to 0 and $\xi \in I_\ell$. This implies that $K_\ell \subseteq I_\ell \subseteq V_\delta(\xi) \subseteq U_{\alpha_0}$. Thus, K_ℓ is covered by a single open set U_{α_0} , which contradicts the assumption that K_ℓ does not have a finite subcover.

Therefore, K must have a finite subcover, and we conclude that K is compact. $\square$

Definition 5.27. A set $K \subseteq \mathbb{R}^n$ is called an *n-cube* (or *box*) if

$$K = [a_1, b_1] \times \cdots \times [a_n, b_n].$$

Define the *diameter* of K as

$$\text{diam}(K) := \max_{1 \leq i \leq j} |b_i - a_i|.$$

Generalised nested interval theorem:

Theorem 5.28. *Let $\{K_i\}$ be a nested decreasing sequence of n -cubes. Then*

$$\bigcap_{i=1}^{\infty} K_i \neq \emptyset.$$

If $\text{diam}(K_i) \rightarrow 0$, then the intersection is a singleton.

By a similar proof, we can show that Heine-Borel theorem holds for $\mathbb{R}^n$ - replace nested interval with nested cube

Theorem 5.29 (Weierstrass). *Any bounded sequence $\{x_n\}$ in $\mathbb{R}^n$ has a convergent subsequence.*

Proof. similar to 1-dimensional case, use nested interval idea

OR consider each coordinate, get subsequence. see lecture notes. $\square$

Theorem 5.30. *Let $\{K_\alpha\}_{\alpha \in \mathcal{A}}$ be a collection of compact subsets of a metric space X such that for any finite subcollection, the intersection is non-empty. Then*

$$\bigcap_{\alpha \in \mathcal{A}} K_\alpha \neq \emptyset.$$

Proof. Suppose, towards a contradiction, that $\bigcap_{\alpha \in \mathcal{A}} K_\alpha = \emptyset$. Fix $\alpha \in \mathcal{A}$. For any $x \in K_\alpha$, since $x \notin \bigcap_{\alpha \in \mathcal{A}} K_\alpha$, there exists $\alpha_x \in \mathcal{A}$ such that $x \notin K_{\alpha_x}$. Denote $G_{\alpha_x} := X \setminus K_{\alpha_x}$. Then $x \in G_{\alpha_x}$.

Since X is a metric space and K_{α_x} is compact, K_{α_x} is closed, so its complement G_{α_x} is open. Thus, the union of all K_{α_x} gives an open cover of K :

$$K_\alpha \subseteq \bigcup_{x \in K_\alpha} G_{\alpha_x}.$$

Since K_α is compact, there exists a finite subcover

$$K_\alpha \subseteq \bigcup_{i=1}^n G_{\alpha_{x_i}}.$$

This implies that

$$K_\alpha \cap \left(\bigcup_{i=1}^n G_{\alpha_{x_i}} \right)^c = \emptyset.$$

But

$$\left(\bigcup_{i=1}^n G_{\alpha_{x_i}} \right)^c = \bigcap_{i=1}^n G_{\alpha_{x_i}}^c = \bigcap_{i=1}^n K_{\alpha_{x_i}}.$$

Hence, $K_\alpha \cap \bigcap_{i=1}^n K_{\alpha_{x_i}} = \emptyset$. This is a finite subcollection with empty intersection, contradicting our assumption that any finite collection has a non-empty intersection. $\square$

An immediate corollary is a generalization of the nested interval theorem (2.27).

Corollary 5.31. *Let $\{K_n\}$ be a nested decreasing sequence of non-empty compact subsets of a metric space X . Then*

$$\bigcap_{n=1}^{\infty} K_n \neq \emptyset.$$

5.5 Sequential compactness

Definition 5.32. Let (X, d) be a metric space. A set $K \subseteq X$ is called *sequentially compact* if every sequence $\{x_n\} \subseteq K$ has a convergent subsequence $\{x_{n_k}\}$ such that $\lim_{k \rightarrow \infty} x_{n_k} = \xi \in K$.

On $\mathbb{R}$, it is easily seen that compact $\iff$ closed and bounded $\iff$ sequentially compact.

Remark. This is equivalent to the fact that any infinite subset $E \subseteq K$ has a limit point ξ in K , i.e., $K \cap E' \neq \emptyset$.

Our goal is to show that compactness is equivalent to sequential compactness.

Theorem 5.33. *Let (X, d) be a metric space, and $K \subseteq X$. Then K is compact if and only if K is sequentially compact.*

Proof. The forward direction is easier. The backward direction is more complicated and will require establishing a few results. Suppose $K \subseteq X$ is compact. We will prove that for any infinite subset $E \subseteq K$, $E' \cap K \neq \emptyset$.

For a contradiction, assume that $E' \cap K = \emptyset$, i.e., for each $x \in K$, $x \notin E'$. Since x is not a limit point of E , there exists $\delta_x > 0$ such that

$$V_{\delta_x}(x) \cap E \subseteq \{x\}.$$

(The intersection is empty if $x \notin E$, and equals $\{x\}$ if $x \in E$.) Consider the collection of neighbourhoods as an open cover of K :

$$K \subseteq \bigcup_{x \in K} V_{\delta_x}(x).$$

Since K is compact, there exists a finite subcover

$$K \subseteq \bigcup_{i=1}^n V_{\delta_{x_i}}(x_i).$$

Note that $E \subseteq K$ implies $E \subseteq \bigcup_{i=1}^n V_{\delta_{x_i}}(x_i)$. Since $V_{\delta_{x_i}}(x_i) \cap E$ is finite (since each neighbourhood contains at most 1 point), the union is also finite. Thus, E must be finite, contradicting the assumption that E is infinite. $\square$

The proof of the backward direction requires the introduction of two auxiliary notions:

Definition 5.34. A space X is *separable* if it admits a countable dense subset.

For example, $\mathbb{R}$ is separable ($\mathbb{Q}$ is countable, and it is dense since every real number is a limit of rationals); for the same reason $\mathbb{R}^n$ is separable (consider all points with only rational coordinates).

Definition 5.35. A collection $\{U_\alpha\}_{\alpha \in \mathcal{A}}$ of open subsets of X is a *base* for X if for every $x \in X$ and for every open set $U \subseteq X$ such that $x \in U$, there exists $\alpha \in \mathcal{A}$ such that $x \in U_\alpha \subseteq U$.

In other words, the U_α 's "generate" all open subsets. The family $\{U_\alpha\}$ almost always contains infinitely many members (the only exception is if X is finite). However, if X happens to be separable, then countably many open subsets are enough to form a base:

Lemma 5.36. *Every separable metric space has a countable base.*

Proof. Suppose X is separable. By definition, X contains a countable dense subset $P = \{p_1, p_2, \dots\}$.

Consider the countable collection of neighborhoods $\{V_r(p_i) : r \in \mathbb{Q}, i = 1, 2, \dots\}$. We show that it is a base by checking the definition.

Consider any open set $U \subseteq X$ and any point $x \in U$. Since U is open, there exists $\delta > 0$ such that $V_\delta(x) \subseteq U$. Decreasing δ if necessary we can assume without loss of generality that δ is rational.

Since P is dense, $V_{\delta/2}(x)$ contains a point $p_i \in P$. Note that $V_{\delta/2}(p_i)$ belongs to the chosen collection, and $V_{\delta/2}(p_i) \subseteq V_{\delta/2}(x) \subseteq V_\delta(x) \subseteq U$. Also, $x \in V_{\delta/2}(p_i)$. Hence, the chosen collection is a base for X . $\square$

Lemma 5.37. *If X is sequentially compact, then X is separable.*

Proof. Fix $\delta > 0$, and choose $x_1 \in X$. Choose $x_2 \in X$ such that $d(x_1, x_2) \geq \delta$, if possible. Having chosen $x_1, \dots, x_j$, choose x_{j+1} (if possible) such that $d(x_i, x_{j+1}) \geq \delta$ for all $i = 1, \dots, j$.

We first notice that this process has to stop after a finite number of iterations. Indeed, otherwise we would obtain an infinite sequence of points x_i mutually distant by at least δ . Since X is sequentially compact the infinite subset $\{x_i : i = 1, 2, \dots\}$ has a limit point $y \in X$. Then $V_{\delta/2}(y)$ would contain infinitely many x_i 's, contradicting the fact that any two of them are distant by at least δ .

(Alternative: Since K is sequentially compact, there exists a convergent subsequence $\{y_k = x_{n_k} : k \in \mathbb{N}\}$; let $y_k \rightarrow y \in K$. There exists $N \in \mathbb{N}$ such that

$$d(y, y_k) < \frac{\delta}{4} \quad \text{for all } k \geq N.$$

By triangle inequality, for all $k \geq N$,

$$\begin{aligned} d(y_k, y_{k+1}) &\leq d(y_k, y) + d(y, y_{k+1}) \\ &< \frac{\delta}{4} + \frac{\delta}{4} < \delta. \end{aligned}$$

This contradicts $d(y_k, y_{k+1}) \geq \delta$.

Hence, after a finite number of iterations we obtain $S_\delta = \{x_1, \dots, x_j\}$ such that $V_\delta(x_1) \cup \dots \cup V_\delta(x_j) = X$ (since every point of X is at distance $< \delta$ from one of the x_i 's).

We now consider this construction for $\delta = \frac{1}{n}$ ($n = 1, 2, \dots$).

- For $n = 1$, the construction gives points $x_{11}, \dots, x_{1j_1}$ such that $V_1(x_{11}) \cup \dots \cup V_1(x_{1j_1}) = X$.
- For $n = 2$, we get $x_{21}, \dots, x_{2j_2}$ such that $V_{1/2}(x_{21}) \cup \dots \cup V_{1/2}(x_{2j_2}) = X$,
- and so on.

Let $S = \bigcup_{n \in \mathbb{N}} S_{1/n}$. Clearly S is countable. We will show that S is dense. Indeed, if $x \in X$ and $r > 0$, the neighborhood $V_r(x)$ always contains at least a point of S (choosing n so that $\frac{1}{n} < r$, one of the x_{ni} 's is at distance less than r from x), so every point of X either belongs to S or is a limit point of S , i.e. $\bar{S} = X$. $\square$

We can now prove the backward direction. Suppose K is sequentially compact. To prove that K is compact, take any open cover $K \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$. We will prove that there exists a finite subcover.

For a **simple case**, suppose the open cover is countable, i.e., $\mathcal{A}$ is countable. Write $K \subseteq \bigcup_{i=1}^{\infty} U_i$. For a contradiction, suppose there does not exist a finite subcover. Then, for every $n \in \mathbb{N}$, we have $K \not\subseteq \bigcup_{i=1}^n U_i$. This means that there exists $x_n \in K$ such that $x_n \notin \bigcup_{i=1}^n U_i$. This gives a sequence $\{x_n\}$ in K . Since K is sequentially compact, there exists a convergent subsequence $\{y_k = x_{n_k}\}$ such that $y_k \rightarrow \xi \in K$.

Since $\xi \in \bigcup_{i=1}^{\infty} U_i$, there exists $m \in \mathbb{N}$ such that $\xi \in U_m$. Since U_m is open, there exists $\delta > 0$ such that $V_\delta(\xi) \subseteq U_m$. Since $y_k \rightarrow \xi$, choose k large enough such that $|y_k - \xi| < \delta$, so $y_k \in V_\delta(\xi) \subseteq U_m$.

Note that $y_k = x_{n_k} \notin U_{m'}$ for all $m' \leq n_k$. In particular, $y_k \notin U_m$ (since $m \leq k$ implies $m \leq n_k$). This gives a contradiction.

Now consider the **general case**, where $\mathcal{A}$ might be uncountable. Take any open cover of K , say $K \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$. By 5.37, K is separable. By 5.36, K has a countable base $\{U_n\}_{n \in \mathbb{N}}$. Then every U_α can be written as $U_\alpha = \bigcup_{n \in S_\alpha} U_n$ for some subset $S_\alpha \subseteq \mathbb{N}$. Thus,

$$K \subseteq \bigcup_{n \in S} U_n, \quad \text{where } S = \bigcup_{\alpha \in \mathcal{A}} S_\alpha \subseteq \mathbb{N}.$$

This is a countable open cover. By the simple case, there exists a finite subcover $K \subseteq \bigcup_{i=1}^N U_{n_i}$. For every U_{n_i} , there exists $\alpha_i \in \mathcal{A}$ such that $U_{n_i} \subseteq U_{\alpha_i}$. Thus, $K \subseteq \bigcup_{i=1}^N U_{\alpha_i}$.

Corollary 5.38. *If X is a compact metric space, then X is complete.*

Proof. Let $\{x_n\}$ be a Cauchy sequence in X . Since X is sequentially compact, $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$ such that $x_{n_k} \rightarrow x$.

We claim that $x_n \rightarrow x$. Note that $d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x)$. $\square$

5.6 Connectedness

Definition 5.39. Let X be a topological space. We say that X is *disconnected* if there exist disjoint non-empty open subsets $A, B \subseteq X$ such that $X = A \cup B$. We call the sets A and B a *separation* of X . If no separation of X exists, then X is *connected*.

A subset $S \subseteq X$ is *connected* if it is connected in the subspace topology. This means there do not exist disjoint non-empty sets $A, B \subseteq S$ such that $S = A \cup B$, and A and B are open in the subspace topology on S . Equivalently, there exist open sets U and V such that $A = S \cap U$, $B = S \cap V$.

Thus, written explicitly: A subset $S \subseteq X$ is connected if there do not exist open sets $U, V \subseteq X$ such that:

- $S \subseteq U \cup V$,
- $S \cap U \neq \emptyset$, $S \cap V \neq \emptyset$,
- $(S \cap U) \cap (S \cap V) = \emptyset$.

The next result characterises all connected subsets of $\mathbb{R}$.

Theorem 5.40. *Let $S \subseteq \mathbb{R}$. Then S is connected if and only if S is an interval, namely, for any $s_1, s_2 \in S$, $s_1 < s_2$,*

$$[s_1, s_2] \subseteq S.$$

Proof.

$\Rightarrow$ Suppose S is connected. For a contradiction, suppose S is not an interval, i.e., there exists $s_1 < s < s_2$ such that $s \notin S$.

Take $A = (-\infty, s)$ and $B = (s, +\infty)$, which are open, disjoint, and $A \cup B$ covers S . The separation is non-trivial; $A \cap S \neq \emptyset$ since $s_1 \in A \cap S$, $B \cap S \neq \emptyset$ since $s_2 \in B \cap S$.

This gives a separation of S , so S is disconnected. This yields the desired contradiction.

$\Leftarrow$ Suppose S is an interval. For a contradiction, assume S is disconnected. Thus, there exist A, B open such that $A \cap S \neq \emptyset$, $B \cap S \neq \emptyset$, $S \subseteq A \cup B$, $S \cap A \cap B = \emptyset$.

Since S is an interval, take $s_1 < s_2 \in S$ such that $s_1 \in A$, $s_2 \in B$. Then $[s_1, s_2] \subseteq S$. Define

$$w = \sup \{s \in [s_1, s_2] : [s_1, s] \subseteq A\}.$$

We consider two cases:

- Case 1: $w \in A$. Since A is open, there exists $\delta > 0$ such that $(w - \delta, w + \delta) \subseteq A$. This implies $[s_1, w + \delta) \subseteq A$, contradicting that w is the supremum.
- Case 2: $w \in B$. Since B is open, there exists $\delta > 0$ such that $(w - \delta, w + \delta) \subseteq B$. This implies there is no $s > w - \delta$ such that $[s_1, s] \subseteq A$, also a contradiction.

□

Remark. simply connected - for any two points, there exists a path between them

5.7 Continuity

Definition 5.41. Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f: X \rightarrow Y$ is continuous at $x \in X$ if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall d_X(x, y) < \delta, \quad d_Y(f(x), f(y)) < \varepsilon.$$

Lemma 5.42. Let (X, d_X) and (Y, d_Y) be metric spaces, $f, g: X \rightarrow Y$ continuous at $x \in X$. Then

$$f \pm g, \quad fg, \quad f/g, \quad \lambda f$$

are continuous at x , provided $g(x) \neq 0$ in $V_\delta(c)$.

Definition 5.43. Let X and Y be topological spaces. A map $f: X \rightarrow Y$ is called *continuous* if for any $U \subseteq Y$ open, $f^{-1}(U)$ is open in X .

Theorem 5.44.

Remark. Given a metric space (X, d) , and a function $f: X \rightarrow \mathbb{R}$, we can define pointwise continuity of f at $c \in X$ similarly to the case $X \subseteq \mathbb{R}$: f is *continuous* at c if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in V_\delta(c), \quad |f(x) - f(c)| < \varepsilon.$$

In this setting, we have that $f: X \rightarrow \mathbb{R}$ continuous (in the sense of maps between topological spaces) if and only if it is continuous at every $c \in X$.

Remark.

Proposition 5.45. *Let X, Y and Z be topological spaces, $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ continuous. Then $g \circ f: X \rightarrow Z$ is continuous.*

The next theorem tells us that continuous functions preserve compactness.

Theorem 5.46. *Let $f: X \rightarrow Y$ be a continuous map. For any compact subset $K \subseteq X$, $f(K)$ is compact.*

Proof. Take any open cover $f(K) \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$. We want to find a finite subcover.

Let $V_\alpha = f^{-1}(U_\alpha)$, which is open in X by the continuity of f .

Claim: $\{V_\alpha\}$ gives an open cover of K , i.e., $K \subseteq \bigcup_{\alpha \in \mathcal{A}} V_\alpha$.

Let $x \in K$. Then $f(x) \in f(K) \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$, so $f(x) \in U_\alpha$ for some $\alpha \in \mathcal{A}$. Thus, $x \in f^{-1}(U_\alpha) = V_\alpha$, so that $x \in \bigcup_{\alpha \in \mathcal{A}} V_\alpha$.

Since K is compact, there exists a finite subcover □

Remark. If $f: [a, b] \rightarrow \mathbb{R}$ is continuous, then $[a, b]$ compact implies $f([a, b])$ is compact, and thus is closed and bounded. This implies the boundedness theorem, and also extreme value theorem since it contains limits points (sup and inf, which are max and min).

The next theorem states that under a continuous map, the image of a connected set is connected.

Theorem 5.47. *Let $f: X \rightarrow Y$ be a continuous map. For any connected subset $C \subseteq X$, $f(C)$ is connected.*

Proof. For a contradiction, suppose $f(C)$ is not connected. Then $f(C)$ has a separation, namely, there exist open sets A and B such that $f(C) \subseteq A \cup B$ and $f(C) \cap A \cap B = \emptyset$, $f(C) \cap A \neq \emptyset$, $f(C) \cap B \neq \emptyset$.

Claim: $f^{-1}(A), f^{-1}(B)$ give a separation of C .

- It is easy to show that $C \subseteq f^{-1}(A) \cup f^{-1}(B)$.
- We only check $C \cap f^{-1}(A) \neq \emptyset$. (The proof of $C \cap f^{-1}(B) \neq \emptyset$ is similar.)
Since $f(C) \cap A \neq \emptyset$, fix $y \in f(C) \cap A$. Then $y = f(x)$ for some $x \in C$. Thus, $x \in C \cap f^{-1}(A)$.
- We check $C \cap f^{-1}(A) \cap f^{-1}(B) = \emptyset$. For a contradiction, suppose not. Fix $x \in C \cap f^{-1}(A) \cap f^{-1}(B)$. Then $f(x) \in f(C) \cap A \cap B$, contradicting the fact that $f(C) \cap A \cap B = \emptyset$.

Hence, C has a separation. This contradicts the connectedness of C . □

Remark. $[a, b]$ connected, so $f([a, b])$ connected thus interval, thus IVT

5.8 Completeness

Definition 5.48. Let (X, d) be a metric space. A sequence $\{x_n\} \subseteq X$ is a *Cauchy sequence* if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \quad d(x_m, x_n) < \varepsilon.$$

Exercise. A convergent sequence is Cauchy.

The Cauchy criterion, as it is called in $\mathbb{R}$, was an “if and only if” statement. In the general metric space setting, however, the converse statement does not always hold.

Definition 5.49. A metric space (X, d) is *complete* if every Cauchy sequence in X converges to an element of X .

Let X be a metric space. If X is not complete, can we somehow “fill in the holes” in a systematic way? That is, can we realize X as a dense subset of a complete metric space (with compatible metrics, of course). More specifically, we seek an isometry (i.e., a metric-preserving map)

$$\phi: X \rightarrow X^*$$

with dense image and X^* complete (note that isometries are always injective).

Definition 5.50. A *completion* of a metric space (X, d) is a pair consisting of a complete metric space (X^*, d^*) and an isometry $\phi: X \rightarrow X^*$ such that $\phi(X)$ is dense in X^* .

Lemma 5.51. Let (X, d) be a metric space and $A \subseteq X$ be dense, such that every Cauchy sequence in A converges in X . Then X is complete.

Theorem 5.52. Every metric space has a completion.

Proof. Let (X, d) be a metric space. Denote by $C(X)$ the collection of all Cauchy sequences in X . Define a relation $\sim$ on $C(X)$ by

$$\{x_n\} \sim \{y_n\} \iff \lim_{n \rightarrow \infty} d(x_n, y_n) = 0.$$

It is easy to see that this is an equivalence relation on $C(X)$. Let X^* be the set of all equivalence classes for $\sim$:

$$X^* := \{[\{x_n\}] : \{x_n\} \in C(X)\}.$$

Define $d^*: X^* \times X^* \rightarrow [0, \infty)$ by

$$d^*([\{x_n\}], [\{y_n\}]) = \lim_{n \rightarrow \infty} d(x_n, y_n),$$

where $[\{x_n\}], [\{y_n\}] \in X^*$. To show that d^* is well-defined, let $\{x'_n\}$ and $\{y'_n\}$ be two Cauchy sequences in X such that $\{x_n\} \sim \{x'_n\}$ and $\{y_n\} \sim \{y'_n\}$. Then

$$\lim_{n \rightarrow \infty} d(x_n, x'_n) = \lim_{n \rightarrow \infty} d(y_n, y'_n) = 0.$$

By triangle inequality,

$$\begin{aligned} d(x_n, y_n) &\leq d(x_n, x'_n) + d(x'_n, y'_n) + d(y'_n, y_n), \\ d(x'_n, y'_n) &\leq d(x'_n, x_n) + d(x_n, y_n) + d(y_n, y'_n). \end{aligned}$$

Hence,

$$|d(x_n, y_n) - d(x'_n, y'_n)| \leq d(x_n, x'_n) + d(y_n, y'_n) \rightarrow 0.$$

This shows that $\lim_{n \rightarrow \infty} d(x_n, y_n) = \lim_{n \rightarrow \infty} d(x'_n, y'_n)$. Thus, d^* is well-defined.

Next, we show that d^* is a metric on X^* . Let $[\{x_n\}], [\{y_n\}], [\{z_n\}] \in X^*$. Then

(1) Non-negativity is obvious. Note that

$$d^*([\{x_n\}], [\{y_n\}]) = 0 \iff \lim_{n \rightarrow \infty} d(x_n, y_n) = 0 \iff \{x_n\} \sim \{y_n\} \iff [\{x_n\}] = [\{y_n\}].$$

(2) Symmetry follows easily:

$$d^*([\{x_n\}], [\{y_n\}]) = \lim_{n \rightarrow \infty} d(x_n, y_n) = \lim_{n \rightarrow \infty} d(y_n, x_n) = d^*([\{y_n\}], [\{x_n\}]).$$

(3) To show triangle inequality, since $d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n)$, taking the limits $n \rightarrow \infty$ yields

$$d^*([\{x_n\}], [\{z_n\}]) \leq d^*([\{x_n\}], [\{y_n\}]) + d^*([\{y_n\}], [\{z_n\}]).$$

After that, we show that $\phi: X \rightarrow X^*$ is an isometry. For each $x \in X$, denote $\hat{x} = [(x, x, \dots)] \in X^*$, the equivalence classes of the constant sequence $(x, x, \dots)$. Define $\phi: X \rightarrow X^*$ by $\phi(x) = \hat{x}$. Then for any $x, y \in X$,

$$d^*(\phi(x), \phi(y)) = d^*(\hat{x}, \hat{y}) = \lim_{n \rightarrow \infty} d(x, y) = d(x, y).$$

To show that $\phi(X)$ is dense in X^* , let $x^* = [(x_n)] \in X^*$ and let $\varepsilon > 0$. Since $\{x_n\}$ is a Cauchy sequence, there exists $N \in \mathbb{N}$ such that

$$d(x_m, x_n) < \frac{\varepsilon}{2} \quad \text{for any } m, n \geq N.$$

Consider the point $z = x_N$. Then $\hat{z} \in \phi(X)$ and

$$d^*(x^*, \hat{z}) = \lim_{n \rightarrow \infty} d(x_n, z) = \lim_{n \rightarrow \infty} d(x_n, x_N) \leq \frac{\varepsilon}{2} < \varepsilon.$$

Hence, $\phi(X)$ is dense in X^* .

Using the lemma, it suffices to show that every Cauchy sequence in the dense subspace $\phi(X)$ converges in X^* . Let $\{\hat{z}_k\}$ be a Cauchy sequence in $\phi(X)$, where each $\hat{z}_k$ is represented by the Cauchy sequence $(z_k, z_k, \dots)$. Since ϕ is an isometry, we have

$$d(z_n, z_m) = d^*(\hat{z}_n, \hat{z}_m) \quad \text{for every } m, n.$$

Thus, $(z_1, z_2, \dots)$ is a Cauchy sequence in X . Consider $z^* = [(z_1, z_2, \dots)] \in X^*$. We will show that $\{\hat{z}_k\}$ converges to z^* .

Let $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that $d(z_k, z_n) < \frac{\varepsilon}{2}$ for any $k, n \geq N$. Hence, for each $k \geq N$,

$$d^*(\hat{z}_k, z^*) = \lim_{n \rightarrow \infty} d(z_k, z_n) \leq \frac{\varepsilon}{2} < \varepsilon.$$

This shows that $\{\hat{z}_k\}$ converges to a point z^* in X^* . Hence, X^* is complete. $\square$

Notation. Since ϕ gives an embedding of X into X^* , we usually write $X \subset X^*$.

Theorem 5.53. *A completion of a metric space is unique up to isometry. More precisely, if $\{\phi_1, (X_1^*, d_1^*)\}$ and $\{\phi_2, (X_2^*, d_2^*)\}$ are two completions of (X, d) , then there is a unique isometry f from X_1^* onto X_2^* such that $f \circ \phi_1 = \phi_2$.*

Proof. Since ϕ_1 is an isometry, ϕ_1 is 1-1. Thus $\phi_1^{-1}: \phi_1(X) \rightarrow X$ is an isometry from $\phi_1(X)$ onto X . Since ϕ_2 is an isometry from X onto $\phi_2(X) \subseteq X_2^*$, it follows that $\phi_2 \circ \phi_1^{-1}: \phi_1(X) \rightarrow \phi_2(X)$ is a surjective isometry. Let $h = \phi_2 \circ \phi_1^{-1}$. Then

$$h \circ \phi_1 = (\phi_2 \circ \phi_1^{-1}) \circ \phi_1 = \phi_2 \circ (\phi_1^{-1} \circ \phi_1) = \phi_2 \circ i_X = \phi_2.$$

Hence there exists a unique isometry f from X_1^* into X_2^* which is an extension of h . For each $x \in X$,

$$f \circ \phi_1(x) = f(\phi_1(x)) = h(\phi_1(x)) = h \circ \phi_1(x) = \phi_2(x).$$

Thus $f \circ \phi_1 = \phi_2$. Similarly, there exists a unique isometry g from X_2^* into X_1^* such that $g \circ \phi_2 = \phi_1$. Therefore

$$g \circ f \circ \phi_1 = g \circ \phi_2 = \phi_1 \quad \text{and} \quad f \circ g \circ \phi_2 = f \circ \phi_1 = \phi_2.$$

Hence $g \circ f = i_{\phi_1(X)}$ and $f \circ g = i_{\phi_2(X)}$. Since $\phi_1(X)$ is dense in X_1^* , we have $g \circ f = i_{X_1^*}$. Similarly, $f \circ g = i_{X_2^*}$. Thus $f = g^{-1}$. Hence, f is a unique isometry from X_1^* onto X_2^* such that $f \circ \phi_1 = \phi_2$. $\square$

$\square$

6 Sequences and Series of Functions

6.1 Sequences of functions

Given a sequence of functions $\{f_n(x) : n \in \mathbb{N}\}$, how do we define the convergence? What can we say about the limit function?

The natural definition is to fix x , then $\{f_n(x)\}$ is a sequence of numbers that converges to $f(x)$. Combine over all x to get a function f .

Definition 6.1. A sequence of functions $\{f_n : A \rightarrow \mathbb{R}\}$ converges pointwise on A to a function f , if for any $x \in A$,

$$\lim_{n \rightarrow \infty} f_n(x) = f(x).$$

In this case, we write $f_n \rightarrow f$.

Example. Consider $f_n : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f_n(x) = \frac{x^2 + nx}{n}.$$

Fix $x \in \mathbb{R}$. Then

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{x^2 + nx}{n} = \lim_{n \rightarrow \infty} \frac{x^2}{n} + x = x$$

is the limit function. Hence, $\{f_n\}$ converges pointwise to $f(x) = x$ on $\mathbb{R}$.

However, pointwise convergence does not preserve continuity.

Example. Let $f_n(x) = x^n$ over $[0, 1]$. For a fixed $x \in [0, 1]$,

$$f(x) = \lim_{n \rightarrow \infty} x^n = \begin{cases} 0 & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x = 1. \end{cases}$$

Hence, the limit function $f(x)$ is not continuous on $[0, 1]$.

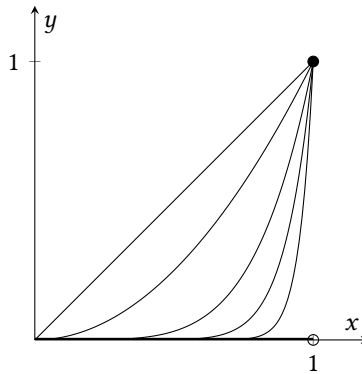


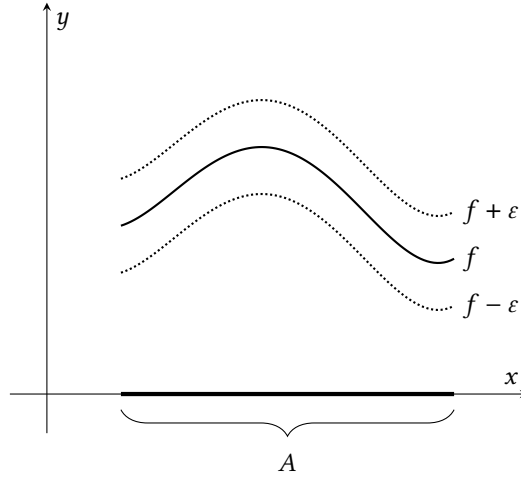
Figure 6.1: $f(x) = \lim_{n \rightarrow \infty} x^n$ is not continuous on $[0, 1]$.

To solve this problem, we define a new, stronger notion of convergence of functions.

Definition 6.2. A sequence of functions $\{f_n : A \rightarrow \mathbb{R}\}$ converges uniformly on A to a function f if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad \forall x \in A, \quad |f_n(x) - f(x)| < \varepsilon.$$

Graphically, if $f_n \rightarrow f$ uniformly, then there exists a point in the sequence after which each f_n is completely contained in an ε -tube around f .



Remark. The difference between pointwise and uniform convergence is similar to that between continuity and uniform continuity.

Example. Consider $\{f_n\}$ defined by

$$f_n(x) = \frac{1}{n(1+x^2)}.$$

For any fixed $x \in \mathbb{R}$, we can see that $\lim_{n \rightarrow \infty} f_n(x) = 0$, so $f(x) = 0$ is the pointwise limit of $\{f_n\}$. Next, we show the convergence is uniform. The observation that $\frac{1}{1+x^2} \leq 1$ for all $x \in \mathbb{R}$ implies that

$$|f_n(x) - f(x)| = \left| \frac{1}{n(1+x^2)} - 0 \right| \leq \frac{1}{n}.$$

Let $\varepsilon > 0$. Choose $N > \frac{1}{\varepsilon}$ (independent of x). If $n \geq N$, then $|f_n(x) - f(x)| < \varepsilon$ for all $x \in \mathbb{R}$.

Definition 6.3. Let $f: A \rightarrow \mathbb{R}$ be a function. The *supremum norm* of f is defined as

$$\|f\| := \sup_{x \in A} |f(x)|.$$

In particular, if f is continuous on $[a, b]$, then $\sup$ norm = maximum.

Let X denote the space of functions on $[a, b]$. Then the induced metric is $d(f, g) = \|f - g\|$. This makes (X, d) a metric space. We prove the triangle inequality: For any functions f, g, h ,

$$d(f, g) = \sup_{x \in [a, b]} |f(x) - g(x)|.$$

For any $x \in [a, b]$, by triangle inequality,

$$\begin{aligned} |f(x) - g(x)| &\leq |f(x) - h(x)| + |h(x) - g(x)| \\ &\leq d(f, h) + d(h, g). \end{aligned}$$

Taking the supremum over $x \in [a, b]$,

$$d(f, g) = \sup_{x \in [a, b]} |f(x) - g(x)| \leq d(f, h) + d(h, g).$$

Corollary 6.4. Let $\{f_n\}$ be a sequence of functions. Then $f_n \rightarrow f$ uniformly if and only if

$$\lim_{n \rightarrow \infty} \|f_n - f\| = 0.$$

Example. In the previous example, f_n does not uniformly converge to f , because for each $n \in \mathbb{N}$,

$$\|f_n - f\| = \sup_{0 \leq x < 1} |x^n - 0| = 1.$$

Hence, $f_n \not\rightarrow f$ uniformly.

One of the most important properties of uniform convergence is that it preserves continuity.

Theorem 6.5. Let $\{f_n: A \rightarrow \mathbb{R}\}$ be a sequence of functions, and $f_n \rightarrow f$ uniformly. If each f_n is continuous on A , then f is continuous on A .

Proof. Fix $c \in A$. We will prove that $f(x)$ is continuous at c , namely,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in V_\delta(c), \quad |f(x) - f(c)| < \varepsilon.$$

Let $\varepsilon > 0$. Since $f_n \rightarrow f$ uniformly, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x \in A$,

$$|f_n(x) - f(x)| < \frac{\varepsilon}{3}.$$

In particular, if $n = N$, then

$$|f_N(x) - f(x)| < \frac{\varepsilon}{3}. \quad (1)$$

Since f_N is continuous, there exists $\delta > 0$ such that for all $x \in V_\delta(c)$,

$$|f_N(x) - f_N(c)| < \frac{\varepsilon}{3}. \quad (2)$$

Then for $x \in V_\delta(c)$, by triangle inequality,

$$\begin{aligned} |f(x) - f(c)| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

□

This result can be interpreted as justifying an “exchange in the order of limits”

$$\lim_{n \rightarrow \infty} \lim_{x \rightarrow c} f_n(x) = \lim_{x \rightarrow c} \lim_{n \rightarrow \infty} f_n(x).$$

Definition 6.6. A sequence of functions $\{f_n\}$ is called *Cauchy* if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \quad \|f_n - f_m\| < \varepsilon.$$

Theorem 6.7. $C[a, b]$ is complete.

Proof. Let $\{f_n\}$ be a Cauchy sequence in $C[a, b]$. Let $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that

$$\|f_n - f_m\| < \varepsilon \quad \text{for all } m, n \geq N.$$

Fix $x \in [a, b]$. Then the sequence of values $\{f_n(x)\}$ is a Cauchy sequence of real numbers, because $|f_n(x) - f_m(x)| \leq \|f_n - f_m\| < \varepsilon$. Since $\mathbb{R}$ is complete, $\{f_n(x)\}$ converges to some value $f(x)$.

Claim: $f_n \rightarrow f$ uniformly on $[a, b]$.

To show uniform convergence, we take the limit as $m \rightarrow \infty$ in the Cauchy inequality:

$$\lim_{m \rightarrow \infty} |f_n(x) - f_m(x)| = |f_n(x) - f(x)| \leq \varepsilon.$$

Since this holds for all $x \in [a, b]$ and all $n \geq N$, it follows that $\|f_n - f\| \leq \varepsilon$, meaning $f_n \rightarrow f$ uniformly.

Since each f_n is continuous and the convergence is uniform, the limit function f is also continuous on $[a, b]$. Hence, $f \in C[a, b]$. $\square$

6.2 Series of functions

Definition 6.8. A series of functions $\sum f_n(x)$ converges uniformly on a set E if the sequence of partial sums $S_n(x) = \sum_{k=1}^n f_k(x)$ converges uniformly to a function $S(x)$ on E .

The next result shows that if the series of norms converges, then the series converges uniformly.

Corollary 6.9. If $\sum \|f_n\|$ converges, then $\sum f_n(x)$ converges uniformly.

Proof. Let $S_n(x) = \sum_{k=1}^n f_k(x)$ denote the sequence of partial sums. We will show that $\{S_n\}$ is a Cauchy sequence in the sup norm. For $m > n$,

$$\begin{aligned} \|s_m - s_n\| &= \left\| \sum_{k=n+1}^m f_k(x) \right\| = \sup_{x \in [a, b]} \left| \sum_{k=n+1}^m f_k(x) \right| \\ &\leq \sup_{x \in [a, b]} \sum_{k=n+1}^m |f_k(x)| \leq \sup_{x \in [a, b]} \sum_{k=n+1}^m \|f_k\| = \sum_{k=n+1}^m \|f_k\|. \end{aligned}$$

Since $\sum \|f_n\|$ converges, its sequence of partial sums is Cauchy. Thus, for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m > n \geq N$,

$$\sum_{k=n+1}^m \|f_k\| < \varepsilon.$$

Combining this with the result above gives

$$\|s_m - s_n\| < \varepsilon$$

which shows that $\{S_k\}$ is a Cauchy sequence in $C[a, b]$. Since $C[a, b]$ is complete, there exists a function $S \in C[a, b]$ such that $S_k \rightarrow S$ uniformly. By definition of S_k , we have

$$S(x) = \sum_{n=1}^{\infty} f_n(x),$$

and the convergence is uniform. Also, $S(x)$ is continuous. $\square$

In practice, this result is often applied by bounding $\|f_n\|$ with a simpler sequence.

Exercise. Let $\{r_n : n \in \mathbb{N}\}$ be the set of rational numbers in $[0, 1]$. Prove that

$$f(x) = \sum_{n=1}^{\infty} \frac{|x - r_n|}{3^n}$$

is continuous on $[0, 1]$.

Solution. Let $f_n(x) = \frac{|x - r_n|}{3^n}$. For any $x \in [0, 1]$, $r_n \in [0, 1]$,

$$f_n(x) = \frac{|x - r_n|}{3^n} \leq \frac{1}{3^n}.$$

Taking sup over $x \in [0, 1]$ gives $\|f_n\| \leq \frac{1}{3^n}$. Since $\sum \frac{1}{3^n}$ converges, by comparison test, $\sum \|f_n\|$ converges. By 6.9, $\sum_{n=1}^{\infty} \frac{|x - r_n|}{3^n}$ converges uniformly. $\square$

A key point is that the bound on $\|f_n\|$ must be independent of x .

Exercise. Prove that

$$f(x) = \sum_{n=1}^{\infty} 2^n \sin\left(\frac{x}{3^n}\right)$$

is continuous on $[0, +\infty)$.

Solution. For infinite interval, show it holds on a finite cut-off. That is, we prove that for any fixed $M > 0$, $\sum_{n=1}^{\infty} 2^n \sin\left(\frac{x}{3^n}\right)$ is uniformly convergent on $[0, M]$.

Then $x \leq M$. Choose $N \in \mathbb{N}$ such that $\frac{M}{3^N} < 1$. Then for $n \geq N$,

$$\left| 2^n \sin\left(\frac{x}{3^n}\right) \right| = 2^n \sin\left(\frac{x}{3^n}\right) \leq \frac{2^n}{3^n} \cdot x \leq \left(\frac{2}{3}\right)^n \cdot M$$

where we used the inequality $\sin x \leq x$ for all $x \in [0, 1]$.

Since $\sum \left(\frac{2}{3}\right)^n M$ is convergent, by comparison test, $\sum \left\| 2^n \sin\left(\frac{x}{3^n}\right) \right\|$ converges (where sup on $[0, M]$, so sup $\leq \left(\frac{2}{3}\right)^n M$).

Hence, $\sum 2^n \sin\left(\frac{x}{3^n}\right)$ converges uniformly on $[0, M]$. But M is arbitrary. $\square$

Corollary 6.10 (Weierstrass M-test). *Let $\{f_n\}$ be a sequence of functions on a set A . Suppose there exists a sequence $\{M_n\}$ such that*

$$|f_n(x)| \leq M_n \quad \text{for all } x \in A.$$

If $\sum M_n$ converges, then $\sum f_n(x)$ converges uniformly on A .

Proof. From $|f_n(x)| \leq M_n$, we obtain $\|f_n\| \leq M_n$. Since $\sum M_n$ converges, by comparison test, $\sum \|f_n\|$ converges. The result now follows from 6.9. $\square$

6.3 Power series

Definition 6.11. A *power series* is a function of the form

$$S(x) = \sum_{n=1}^{\infty} a_n x^n.$$

Example. Taylor series

$$\begin{aligned} \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\ e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \end{aligned}$$

Proposition 6.12. *Let $L = \limsup_{n \rightarrow \infty} |a_n|^{1/n}$.*

(1) *If $L = +\infty$, then $\sum a_n x^n$ diverges for any $x \in \mathbb{R}$.*

(2) *If $L < +\infty$, then $\sum a_n x^n$ absolutely converges for $|x| < \frac{1}{L}$, and diverges for $|x| > \frac{1}{L}$.*

Convention: $1/0 = +\infty$.

Proof. The proof follows from the root test. Consider the term $u_n = a_n x^n$. We evaluate

$$\limsup_{n \rightarrow \infty} |u_n|^{1/n} = \limsup_{n \rightarrow \infty} |a_n x^n|^{1/n} = |x| \cdot \limsup_{n \rightarrow \infty} |a_n|^{1/n} = |x| \cdot L.$$

By root test, the series converges absolutely if $|x| \cdot L < 1$ and diverges if $|x| \cdot L > 1$.

Defining $R = 1/L$ establishes the interval of convergence $(-R, R)$. $\square$

Definition 6.13. $R = 1/L$ is called the *radius of convergence* of the power series $\sum a_n x^n$.

Then 6.12 states that a power series converges pointwise inside the radius, diverges outside the radius, and convergence is not clear on the boundary.

Theorem 6.14. Let $S(x) = \sum_{n=1}^{\infty} a_n x^n$ be a power series with radius of convergence $R > 0$. Then $S(x)$ is continuous on $(-R, R)$.

Proof. Let $0 < R' < R$. We will show that $\sum a_n x^n$ converges uniformly on $[-R', R']$, using 6.9. (This is enough to prove the statement on $(-R, R)$.)

For any $x \in [-R', R']$, we have $|x| \leq R'$, so $|a_n x^n| \leq |a_n|(R')^n$. Thus,

$$\|a_n x^n\| \leq |a_n|(R')^n.$$

Since $R' < R$, by 6.12, $\sum |a_n|(R')^n$ absolutely converges (inside the radius of convergence) and thus converges, so $\sum \|a_n x^n\|$ converges. By 6.9, $\sum a_n x^n$ converges uniformly on $[-R', R']$, so it is continuous. $\square$

Theorem 6.15 (Simple case). Let $S(x) = \sum a_n x^n$ be a power series with radius of convergence R . If the series converges absolutely at $x = R$, then

$$\lim_{x \rightarrow R^-} S(x) = S(R).$$

Proof. Let $\varepsilon > 0$. We want to find $\delta > 0$ such that $|S(x) - S(R)| < \varepsilon$ for all $x \in (R - \delta, R)$.

Let $S_N(x) = \sum_{n=1}^N a_n x^n$ denote the partial sum. By triangle inequality,

$$|S(x) - S(R)| \leq |S(x) - S_N(x)| + |S_N(x) - S_N(R)| + |S_N(R) - S(R)|.$$

By assumption, since $\sum a_n x^n$ converges absolutely at $x = R$, i.e., $\sum |a_n R^n|$ converges, there exists $N \in \mathbb{N}$ such that

$$\sum_{n=N+1}^{\infty} |a_n R^n| < \frac{\varepsilon}{3}.$$

For any $x \in [0, R]$, we have $|x|^n \leq R^n$, so

$$|S(x) - S_N(x)| = \left| \sum_{n=N+1}^{\infty} a_n x^n \right| \leq \sum_{n=N+1}^{\infty} |a_n| |x|^n \leq \sum_{n=N+1}^{\infty} |a_n| R^n < \frac{\varepsilon}{3}. \quad (1)$$

In particular, at the point $x = R$,

$$|S(R) - S_N(R)| < \frac{\varepsilon}{3}. \quad (2)$$

Since $S_N(x)$ is a polynomial, it is continuous at $x = R$. Thus, there exists $\delta > 0$ such that for all $x \in (R - \delta, R)$,

$$|S_N(x) - S_N(R)| < \frac{\varepsilon}{3}. \quad (3)$$

Combining the estimates (1), (2) and (3) gives $|S(x) - S(R)| < \varepsilon$, which proves the result. $\square$

Thus, if a power series converges (absolutely) at an endpoint, the sum function is continuous there from within the interval of convergence.

Theorem 6.16 (Abel's theorem). Let $S(x) = \sum a_n x^n$ be a power series with radius of convergence R . If the series $\sum a_n R^n$ converges (not necessarily absolutely), then

$$\lim_{x \rightarrow R^-} S(x) = S(R).$$

Proof. Without loss of generality, assume $R = 1$. Let $S(1) = \sum_{n=1}^{\infty} a_n$, $s_n = \sum_{k=1}^n a_k$, and $\tilde{s}_n = s_n - S(1)$. Then $\tilde{s}_n \rightarrow 0$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|\tilde{s}_n| < \frac{\varepsilon}{4}.$$

Using Abel's summation formula for $m > N$ and $x \in (0, 1)$,

$$\begin{aligned} \left| \sum_{n=N+1}^m a_n x^n \right| &= \left| \tilde{s}_m x^m - \tilde{s}_N x^{N+1} + \sum_{n=N+1}^{m-1} \tilde{s}_n (x^n - x^{n+1}) \right| \\ &\leq |\tilde{s}_m| x^m + |\tilde{s}_N| x^{N+1} + \sum_{n=N+1}^{m-1} |\tilde{s}_n| (x^n - x^{n+1}) \\ &< \frac{\varepsilon}{4} x^m + \frac{\varepsilon}{4} x^{N+1} + \frac{\varepsilon}{4} \sum_{n=N+1}^{m-1} (x^n - x^{n+1}) \\ &= \frac{\varepsilon}{4} x^m + \frac{\varepsilon}{4} x^{N+1} + \frac{\varepsilon}{4} (x^{N+1} - x^m) \\ &= \frac{\varepsilon}{2} x^{N+1} < \frac{\varepsilon}{2}. \end{aligned}$$

Taking the limit as $m \rightarrow \infty$, we obtain

$$|S(x) - S_N(x)| \leq \frac{\varepsilon}{2}. \quad (1)$$

By triangle inequality, write

$$|S(x) - S(1)| \leq |S(x) - S_N(x)| + |S_N(x) - S_N(1)| + |S_N(1) - S(1)|.$$

Note that

$$|S_N(1) - S(1)| = |\tilde{s}_N| < \frac{\varepsilon}{4}. \quad (2)$$

Since S_N is a polynomial, it is continuous at $x = 1$. Thus, there exists $\delta > 0$ such that for $x \in (1 - \delta, 1)$,

$$|S_N(x) - S_N(1)| < \frac{\varepsilon}{4}. \quad (3)$$

Combining the bounds (1), (2) and (3) yields $|S(x) - S(1)| < \varepsilon$. This proves that $\lim_{x \rightarrow 1^-} S(x) = S(1)$. $\square$

Example. $S(x) = \frac{x^n}{n}$ where radius of convergence is 1. Thus, $S(x)$ is continuous on $(-1, 1)$. At $x = 1$, $S(1) = \sum \frac{1}{n}$ divergent, so cannot deduce continuity. At $x = -1$, $S(-1) = \sum \frac{(-1)^n}{n}$ converges, so $\lim_{x \rightarrow -1^-} S(x) = S(-1)$.

Generally, pointwise convergence does not imply uniform convergence. However, this is one of the few situations where pointwise convergence implies uniform convergence; the key is the greater control implied by the monotonicity.

Theorem 6.17 (Dini's theorem). *Let $f_n: [a, b] \rightarrow \mathbb{R}$ be a sequence of continuous functions such that $f_1 \leq f_2 \leq \dots$ (i.e., for each $x \in [a, b]$, $\{f_n(x)\}$ is monotonically increasing) and converges pointwise to a continuous function f . Then $f_n \rightarrow f$ uniformly on $[a, b]$.*

We can prove a more general result.

Theorem 6.18 (Dini's theorem, general). *Let X be a compact topological space. If $\{f_n: X \rightarrow \mathbb{R}\}$ is monotonically increasing and continuous, which converges pointwise to a continuous function $f: X \rightarrow \mathbb{R}$, then the convergence is uniform.*

Proof. Let $\varepsilon > 0$. For each $n \in \mathbb{N}$, let $g_n = f - f_n$, and consider the set

$$E_n = \{x \in X : g_n(x) < \varepsilon\}.$$

Each g_n is continuous, and so each E_n is open (because each E_n is the pre-image of the open set $(-\infty, \varepsilon)$ under g_n , a continuous function).

Since $\{f_n\}$ is monotonically increasing, i.e., $f_{n+1}(x) \geq f_n(x)$, this implies $f(x) - f_{n+1}(x) \leq f(x) - f_n(x)$, so $\{g_n\}$ is monotonically decreasing. Thus, the sequence E_n is ascending, namely, $E_n \subseteq E_{n+1}$ for all $n \in \mathbb{N}$.

We claim that $\{E_n\}$ is an open cover of X . Let $x \in X$. Since $f_n \rightarrow f$ pointwise, this means $\lim_{n \rightarrow \infty} f_n(x) = f(x)$. Thus, there exists some $n \in \mathbb{N}$ such that $|f_n(x) - f(x)| = f(x) - f_n(x) = g_n(x) < \varepsilon$, meaning $x \in E_n$.

Since X is compact, there is a finite subcover $\{E_{n_i}\}_{i=1}^k$. Let $N = \max\{n_1, \dots, n_k\}$. Since E_n are ascending, we have $E_N = X$. This means that for all $x \in X$, $f(x) - f_N(x) < \varepsilon$.

To complete the proof, choose this N . Then for all $n \geq N$ and $x \in X$, by monotonicity, we have

$$|f(x) - f_n(x)| \leq |f(x) - f_N(x)| < \varepsilon,$$

as desired. This proves that the convergence is uniform. □

A Tutorials

A.1 Week 3

Problem A.1.1. Let $\mathcal{F}$ be the collection of all open intervals (a, b) such that $a, b \in \mathbb{Q}$. Determine whether $\mathcal{F}$ is countable and prove your claim.

Solution. Since $\mathbb{Q}$ is countable, we can list its elements as

$$\mathbb{Q} = \{r_1, r_2, \dots, r_n, \dots\}.$$

Then every open interval in $\mathcal{F}$ is of the form (r_i, r_j) , where $i, j \in \mathbb{N}$. Define

$$F_n := \{(r_i, r_j) : r_i < r_j, i + j = n\}.$$

Then F_n is finite, and any open interval of the form (r_i, r_j) belongs to F_{i+j} . Hence, $\mathcal{F} = \bigcup_{n=1}^{\infty} F_n$. $\square$

Alternative solution. Define

$$F_n = \left\{ \left(\frac{p}{q}, \frac{p'}{q'} \right) : |p| + |q| + |p'| + |q'| = n \right\}.$$

$\square$

Problem A.1.2. Let $\{I_\alpha\}_{\alpha \in \mathcal{A}}$ be a collection of disjoint finite open intervals, namely, every $I_\alpha = (a, b)$, and for $\alpha_1 \neq \alpha_2$, $I_{\alpha_1} \cap I_{\alpha_2} = \emptyset$. Prove that if $\mathcal{A}$ is infinite, then $\mathcal{A}$ is countable.

Solution. For each $\alpha \in \mathcal{A}$, choose a rational $r_\alpha \in I_\alpha$. This gives a map $f: \mathcal{A} \rightarrow \mathbb{Q}$. Since the I_α 's are disjoint, f is injective. Since $\mathbb{Q}$ is countable,

$$\mathbb{Q} = \{r_1, r_2, \dots, r_n, \dots\}.$$

Define

$$A_n = \{\alpha \in \mathcal{A} : f(\alpha) = r_n\}.$$

Then A_n is finite. For any $\alpha \in \mathcal{A}$, $f(\alpha) = r_n$ for some $n \in \mathbb{N}$, so $\alpha \in A_n$. Hence, $\mathcal{A} = \bigcup_{n=1}^{\infty} A_n$.

Alternatively, define

$$A_n = \left\{ I_\alpha : |I_\alpha| \geq \frac{1}{n} \text{ and } I_\alpha \subseteq [-n, n] \right\}.$$

Then prove that A_n is finite, and $\bigcup_{n=1}^{\infty} A_n = \{I_\alpha\}_{\alpha \in \mathcal{A}}$. $\square$

Problem A.1.3. $\alpha \in \mathbb{R}$ is called *algebraic* if there exists a polynomial

$$P(x) = a_n x^n + \dots + a_1 x + a_0$$

with integer coefficients such that $P(\alpha) = 0$. Let $\bar{\mathbb{Q}}$ denote the set of algebraic numbers. Prove that $\bar{\mathbb{Q}}$ is countable.

Solution. Define F_N as the set of roots α of $P(x) \in \mathbb{Z}[x]$ such that $P(x) = a_n x^n + \dots + a_1 x + a_0$, with $|a_n| + |a_{n-1}| + \dots + |a_1| + |a_0| + n \leq N$. Then F_N is finite, and $\bigcup_{n=1}^{\infty} F_n = \bar{\mathbb{Q}}$. $\square$

Problem A.1.4. Let $f: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ be a bounded function. For every $m \in \mathbb{N}$, define $X(m) := \sup\{f(m, n) : n \in \mathbb{N}\}$. For every $n \in \mathbb{N}$, define $Y(n) := \inf\{f(m, n) : m \in \mathbb{N}\}$. Prove that $\inf\{X(m) : m \in \mathbb{N}\} \geq \sup\{Y(n) : n \in \mathbb{N}\}$.

Solution. For all $m, n \in \mathbb{N}$,

$$X(m) \geq f(m, n) \geq Y(n)$$

where the first inequality holds because $X(m)$ is the supremum of $f(m, n)$ over $n \in \mathbb{N}$, and the second inequality holds because $Y(n)$ is the infimum of $f(m, n)$ over $m \in \mathbb{N}$.

Thus, $X(m) \geq Y(n)$ for any $m, n \in \mathbb{N}$. This implies $X(m)$ is an upper bound of $\{Y(n) : n \in \mathbb{N}\}$, so

$$X(m) \geq \sup\{Y(n) : n \in \mathbb{N}\}$$

for all $m \in \mathbb{N}$. This, in turn, implies that $\sup\{Y(n) : n \in \mathbb{N}\}$ is a lower bound of $\{X(m) : m \in \mathbb{N}\}$, so

$$\sup\{Y(n) : n \in \mathbb{N}\} \leq \inf\{X(m) : m \in \mathbb{N}\}.$$

□

Problem A.1.5. Let $\{I_n = [a_n, b_n] : n \in \mathbb{N}\}$ be a sequence of nested intervals such that $|I_n| < 1/n$. Prove that the intersection $\bigcap_{n=1}^{\infty} I_n$ is a singleton.

Solution. By the nested interval property, there exists $x \in \mathbb{R}$ such that $x \in \bigcap_{n=1}^{\infty} I_n$.

Suppose, towards a contradiction, that there exists $y \neq x$ such that $y \in \bigcap_{n=1}^{\infty} I_n$. Then $\varepsilon = |x - y| > 0$. For all $n \in \mathbb{N}$,

$$\varepsilon = |x - y| < |I_n| = \frac{1}{n}.$$

But by the Archimedean property, there exists $m \in \mathbb{N}$ such that

$$\frac{1}{m} < \varepsilon.$$

This yields the desired contradiction. □

Problem A.1.6. Prove that for any $n \in \mathbb{N}$,

$$\sum_{k=1}^n \frac{1}{k^2} < 2.$$

Solution. Using telescoping sums,

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k^2} &= 1 + \sum_{k=2}^n \frac{1}{k^2} \\ &< 1 + \sum_{k=2}^n \frac{1}{k(k-1)} \\ &= 1 + \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) \\ &= 1 + \left(\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{n-1} - \frac{1}{n} \right) = 2 - \frac{1}{n} < 2. \end{aligned}$$

□

Problem A.1.7. Prove that for any $n \in \mathbb{N}$, $n > 2$,

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} > \sqrt{n+1}.$$

Solution. Again, using telescoping sums,

$$\begin{aligned} \sum_{k=1}^n \frac{1}{\sqrt{k}} &= 2 \sum_{k=1}^n \frac{1}{2\sqrt{k}} \\ &> 2 \sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k+1}} \\ &= 2 \sum_{k=1}^n (\sqrt{k+1} - \sqrt{k}) \\ &= 2(\sqrt{n+1} - 1) = \sqrt{n+1} + (\sqrt{n+1} - 2) > \sqrt{n+1} \end{aligned}$$

for $n \geq 3$. □

Problem A.1.8. Prove that for $n = 2^m$, $m \in \mathbb{N}$,

$$\sum_{k=1}^n \frac{1}{k} > \frac{m}{2}.$$

Solution. We proceed by dyadic partition, where we group the terms of the summation as follows:

$$\sum_{k=1}^n \frac{1}{k} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^{k-1}+1} + \cdots + \frac{1}{2^k}\right) + \cdots + \frac{1}{2^m}.$$

Note that

$$\frac{1}{2^{k-1}+1} + \cdots + \frac{1}{2^k} > 2^{k-1} \cdot \frac{1}{2^k} = \frac{1}{2}.$$

Hence,

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k} &> \sum_{M=1}^m \sum_{k=2^{M-1}}^{2^M-1} \frac{1}{k} \\ &> \sum_{M=1}^m \sum_{k=2^{M-1}}^{2^M-1} \frac{1}{2^M} = \sum_{M=1}^m \frac{1}{2} = \frac{m}{2}. \end{aligned}$$

□

A.2 Week 4

Problem A.2.1. Prove that

$$\lim_{n \rightarrow \infty} \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots 2n} = 0.$$

Solution. Let

$$a_n = \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2n-1}{2n}.$$

Since $\frac{1}{2} < \frac{2}{3}$, $\frac{3}{4} < \frac{4}{5}$, ..., $\frac{2n-3}{2n-2} < \frac{2n-2}{2n-1}$, $\frac{2n-1}{2n} < \frac{2n}{2n+1}$,

$$a_n < \frac{2}{3} \cdot \frac{4}{5} \cdot \frac{6}{7} \cdots \frac{2n-2}{2n-1} \cdot \frac{2n}{2n+1}.$$

Hence,

$$\begin{aligned} a_n \cdot a_n &< \left(\frac{1}{2} \cdot \frac{3}{4} \cdots \frac{2n-1}{2n}\right) \left(\frac{2}{3} \cdot \frac{4}{5} \cdots \frac{2n}{2n+1}\right) \\ &= \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} \cdots \frac{2n-1}{2n} \cdot \frac{2n}{2n+1} \\ &= \frac{1}{2n+1}. \end{aligned}$$

Thus, $a_n < \sqrt{\frac{1}{2n+1}}$. By the squeeze theorem, we are done. □

Problem A.2.2. Find

$$\lim_{n \rightarrow \infty} \sqrt[n]{2^n + 3^n + 5^n}.$$

Solution. For all $n \in \mathbb{N}$, we bound

$$\sqrt[n]{5^n} \leq \sqrt[n]{2^n + 3^n + 5^n} \leq \sqrt[n]{3(5^n)}.$$

Note that $\lim_{n \rightarrow \infty} \sqrt[n]{5^n} = 5$ and $\lim_{n \rightarrow \infty} \sqrt[n]{3(5^n)} = 5 \lim_{n \rightarrow \infty} \sqrt[n]{3} = 5$. By squeeze theorem, $\lim_{n \rightarrow \infty} \sqrt[n]{2^n + 3^n + 5^n} = 5$. □

Problem A.2.3 (Cauchy's first limit theorem). Suppose $\lim_{n \rightarrow \infty} a_n = A$. Prove that

$$\lim_{n \rightarrow \infty} \frac{a_1 + \cdots + a_n}{n} = A.$$

Solution. Let $\varepsilon > 0$. There exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|a_n - A| < \frac{\varepsilon}{2}$.

We want to find $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\left| \frac{\sum_{k=1}^n a_k}{n} - A \right| < \varepsilon.$$

Observe that

$$\left| \frac{\sum_{k=1}^n a_k}{n} - A \right| = \left| \frac{\sum_{k=1}^n (a_k - A)}{n} \right| \leq \frac{\sum_{k=1}^{N_1-1} |a_k - A|}{n} + \frac{\sum_{k=N_1}^n |a_k - A|}{n}.$$

Let $C = \sum_{k=1}^{N_1-1} |a_k - A|$. Note that C is a constant independent of n . Substituting this into our inequality gives

$$\left| \frac{\sum_{k=1}^n a_k}{n} - A \right| \leq \frac{C}{n} + \frac{(n - N_1 + 1)\frac{\varepsilon}{2}}{n} < \frac{C}{n} + \frac{\varepsilon}{2}.$$

Thus, for the first term $\frac{C}{n} < \frac{\varepsilon}{2}$, we need $n > \frac{2C}{\varepsilon}$. Hence, we choose $N = \max \left\{ N_1, \left\lceil \frac{2C}{\varepsilon} \right\rceil + 1 \right\}$. Then for all $n \geq N$,

$$\left| \frac{\sum_{k=1}^n a_k}{n} - A \right| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Problem A.2.4. Let $a_1 > 0$ and $a_{n+1} = a_n + \frac{1}{a_n}$. Determine whether $\{a_n\}$ is convergent. If it is convergent, find the limit.

Solution. The sequence $\{a_n\}$ is divergent. Suppose, towards a contradiction, that $\{a_n\}$ is convergent; let $L = \lim_{n \rightarrow \infty} a_n$. Then $L > 0$ since $a_1 > 0$ and $\{a_n\}$ is increasing. Now take limits on both sides to find L :

$$\begin{aligned} \lim_{n \rightarrow \infty} a_{n+1} &= \lim_{n \rightarrow \infty} \left(a_n + \frac{1}{a_n} \right) \\ L &= L + \frac{1}{L} \\ \frac{1}{L} &= 0. \end{aligned}$$

But L has no real solution, a contradiction. □

Remark. $\lim_{n \rightarrow \infty} a_n = +\infty$.

Problem A.2.5. Let $a_1 = 3$ and $a_{n+1} = 3 - \frac{4}{a_n^2}$. Determine whether $\{a_n\}$ is convergent. If it is convergent, find the limit.

Solution. We first show that $\{a_n\}$ is bounded. We claim that $2 \leq a_n \leq 3$ for all $n \in \mathbb{N}$ by induction. The base case $a_1 = 3$ is clear. Assume $2 \leq a_k \leq 3$. Then one can show that $2 \leq a_{k+1} \leq 3$.

Next, we show that $\{a_n\}$ is decreasing, i.e., $a_n > a_{n+1}$ for all n , by induction.

Since $\{a_n\}$ is monotonic and bounded, by the monotone convergence theorem, $\{a_n\}$ converges; let $L = \lim_{n \rightarrow \infty} a_n$.

Taking the limit of the recurrence relation,

$$L = 3 - \frac{4}{L^2} \implies L^3 - 3L^2 + 4 = 0$$

Factoring gives $(L - 2)^2(L + 1) = 0$. Since $a_n \geq 2$, the limit must be $L = 2$. $\square$

Problem A.2.6. Let $a_1 = 3$ and $a_{n+1} = \frac{3 + a_n}{1 + a_n}$. Prove that $\{a_n\}$ is convergent and find the limit.

Solution. Since $a_1 = 3 > 0$, it follows by induction that all $a_n > 0$. From the recurrence $a_{n+1} = 1 + \frac{2}{a_{n+1}}$, it is immediate that:

$$a_n > 1 \quad \text{for all } n > 1$$

We evaluate the distance between successive terms to show the sequence is Cauchy. Consider the difference:

$$\begin{aligned} |a_{n+1} - a_n| &= \left| \left(1 + \frac{2}{a_{n+1}}\right) - \left(1 + \frac{2}{a_n}\right) \right| \\ &= \left| \frac{2}{a_{n+1}} - \frac{2}{a_n} \right| \\ &= \frac{2|a_n - a_{n+1}|}{(a_{n+1})(a_n)} \end{aligned}$$

Using our bound $a_n > 1$, we know $(a_n + 1) > 2$ and $(a_{n-1} + 1) > 2$. Thus:

$$|a_{n+1} - a_n| < \frac{2}{2 \cdot 2} |a_n - a_{n-1}| = \frac{1}{2} |a_n - a_{n-1}|$$

Since $|a_{n+1} - a_n| < \frac{1}{2} |a_n - a_{n-1}|$, the sequence $\{a_n\}$ is contractive, so it is a Cauchy sequence, and thus converges to a limit L . Taking limits as $n \rightarrow \infty$ on both sides and solving $L = \frac{3+L}{1+L}$ yields $L = \sqrt{3}$, since $a_n > 1$. $\square$

Problem A.2.7. Alice and Bob play the following game: Alice first chooses a closed interval A_1 and Bob chooses a closed interval $B_1 \subset A_1$ such that $|B_1| = |A_1|/3$. In step k , Alice chooses a closed interval $A_k \subseteq B_{k-1}$, and Bob chooses $B_k \subset A_k$ with $|B_k| = |A_k|/3$.

The choices of Alice give a sequence of nested intervals:

$$A_1 \supset A_2 \supset \cdots \supset A_n \supset \cdots$$

Note that $\bigcap_{n=1}^{\infty} A_n = \{\xi\}$, since $|A_n| \rightarrow 0$ as $n \rightarrow \infty$. If $\xi \in \mathbb{Q}$, then Alice wins; otherwise, Bob wins. Determine who has a winning strategy and prove it.

Solution. Bob has a winning strategy.

Since the set of rational numbers $\mathbb{Q}$ is countable, we can enumerate its elements as $\mathbb{Q} = \{r_1, r_2, r_3, \dots\}$.

The lengths of the nested intervals decrease geometrically because $|A_{k+1}| \leq |B_k| = \frac{|A_k|}{3}$, so $|A_k| \rightarrow 0$. By Problem A.1.5, the intersection of this nested sequence of closed intervals is a singleton:

$$\bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} B_n = \{\xi\}.$$

Bob's strategy is to ensure that $r_k \notin B_k$ at each step k , i.e., avoid every rational number in each step. When Alice chooses A_k , Bob divides this interval into three disjoint closed sub-intervals of equal length, L_k , M_k , and R_k :

$$A_k = L_k \cup M_k \cup R_k, \quad \text{where } |L_k| = |M_k| = |R_k| = \frac{1}{3} |A_k|$$

Bob determines his choice for B_k based on the position of the rational r_k . If $r_k \in L_k$, Bob chooses $B_k = R_k$. If $r_k \notin L_k$, Bob chooses $B_k = L_k$. Since L_k and R_k are separated by the interval M_k , r_k cannot be in the interior of both. Even if r_k lies on a boundary, Bob can always select the sub-interval that excludes it from all future nested choices.

Because $\xi \in B_n$ for all $n \in \mathbb{N}$, it follows that for every specific k , $\xi \in B_k$. However, Bob's strategy guaranteed that $r_k \notin B_k$ at step k , which implies $\xi \neq r_k$ for all $k \in \mathbb{N}$. Since the sequence $\{r_k\}$ includes every rational number, ξ

must be irrational. Consequently, Bob wins the game. $\square$

Remark. $\frac{1}{3}$ cannot be changed to $\frac{2}{3}$; otherwise, Alice can win.

A.3 Week 5

Problem A.3.1. Find

$$\sum_{n=1}^{\infty} \frac{n}{3^n}.$$

Solution. We first express the partial sum:

$$\begin{aligned} \sum_{k=1}^n \frac{k}{3^k} &= \frac{1}{3} + \frac{2}{3^2} + \frac{3}{3^3} + \cdots + \frac{n}{3^n} \\ &= \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \cdots + \frac{1}{3^n} \right) + \left(\frac{1}{3^2} + \frac{1}{3^3} + \cdots + \frac{1}{3^n} \right) + \left(\frac{1}{3^3} + \cdots + \frac{1}{3^n} \right) + \cdots + \frac{1}{3^n} \\ &= \frac{\frac{1}{3} - \frac{1}{3^{n+1}}}{1 - \frac{1}{3}} + \frac{\frac{1}{3^2} - \frac{1}{3^{n+1}}}{1 - \frac{1}{3}} + \cdots + \frac{\frac{1}{3^n} - \frac{1}{3^{n+1}}}{1 - \frac{1}{3}} \\ &= \frac{3}{2} \left(\frac{1}{3} + \frac{1}{3^2} + \cdots + \frac{1}{3^n} \right) - \frac{3}{2} \cdot n \cdot \frac{1}{3^{n+1}} \\ &= \frac{3}{2} \cdot \frac{\frac{1}{3} - \frac{1}{3^{n+1}}}{\frac{2}{3}} - \frac{3}{2} n \cdot \frac{1}{3^{n+1}}. \end{aligned}$$

Taking the limit as $n \rightarrow \infty$, the series equals $\frac{3}{4}$. $\square$

Problem A.3.2 (Stolz theorem). Let $\{a_n\}$ be a sequence and $\{b_n\}$ be a strictly increasing and divergent sequence. Suppose

$$\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = A.$$

Prove that $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = A$.

Solution. Let $\varepsilon > 0$ be given. There exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\left| \frac{a_{n+1} - a_n}{b_{n+1} - b_n} - A \right| < \frac{\varepsilon}{2}.$$

Let $C_n = (a_{n+1} - a_n) - A(b_{n+1} - b_n)$. Then $|C_n| < \frac{\varepsilon}{2}(b_{n+1} - b_n)$. Note that $a_n = a_N + \sum_{k=N}^{n-1} (a_{k+1} - a_k)$, so

$$\begin{aligned} \frac{a_n}{b_n} &= \frac{a_N}{b_n} + \frac{\sum_{k=N}^{n-1} (a_{k+1} - a_k)}{b_n} \\ &= \frac{a_N}{b_n} + \frac{\sum_{k=N}^{n-1} [A(b_{k+1} - b_k) + c_k]}{b_n} \\ &= \frac{a_N}{b_n} + \frac{A \sum_{k=N}^{n-1} (b_{k+1} - b_k) + A \sum_{k=N}^{n-1} c_k}{b_n} \\ &= \frac{a_N}{b_n} + \frac{A(b_n - b_N)}{b_n} + \frac{\sum_{k=N}^{n-1} c_k}{b_n} \\ &= A + \frac{a_N}{b_n} - \frac{Ab_N}{b_n} + \frac{\sum_{k=N}^{n-1} c_k}{b_n}. \end{aligned}$$

Finally,

$$\begin{aligned}
 \left| \frac{a_n}{b_n} - A \right| &= \left| \frac{a_N}{b_n} - \frac{Ab_N}{b_n} + \sum_{k=N}^{n-1} \frac{c_k}{b_n} \right| \\
 &\leq \frac{|a_N|}{b_n} + |A| \frac{|b_N|}{b_n} + \sum_{k=N}^{n-1} \frac{|c_k|}{b_n} \\
 &\leq \frac{|a_N|}{b_n} + |A| \frac{|b_N|}{b_n} + \frac{\varepsilon}{2} \sum_{k=N}^{n-1} \frac{b_{k+1} - b_k}{b_n} \\
 &= \frac{|a_N|}{b_n} + |A| \frac{|b_N|}{b_n} + \frac{\varepsilon}{2} \frac{b_n - b_N}{b_n} \\
 &\leq \frac{|a_N|}{b_n} + |A| \frac{|b_N|}{b_n} + \frac{\varepsilon}{2}.
 \end{aligned}$$

Since $b_n \rightarrow +\infty$, there exists $N_2 \in \mathbb{N}$ such that for $n \geq N_2$,

$$\frac{|a_N| + |A||b_N|}{b_n} < \frac{\varepsilon}{2}.$$

□

Stolz theorem is useful to evaluate the limit of a quotient, where either the numerator or denominator is a summation. For instance, to evaluate $\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n \frac{1}{k}}{\log n}$, let $a_n = \sum_{k=1}^n \frac{1}{k}$ and $b_n = \log n$. Then

$$\begin{aligned}
 a_{n+1} - a_n &= \frac{1}{n+1} \\
 b_{n+1} - b_n &= \log(n+1) - \log n = \log \frac{n+1}{n} = \log \left(1 + \frac{1}{n} \right)
 \end{aligned}$$

Thus,

$$\frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \frac{1}{(n+1) \log \left(1 + \frac{1}{n} \right)} = \frac{1}{\log \left(1 + \frac{1}{n} \right)^{n+1}} \rightarrow \frac{1}{\log e} = 1,$$

where we used the continuity of the logarithmic function to interchange log and lim.

Problem A.3.3. Prove that

$$\sum_{n=1}^{\infty} \sin n$$

is divergent.

Solution. We apply the divergence test. For the series to converge, it is necessary that $\lim_{n \rightarrow \infty} \sin n = 0$. We will show this limit does not exist by identifying a subsequence bounded away from zero.

Consider the sequence of intervals:

$$I_k = \left(2\pi k + \frac{\pi}{6}, 2\pi k + \frac{5\pi}{6} \right), \quad k \in \mathbb{Z}.$$

For any $x \in I_k$, we have $\sin x > \sin \frac{\pi}{6} = \frac{1}{2}$. The length of each interval is

$$\Delta I = \frac{5\pi}{6} - \frac{\pi}{6} = \frac{2\pi}{3} \approx 2.09.$$

Since the length of each interval I_k is greater than 1, each interval must contain at least one integer $n_k \in \mathbb{Z}$.

Consequently, there exists a subsequence $\{n_k\}_{k=1}^{\infty}$ such that $\sin(n_k) > 1/2$ for all k . This implies that $\sin n$ does not approach 0 as $n \rightarrow \infty$. By the divergence test, the series $\sum \sin n$ diverges. □

Problem A.3.4. Find

$$\lim_{n \rightarrow \infty} n(e^{1/n} - 1).$$

Solution. Recall from lecture that

$$a_n = \left(1 + \frac{1}{n}\right)^n \nearrow e, \quad b_n = \left(1 + \frac{1}{n}\right)^{n+1} \searrow e.$$

Then $a_n \leq e \leq b_{n-1}$. Take n -th root on both sides:

$$\begin{aligned} 1 + \frac{1}{n} &< e^{1/n} < 1 + \frac{1}{n-1} \\ \frac{1}{n} &< e^{1/n} - 1 < \frac{1}{n-1} \\ 1 &< n(e^{1/n} - 1) < \frac{n}{n-1} \end{aligned}$$

By squeeze theorem, $\lim_{n \rightarrow \infty} n(e^{1/n} - 1) = 1$. □

Problem A.3.5. Let $a_n = \sum_{k=1}^n \frac{1}{k} - \log n$. Prove that $\{a_n\}$ is convergent.

Solution. We aim to show that the sequence $\{a_n\}$ is monotonic and bounded.

To determine if the sequence is decreasing, we examine the difference $a_{n+1} - a_n$:

$$\begin{aligned} a_{n+1} - a_n &= \left(\sum_{k=1}^{n+1} \frac{1}{k} - \log(n+1) \right) - \left(\sum_{k=1}^n \frac{1}{k} - \log n \right) \\ &= \frac{1}{n+1} - (\log(n+1) - \log n) \end{aligned}$$

Recall the inequality $e < (1 + \frac{1}{n})^{n+1}$, which implies $1 < (n+1) \log\left(\frac{n+1}{n}\right)$. Rearranging this gives

$$\frac{1}{n+1} < \log(n+1) - \log n$$

Therefore, $a_{n+1} - a_n < 0$, proving that $\{a_n\}$ is decreasing.

We now show the sequence is bounded below by zero. Using the inequality $\frac{1}{k} > \log(k+1) - \log k$ (derived from $e > (1 + \frac{1}{k})^k$), we have:

$$\begin{aligned} a_n &= \sum_{k=1}^n \frac{1}{k} - \log n > \sum_{k=1}^n (\log(k+1) - \log k) - \log n \\ &= (\log(n+1) - \log 1) - \log n \\ &= \log(n+1) - \log n > 0. \end{aligned}$$

Since $a_n > 0$ for all n , the sequence is bounded below.

By the monotone convergence theorem, since $\{a_n\}$ is decreasing and bounded below, the sequence $\{a_n\}$ is convergent. □

Remark. The limit $\gamma = \lim_{n \rightarrow \infty} a_n$ is known as *Euler's constant*.

Problem A.3.6. Let $\{a_n\}$ be a positive sequence such that

$$\sum_{n=1}^{\infty} a_n \text{ is divergent.}$$

Let $s_n = \sum_{k=1}^n a_k$. Prove that

$$\sum_{n=1}^{\infty} \frac{a_n}{s_n} \text{ is divergent.}$$

Solution. To prove that $\sum \frac{a_n}{s_n}$ is divergent, we will prove that it is not Cauchy, i.e., for any $N \in \mathbb{N}$, there exists $m > n \geq N$ such that $\sum_{k=n+1}^m \frac{a_k}{s_k} > \frac{1}{2}$. Note that

$$\sum_{k=n+1}^m \frac{a_k}{s_k} \geq \sum_{k=n+1}^m \frac{a_k}{s_m} = \frac{1}{s_m} \sum_{k=n+1}^m a_k = \frac{s_m - s_n}{s_m} = 1 - \frac{s_n}{s_m}.$$

Since $\sum a_n$ is divergent and $a_n > 0$, we know $s_m \rightarrow \infty$ as $m \rightarrow \infty$. For a fixed $n = N$, we can choose $m = N_2$ large enough such that $\frac{s_N}{s_{N_2}} < \frac{1}{2}$. Then

$$\sum_{k=N+1}^{N_2} \frac{a_k}{s_k} \geq 1 - \frac{s_N}{s_{N_2}} > 1 - \frac{1}{2} = \frac{1}{2}.$$

□

Problem A.3.7. Let $\{a_n\}$ be a positive sequence such that

$$\sum_{n=1}^{\infty} a_n \text{ is convergent.}$$

Prove that

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n} \text{ is convergent.}$$

Solution. Let $S_N = \sum_{n=1}^N \frac{\sqrt{a_n}}{n}$ be the N -th partial sum. By the Cauchy-Schwarz inequality,

$$S_N^2 = \left(\sum_{n=1}^N \sqrt{a_n} \cdot \frac{1}{n} \right)^2 \leq \left(\sum_{n=1}^N a_n \right) \left(\sum_{n=1}^N \frac{1}{n^2} \right).$$

Since $\sum_{n=1}^{\infty} a_n$ converges to L and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges to $\frac{\pi^2}{6}$, we have:

$$S_N^2 \leq L \cdot \frac{\pi^2}{6} < \infty$$

Thus, the sequence of partial sums $\{S_N\}$ is monotonic (since $a_n > 0$) and bounded above. By the monotone convergence theorem, $\sum \frac{\sqrt{a_n}}{n}$ converges. □

Remark. Convergence does not imply ratio test. Counterexample: $\sum \frac{1}{n^2}$.

A.4 Week 6

Problem A.4.1. Let $\{a_n\}$ be a positive sequence satisfying that for any positive decreasing sequence $\{b_n\}$ such that $\lim_{n \rightarrow \infty} b_n = 0$,

$$\sum_{n=1}^{\infty} a_n b_n \text{ is convergent.}$$

Prove that $\sum_{n=1}^{\infty} a_n$ is convergent.

Solution. For a contradiction, suppose that $\sum a_n$ is divergent. We want to find a positive decreasing sequence $\{b_n\}$ such that $b_n \rightarrow 0$ but $\sum a_n b_n$ is divergent.

Choose $b_n = \frac{1}{A_n}$, where $A_n = \sum_{k=1}^n a_k$ is the partial sum.

- It is easily seen that $A_n > 0$, so $b_n > 0$.

- Since A_n is increasing and divergent, we have $\lim_{n \rightarrow \infty} A_n = \infty$, so $\lim_{n \rightarrow \infty} b_n = 0$.
- Also, $\{b_n\}$ is increasing:

$$b_n - b_{n+1} = \frac{1}{A_n} - \frac{1}{A_{n+1}} = \frac{a_{n+1}}{A_n A_{n+1}} > 0.$$

- By Problem A.3.6, $\sum a_n b_n = \sum \frac{a_n}{A_n}$ is divergent.

□

Problem A.4.2. Find

$$\sum_{n=1}^{\infty} (\sqrt{n} + \sqrt{n+2} - 2\sqrt{n+1}).$$

Solution. Using telescoping sum, the partial sum evaluates to

$$\begin{aligned} \sum_{n=1}^N (\sqrt{n} + \sqrt{n+2} - 2\sqrt{n+1}) &= \sum_{n=1}^N (\sqrt{n+2} - \sqrt{n+1}) - \sum_{n=1}^N (\sqrt{n+1} - \sqrt{n}) \\ &= (\sqrt{N+2} - \sqrt{2}) - (\sqrt{N+1} - \sqrt{1}) \\ &= 1 - \sqrt{2} - \sqrt{N+1} + \sqrt{N+2}. \end{aligned}$$

Note that

$$\sqrt{N+2} - \sqrt{N+1} = \frac{1}{\sqrt{N+2} + \sqrt{N+1}} \rightarrow 0.$$

Hence, $\sum_{n=1}^{\infty} (\sqrt{n} + \sqrt{n+2} - 2\sqrt{n+1}) = 1 - \sqrt{2}$.

□

Problem A.4.3 (Fekete's Subadditive Lemma). Let $\{a_n\}$ be a positive sequence satisfying that

$$a_{m+n} \leq a_m + a_n \quad \text{for all } m, n \in \mathbb{N}.$$

Prove that $\{\frac{a_n}{n}\}$ is convergent.

Solution. Fix an arbitrary $m \in \mathbb{N}$. For any $n \in \mathbb{N}$ such that $n > m$, by the division algorithm,

$$n = qm + r, \quad \text{where } 0 \leq r < m.$$

By the sub-additive property, we can decompose a_n as follows:

$$\begin{aligned} a_n &= a_{qm+r} \leq a_{qm} + a_r \\ &\leq \underbrace{a_m + a_m + \cdots + a_m}_{q \text{ times}} + a_r \\ &= qa_m + a_r \end{aligned}$$

Dividing both sides by n ,

$$\frac{a_n}{n} \leq \frac{qa_m}{n} + \frac{a_r}{n}.$$

Since $n = qm + r$, it follows that $qm \leq n$, which implies $\frac{q}{n} \leq \frac{1}{m}$. Substituting this into the inequality gives

$$\frac{a_n}{n} \leq \frac{a_m}{m} + \frac{a_r}{n}.$$

Let $C = \max\{a_1, a_2, \dots, a_{m-1}\}$ (with $a_0 = 0$). Since $0 \leq r < m$, we have $a_r \leq C$. Thus,

$$\frac{a_n}{n} \leq \frac{a_m}{m} + \frac{C}{n}$$

Taking the limit superior as $n \rightarrow \infty$, and noting that $\lim_{n \rightarrow \infty} \frac{c}{n} = 0$,

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} \leq \frac{a_m}{m}.$$

Since this inequality holds for every $m \in \mathbb{N}$, we can take the limit inferior as $m \rightarrow \infty$ on the RHS:

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} \leq \liminf_{m \rightarrow \infty} \frac{a_m}{m}$$

By the definition of limits, we know that $\liminf \leq \limsup$ for any sequence. Hence, we must have

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} = \liminf_{n \rightarrow \infty} \frac{a_n}{n}.$$

This equality proves that the limit exists, and thus the sequence $\{\frac{a_n}{n}\}$ is convergent. $\square$

Problem A.4.4. Determine whether the following series is convergent and prove it:

$$\sum_{n=2}^{\infty} \frac{1}{(\log n)^{\log n}}.$$

Solution. We write

$$\frac{1}{(\log n)^{\log n}} = \left(\frac{1}{\log n}\right)^{\log n} = \left(e^{-\log \log n}\right)^{\log n} = \left(e^{\log n}\right)^{-\log \log n} = \frac{1}{n^{\log \log n}}.$$

Since $\log \log n \rightarrow +\infty$, there exists $N \in \mathbb{N}$ such that $\log \log n \geq 2$ for all $n \geq N$. Thus, for all $n \geq N$,

$$\frac{1}{(\log n)^{\log n}} = \frac{1}{n^{\log \log n}} \leq \frac{1}{n^2}.$$

Since $\sum \frac{1}{n^2}$ converges, by comparison test, $\sum_{n=2}^{\infty} \frac{1}{(\log n)^{\log n}}$ is convergent. $\square$

Alternatively, use Cauchy condensation test.

Problem A.4.5. Determine whether the following series is convergent and prove it:

$$\sum_{n=100}^{\infty} \frac{1}{n \log n (\log \log n)}.$$

Solution. Let

$$a_n = \frac{1}{n \log n \log \log n}.$$

To check convergence of $\sum a_n$, by Cauchy Condensation Test, it suffices to check convergence of $\sum b_k$, where

$$b_k = 2^k a_{2^k} = \frac{2^k}{2^k k \log 2 \log(k \log 2)}.$$

To check convergence of $\sum b_k$, by Cauchy Condensation Test, it suffices to check convergence of $\sum c_r$, where

$$c_r = 2^r b_{2^r} = \frac{2^r}{2^r \log 2 [\log 2^n + \log 2]} = \frac{1}{\log 2 (r \log 2 + \log 2)}.$$

Since $\sum c_r$ diverges, we conclude that the given series $\sum_{n=100}^{\infty} \frac{1}{n \log n (\log \log n)}$ diverges. $\square$

Problem A.4.6. Let $\{a_n\}$ be a positive sequence such that $\sum_{n=1}^{\infty} a_n$ is convergent. Let $r_n = \sum_{k=n}^{\infty} a_k$. Prove that

$$\sum_{n=1}^{\infty} \frac{a_n}{\sqrt{r_n}} \quad \text{is convergent.}$$

Solution. Since $a_n > 0$, it follows that $r_n > 0$ for all $n \in \mathbb{N}$. Furthermore,

$$r_n - r_{n+1} = \sum_{k=n}^{\infty} a_k - \sum_{k=n+1}^{\infty} a_k = a_n > 0$$

implies that $\{r_n\}$ is a decreasing sequence. Since $\{r_n\}$ is decreasing, we have $\sqrt{r_n} > \sqrt{r_{n+1}}$, so

$$2\sqrt{r_n} > \sqrt{r_n} + \sqrt{r_{n+1}}.$$

Taking the reciprocal and multiplying by $2a_n$, we get

$$\frac{a_n}{\sqrt{r_n}} < \frac{2a_n}{\sqrt{r_n} + \sqrt{r_{n+1}}} = \frac{2(r_n - r_{n+1})}{\sqrt{r_n} + \sqrt{r_{n+1}}} = 2(\sqrt{r_n} - \sqrt{r_{n+1}}).$$

Now, consider the partial sum

$$\sum_{k=1}^n \frac{a_k}{\sqrt{r_k}} < \sum_{k=1}^n 2(\sqrt{r_k} - \sqrt{r_{k+1}}) = 2(\sqrt{r_1} - \sqrt{r_{n+1}}) < 2\sqrt{r_1}.$$

Since the sequence of partial sums is monotonically increasing (each term is positive) and bounded above, by the Monotone Convergence Theorem, the series $\sum \frac{a_n}{\sqrt{r_n}}$ is convergent. $\square$

Problem A.4.7. Let $\{a_n\}$ be a positive sequence such that $\sum_{n=1}^{\infty} a_n$ is convergent. Let $r_n = \sum_{k=n}^{\infty} a_k$. Determine whether

$$\sum_{n=1}^{\infty} \frac{a_n}{r_n}$$

is convergent and prove it.

Solution. We will prove that $\sum \frac{a_n}{r_n}$ is **divergent** by showing it is not Cauchy.

Claim: For any $N \in \mathbb{N}$, there exist $m, n > N$ such that $\sum_{i=n}^m \frac{a_i}{r_i} > \frac{1}{2}$.

Since $a_i > 0$, the sequence of remainders $\{r_i\}$ is strictly decreasing. For a fixed n ,

$$\sum_{i=n}^m \frac{a_i}{r_i} > \frac{1}{r_n} \sum_{i=n}^m a_i.$$

The sum of the terms from n to m can be expressed in terms of remainders:

$$\sum_{i=n}^m a_i = r_n - r_{m+1}.$$

Taking the limit as $m \rightarrow \infty$,

$$\lim_{m \rightarrow \infty} \sum_{i=n}^m a_i = \sum_{i=n}^{\infty} a_i = r_n.$$

Since the sequence of partial sums $b_m = \sum_{i=n}^m a_i$ is strictly increasing toward r_n , there must exist some m_0 large enough such that

$$\sum_{i=n}^{m_0} a_i > \frac{1}{2} r_n.$$

Substituting this back into our inequality,

$$\sum_{i=n}^{m_0} \frac{a_i}{r_i} > \frac{\frac{1}{2} r_n}{r_n} = \frac{1}{2}.$$

$\square$

A.5 Week 8 (Mid-term Review)

Problem A.5.1. Find the following limits:

$$(1) \lim_{n \rightarrow \infty} (\sqrt{n^2 + 7n + 3} - n);$$

$$(2) \lim_{n \rightarrow \infty} \left(1 + \frac{2}{n^2}\right)^{2n}.$$

Solution.

(1) Take the conjugate:

$$\sqrt{n^2 + 7n + 3} - n = \frac{(\sqrt{n^2 + 7n + 3}) - n^2}{\sqrt{n^2 + 7n + 3} + n} = \frac{7n + 3}{\sqrt{n^2 + 7n + 3} + n} = \frac{7 + \frac{3}{n}}{\sqrt{1 + \frac{7}{n} + \frac{3}{n^2}} + 1}.$$

(2) Note that

$$\left(1 + \frac{2}{n^2}\right)^{2n} = \left(\left(1 + \frac{1}{n^2/2}\right)^{n^2/2}\right)^{4/n}.$$

Bound $m \leq n^2/2 \leq m + 1$, then

$$\left(\left(1 + \frac{1}{n^2/2}\right)^{n^2/2}\right)^{4/n} \leq \left(1 + \frac{1}{m}\right)^{m+1} < 4.$$

Squeeze

$$1 < \left(\left(1 + \frac{1}{n^2/2}\right)^{n^2/2}\right)^{4/n} < 4^{4/n}$$

where $4^{4/n} \rightarrow 1$.

□

Problem A.5.2. Find

$$\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n \sqrt{k}}{\sqrt{n^3}}.$$

Solution. Since a summation is involved, the idea is to use Stolz theorem. Let $a_n = \sum_{k=1}^n \sqrt{k}$ and $b_n = \sqrt{n^3}$. Then

$$\begin{aligned} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} &= \frac{\sqrt{n+1}}{\sqrt{(n+1)^3} - \sqrt{n^3}} \\ &= \frac{\sqrt{n+1}(\sqrt{(n+1)^3} + \sqrt{n^3})}{(n+1)^3 - n^3} \\ &= \frac{(n+1)^2 + n\sqrt{n^2+n}}{3n^2 + 3n + 1} \\ &= \frac{\left(1 + \frac{1}{n}\right)^2 + \sqrt{1 + \frac{1}{n}}}{3 + \frac{3}{n} + \frac{1}{n^2}} \rightarrow \frac{2}{3}. \end{aligned}$$

By Stolz theorem,

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \frac{2}{3}.$$

□

Problem A.5.3. Let $a_1 = 2$, $a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$ for $n \geq 1$. Prove that $\{a_n\}$ is convergent and find the limit.

Solution. This is a typical recursive sequence, so we use the monotone convergence theorem. Consider the difference

$$\begin{aligned} a_{n+1} - a_n &= \frac{1}{2} \left(a_n + \frac{2}{a_n} \right) - a_n \\ &= \frac{1}{2} \left(\frac{2}{a_n} - a_n \right) \\ &= \frac{1}{2} \cdot \frac{2 - a_n^2}{a_n} \end{aligned}$$

Thus, we want to determine the sign of $2 - a_n^2$. By AM-GM inequality,

$$a_{n+1} = \frac{1}{2} \left(a_n + \frac{2}{a_n} \right) = \frac{1}{2} \cdot 2\sqrt{2} = \sqrt{2}.$$

Thus, $2 - a_n^2 \leq 0$, so $a_{n+1} - a_n \leq 0$. Hence, $\{a_n\}$ is decreasing.

Since $\{a_n\}$ is decreasing and bounded below by $\sqrt{2}$, by monotone convergence theorem, $\{a_n\}$ converges. Let $\lambda = \lim_{n \rightarrow \infty} a_n$. Take limits on both sides of the recursive formula to get $\lambda = \frac{1}{2} \left(\lambda + \frac{2}{\lambda} \right)$, so $\lambda = \pm\sqrt{2}$. Since every $a_n \geq \sqrt{2}$, we must have $\lambda \geq \sqrt{2}$. Hence, $\lambda = \sqrt{2}$. $\square$

Problem A.5.4. Let $\{a_n\}$ be a bounded sequence and let

$$b_n = \frac{a_1 + \cdots + a_n}{n}.$$

Prove that

$$\limsup_{n \rightarrow \infty} b_n \leq \limsup_{n \rightarrow \infty} a_n.$$

Solution. Denote $L = \limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup\{a_k : k \geq n\}$. Denote $M_n = \sup\{a_k : k \geq n\}$. For n large, M_n is closed to L , i.e., $M_n \leq L + \varepsilon$.

For $m > n$,

$$\begin{aligned} b_m &= \frac{a_1 + a_2 + \cdots + a_n + a_{n+1} + \cdots + a_m}{m} \\ &\leq \frac{\overbrace{a_1 + \cdots + a_{m-1}}^{\text{fixed}} + (m - n + 1)(L + \varepsilon)}{m} \\ &= \frac{C}{m} + \frac{m - n + 1}{m} (L + \varepsilon) \\ &\leq \frac{C}{m} + \frac{m}{m} (L + \varepsilon) = \frac{C}{m} + (L + \varepsilon) \end{aligned}$$

where we used the fact that $a_n, a_{n+1}, \dots, a_m \leq M_n < L + \varepsilon$. Since this holds for all $m > n$, taking limsup on both sides gives

$$\begin{aligned} \limsup_{n \rightarrow \infty} b_n &\leq \limsup_{n \rightarrow \infty} \left(\frac{C}{m} + L + \varepsilon \right) && C/m \rightarrow 0, L + \varepsilon \text{ is a constant} \\ &= \lim_{m \rightarrow \infty} \left(\frac{C}{m} + L + \varepsilon \right) = L + \varepsilon. \end{aligned}$$

Since ε was arbitrary, we conclude that $\limsup_{n \rightarrow \infty} b_n \leq L = \limsup_{n \rightarrow \infty} a_n$.

(Exercise: If $a \leq b + \varepsilon$ for all $\varepsilon > 0$, then $a \leq b$.) $\square$

Alternative solution. Write

$$b_n = \frac{a_1 + a_2 + \cdots + a_n}{n} \leq \frac{c_1 + c_2 + \cdots + c_n}{n},$$

where $c_n = \sup\{a_k : k \geq n\}$. Note that $\lim_{n \rightarrow \infty} c_n = \limsup_{n \rightarrow \infty} a_n$, by definition. Take limsup on both sides. By order preserving property,

$$\begin{aligned} \limsup_{n \rightarrow \infty} b_n &\leq \limsup_{n \rightarrow \infty} \frac{c_1 + c_2 + \cdots + c_n}{n} \\ &= \lim_{n \rightarrow \infty} \frac{c_1 + c_2 + \cdots + c_n}{n} \\ &= \lim_{n \rightarrow \infty} c_n && \text{by Cauchy's first limit theorem - tutorial 2} \\ &= \limsup_{n \rightarrow \infty} a_n. \end{aligned}$$

□

Problem A.5.5. Let $\{a_n\}$ be a positive and decreasing sequence such that

$$\sum_{n=1}^{\infty} n^2 a_{n^3} \quad \text{is convergent.}$$

Prove that $\sum a_n$ is convergent.

Solution. This appears to be a variation of the Cauchy condensation test, so we try to proceed by a similar proof, using dyadic partition. Suppose $N \leq (m+1)^3 - 1$. Then we group terms:

$$\begin{aligned} \sum_{n=1}^N a_n &\leq \sum_{n=1}^{(m+1)^3-1} a_n \\ &= \sum_{k=1}^m \sum_{n=k^3}^{(k+1)^3-1} a_n \\ &\leq \sum_{k=1}^m a_{k^3} [(k+1)^3 - k^3] && \text{since } a_n \text{ is decreasing, replace with the first term} \\ &= \sum_{k=1}^m (3k^2 + 3k + 1) a_{k^3} && \text{replace } k \text{ to } k^3, 1 \text{ to } k^2 \\ &\leq \sum_{k=1}^m 7k^2 a_{k^3}. \end{aligned}$$

Since $\sum_{k=1}^{\infty} k^2 a_{k^3}$ is convergent (by assumption), it follows that the partial sums are bounded above: $\sum_{k=1}^m k^2 a_{k^3} \leq C$. Then $\sum_{n=1}^N a_n \leq 7C$ for all $N \in \mathbb{N}$. □

Problem A.5.6. Let $a_1 > 0$ and $a_{n+1} = a_n + \frac{1}{a_n}$ for $n \geq 1$. Determine whether $\left\{\frac{a_n}{\sqrt{n}}\right\}$ is convergent and prove your claim. If it is convergent, find the limit.

Solution. Recall that we have shown in tutorial that $\{a_n\}$ is increasing and divergent, so $\lim_{n \rightarrow \infty} a_n = +\infty$.

The square root $\sqrt{n}$ is not easy to handle, so we want to deal with $\left\{\frac{a_n^2}{n}\right\}$ instead. We will use Stolz theorem to find the limit.

Since $a_{n+1} = a_n + \frac{1}{a_n}$,

$$a_{n+1}^2 = \left(a_n + \frac{1}{a_n}\right)^2 = a_n^2 + 2 + \frac{1}{a_n^2},$$

so $a_{n+1}^2 - a_n^2 = 2 + \frac{1}{a_n^2} \rightarrow 2$. Let $b_n = n$. By Stolz theorem,

$$\lim_{n \rightarrow \infty} \frac{a_n^2}{n} = \lim_{n \rightarrow \infty} \frac{a_{n+1}^2 - a_n^2}{(n+1) - n} = \lim_{n \rightarrow \infty} (a_{n+1}^2 - a_n^2) = 2.$$

Take square root to get $\lim_{n \rightarrow \infty} \frac{a_n}{\sqrt{n}} = \sqrt{2}$. □

Problem A.5.7. It can be shown that the Dirichlet function is discontinuous on $\mathbb{R}$. The Riemann function is defined as

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ 1 & \text{if } x = \frac{p}{q}, (p, q) = 1 \end{cases}$$

Prove that the Riemann function is continuous on the set of irrational numbers.

Solution. Let x_0 be an irrational point. We want to show $\lim_{x \rightarrow x_0} f(x) = f(x_0) = 0$.

Let $\varepsilon > 0$. Choose $n > \frac{1}{\varepsilon}$. problematic terms are with small denominator. Consider

$$C = \left\{ \frac{p}{q} : q \leq n, \frac{p}{q} \in (x_0 - 1, x_0 + 1) \right\}.$$

There are finitely many choices for denominator, so there are finitely many numbers $\frac{p}{q}$, so we can take the minimum.

Take

$$\delta = \min_{\frac{p}{q} \in C} \left| \frac{p}{q} - x_0 \right|.$$

If a rational number $\frac{p'}{q'} \in V_\delta(x_0)$, then $q' > n$,

$$f\left(\frac{p'}{q'}\right) = \frac{1}{q'} < \frac{1}{n} < \varepsilon.$$

□

Problem A.5.8. Let $f: [0, 1] \rightarrow \mathbb{R}$ be a function satisfying that for any $x \in [0, 1]$, there exist $C > 0$ and $\delta > 0$ such that

$$|f(y)| \leq C \quad \text{for all } y \in V_\delta(x) \cap [0, 1].$$

Prove that f is bounded.

This tells us that local boundedness implies a uniform bound.

Solution. Suppose, for a contradiction, that f is not bounded on $[0, 1]$. Then for every $n \in \mathbb{N}$, there exists $x_n \in [0, 1]$ such that $|f(x_n)| > n$.

Since each x_n lies in $[0, 1]$, the sequence $\{x_n\}$ is bounded. By Bolzano-Weierstrass theorem, $\{x_n\}$ has a convergent subsequence $y_k = x_{n_k}$ such that $\lim_{k \rightarrow \infty} y_k = \xi \in [0, 1]$.

By local boundedness near ξ , there exist $C > 0$ and $\delta > 0$ such that

$$|f(x)| \leq C \quad \text{for all } x \in V_\delta(\xi).$$

Since $y_k \rightarrow \xi$, there exists $N \in \mathbb{N}$ such that

$$y_k \in V_\delta(\xi) \quad \text{for all } k \geq N.$$

Hence, $|f(y_k)| \leq C$ for all $k \geq N$, contradicting $\lim_{k \rightarrow \infty} |f(y_k)| = +\infty$.

□

Remark. Alternatively, use compactness of $[0, 1]$ to get a finite cover of $[0, 1]$.

A.6 Week 9

Problem A.6.1.

(1) Find $\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x}{1 - \cos x}$.

(2) Find $\lim_{x \rightarrow 0} \left(\frac{2^x + 5^x}{2} \right)^{1/x}$.

Solution.

- (1) To find the limit, we first rewrite the numerator to isolate terms involving the standard limit $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$.

$$\begin{aligned} \frac{1 - \cos x \cos(2x)}{1 - \cos x} &= \frac{1 - \cos x + \cos x - \cos x \cos(2x)}{1 - \cos x} \\ &= \frac{(1 - \cos x) + \cos x(1 - \cos(2x))}{1 - \cos x} \\ &= \frac{1 - \cos x}{1 - \cos x} + \frac{\cos x(1 - \cos(2x))}{1 - \cos x} \\ &= 1 + \cos x \cdot \frac{1 - \cos(2x)}{1 - \cos x} \end{aligned}$$

We now examine the second term. To utilize known limits, we multiply and divide by x^2 :

$$\frac{1 - \cos(2x)}{1 - \cos x} = \frac{\frac{1 - \cos(2x)}{x^2}}{\frac{1 - \cos x}{x^2}}$$

Applying the substitution $u = 2x$ for the numerator:

$$\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x^2} = \lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{(2x)^2} \cdot 4 = \frac{1}{2} \cdot 4 = 2$$

Given that $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$, we have:

$$\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{1 - \cos x} = \frac{2}{1/2} = 4$$

Combining these results:

$$\begin{aligned} \lim_{x \rightarrow 0} \left(1 + \cos x \cdot \frac{1 - \cos(2x)}{1 - \cos x} \right) &= 1 + \left(\lim_{x \rightarrow 0} \cos x \right) \cdot \left(\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{1 - \cos x} \right) \\ &= 1 + 1 \cdot 4 \\ &= 5. \end{aligned}$$

- (2) To deal with power functions, rewrite it as an exponential function then take logarithm.

$$\left(\frac{2^x + 5^x}{2} \right)^{1/x} = \exp \left[\log \left(\frac{2^x + 5^x}{2} \right)^{1/x} \right].$$

By continuity of exponential function, it suffices to find the limit of the log as $x \rightarrow 0$.

Recall $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$, and note that $\frac{2^x + 5^x}{2} - 1 \rightarrow 0$ as $x \rightarrow 0$. Thus,

$$\begin{aligned} \lim_{x \rightarrow 0} \log \left(\frac{2^x + 5^x}{2} \right)^{1/x} &= \lim_{x \rightarrow 0} \frac{\log \left(\frac{2^x + 5^x}{2} \right)}{x} \\ &= \lim_{x \rightarrow 0} \frac{\log \left(\frac{2^x + 5^x}{2} \right)}{\frac{2^x + 5^x}{2} - 1} \cdot \frac{\frac{2^x + 5^x}{2} - 1}{x} \end{aligned}$$

The first limit term is 1. For the second term,

$$\lim_{x \rightarrow 0} \frac{\frac{2^x + 5^x}{2} - 1}{x} = \lim_{x \rightarrow 0} \frac{1}{2} \left(\frac{2^x - 1}{x} + \frac{5^x - 1}{x} \right).$$

We know $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$. Change base by replacing x : $\lim_{x \rightarrow 0} \frac{e^{x \log 2} - 1}{x \log 2} = 1$, so $\lim_{x \rightarrow 0} \frac{2^x - 1}{x} = \log 2$. Similarly, $\lim_{x \rightarrow 0} \frac{5^x - 1}{x} = \log 5$. Thus,

$$\lim_{x \rightarrow 0} \frac{\frac{2^x + 5^x}{2} - 1}{x} = \frac{1}{2} (\log 2 + \log 5) = \log \sqrt{10}.$$

Hence, the given limit equals $\exp(\log \sqrt{10}) = \sqrt{10}$.

□

Problem A.6.2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a uniformly continuous function on $\mathbb{R}$. Prove that there exist constants $A, B > 0$ such that for any $x \in \mathbb{R}$,

$$|f(x)| \leq A|x| + B.$$

Solution. Since f is uniformly continuous, there exists $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < 1. \quad (*)$$

Fix $x \in \mathbb{R}$. Assume $x > 0$ (the case $x < 0$ is symmetric). The idea is to partition $[0, x]$ into n subintervals of length $< \delta$, i.e., $\frac{x}{n} < \delta$, so that we can apply $(*)$ on each subinterval.

Formally, choose $n \in \mathbb{N}$ such that $n - 1 \leq \frac{x}{\delta} < n$. This gives $n + 1$ points $x_k = k \cdot \frac{x}{n}$, where $k = 0, 1, \dots, n$.

Then $|x_k - x_{k-1}| = \frac{x}{n} < \delta$. By $(*)$, $|f(x_k) - f(x_{k-1})| < 1$.

Since $f(0)$ is constant, consider

$$|f(x) - f(0)| \leq \sum_{k=0}^n |f(x_k) - f(x_{k-1})| < n + 1 \leq \frac{x}{\delta} + 2.$$

Adding $|f(0)|$ on both sides gives

$$|f(x)| < \frac{x}{\delta} + 2 + |f(0)|.$$

Setting $A = \frac{1}{\delta}$ and $B = |f(0)| + 2$ completes the proof.

□

Problem A.6.3. Let $f: [0, +\infty) \rightarrow \mathbb{R}$ be a continuous function such that $\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = 0$. Let $\{a_n\}$ be a non-negative sequence such that $\{\frac{a_n}{n}\}$ is bounded. Prove that

$$\lim_{n \rightarrow \infty} \frac{f(a_n)}{n} = 0.$$

Solution. Since $\{\frac{a_n}{n}\}$ is bounded, there exists $C > 0$ such that $|\frac{a_n}{n}| \leq C$ for all $n \in \mathbb{N}$.

Let $\varepsilon > 0$ be given. There exists $M > 0$ such that $|\frac{f(x)}{x}| < \frac{\varepsilon}{2C}$ for all $x \geq M$.

Case 1: If $a_n > M$, then

$$\left| \frac{f(a_n)}{n} \right| = \left| \frac{f(a_n)}{a_n} \right| \left| \frac{a_n}{n} \right| < \frac{\varepsilon}{2C} \cdot C = \frac{\varepsilon}{2} < \varepsilon.$$

Case 2: If $a_n \leq M$, let $L = \max_{x \in [0, M]} |f(x)|$, which exists by continuity of f .

Since L is a constant, choose N large enough such that $\frac{L}{N} < \frac{\varepsilon}{2}$. Then for $n \geq N$,

$$\left| \frac{f(a_n)}{n} \right| \leq \frac{L}{N} < \frac{\varepsilon}{2} < \varepsilon.$$

In both cases, $\left| \frac{f(a_n)}{n} \right| < \varepsilon$ for all $n \geq N$. □

Problem A.6.4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that for any $x, y \in \mathbb{R}$,

$$|f(x) - f(y)| \leq |x - y|^2.$$

Prove that $f(x)$ is a constant function.

Holder, but $\alpha > 1$ implies that the function must be constant.

Similar to Q2, cut interval $[x, y]$ into n parts

Solution. Let $x, y \in \mathbb{R}$, $x < y$. Partition $[x, y]$ into n intervals of length $\frac{y-x}{n}$ with points $x_k = x + k \frac{y-x}{n}$. By the triangle inequality and the given condition,

$$|f(y) - f(x)| \leq \sum_{k=1}^n |f(x_k) - f(x_{k-1})| \leq \sum_{k=1}^n |x_k - x_{k-1}|^2 = \sum_{k=1}^n \left(\frac{y-x}{n} \right)^2.$$

This simplifies to $|f(y) - f(x)| \leq n \cdot \frac{(y-x)^2}{n^2} = \frac{(y-x)^2}{n}$. Taking the limit as $n \rightarrow \infty$, we find $|f(y) - f(x)| \leq 0$. Therefore, $f(x) = f(y)$, and so f is constant. □

Problem A.6.5. Let (X, d) be a metric space and $E \subseteq X$ be a non-empty subset. For $x \in X$, define

$$f(x) := \inf \{d(x, y) : y \in E\}.$$

Prove:

- (1) $f(x) = 0$ if and only if $x \in \bar{E}$;
- (2) $f: X \rightarrow \mathbb{R}$ is continuous.

$f(x)$ is the distance between x and E (which exists since each $d(x, y) \geq 0$)

Solution.

- (1) $\Rightarrow$ Suppose $f(x) = 0$. Then $\inf\{d(x, y) : y \in E\} = 0$. This means that every $\varepsilon > 0$ is an upper bound of the set, i.e., for each $\varepsilon > 0$, there exists $y \in E$ such that $d(x, y) < \varepsilon$.

Then $V_\varepsilon(x) \neq \emptyset$. Since ε is arbitrary, if $x \in E$ then we are done. If $x \notin E$, then x is a limit point of E . In both cases, $x \in \bar{E}$.

$\Leftarrow$ Let $x \in \bar{E} = E \cup E'$.

If $x \in E$, then $0 \in \{d(x, y) : y \in E\}$, so $\inf\{d(x, y) : y \in E\} = 0$. Thus, $f(x) = 0$.

If $x \in E'$, then for each $\varepsilon > 0$, there exists $y \in E$ such that $d(x, y) < \varepsilon$. Thus, $\inf\{d(x, y) : y \in E\} < \varepsilon$. Since ε was arbitrary, we conclude that $\inf\{d(x, y) : y \in E\} = 0$, i.e., $f(x) = 0$.

- (2) Fix $x \in X$. We want to show that f is continuous at x , namely

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall y \in V_\delta(x), \quad |f(x) - f(y)| < \varepsilon.$$

Let $\varepsilon > 0$ be given. There exists $z \in E$ such that

$$f(x) \leq d(x, z) < f(x) + \frac{\varepsilon}{2}.$$

By triangle inequality,

$$d(z, y) \leq d(z, x) + d(x, y) < f(x) + \frac{\varepsilon}{2} + \delta.$$

Thus,

$$f(y) \leq d(z, y) < f(x) + \frac{\varepsilon}{2} + \delta = f(x) + \varepsilon$$

where we choose $\delta = \frac{\varepsilon}{2}$. Thus, $f(y) < f(x) + \varepsilon$. Interchange the roles of x and y , we get $f(x) < f(y) + \varepsilon$. Hence, $|f(x) - f(y)| < \varepsilon$.

□

Notation: we define $d(x, E) := \inf_{y \in E} d(x, y)$.

Problem A.6.6. Let (X, d) be a metric space. Let K be a non-empty compact subset of X , and F be a non-empty closed subset of X such that $K \cap F = \emptyset$. Prove that there exists $\delta > 0$ such that $d(x, y) \geq \delta$ for any $x \in K, y \in F$.

Solution. For a contradiction, suppose that for all $\delta > 0$, there exist $x \in K$ and $y \in F$ such that $d(x, y) < \delta$.

For each $n \in \mathbb{N}$, let $\delta_n = \frac{1}{n}$. Then there exist $x_n \in K$ and $y_n \in F$ such that $d(x_n, y_n) < \frac{1}{n}$.

Since K is compact, K is sequentially compact. Thus, $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$; let $\lim_{k \rightarrow \infty} x_{n_k} = x \in K$.

Claim: $x \in F$. (We will show that $x \in F'$, i.e., x is a limit point of F .)

Note that $d(x, y_{n_k}) \leq d(x, x_{n_k}) + d(x_{n_k}, y_{n_k}) \rightarrow 0$ as $k \rightarrow \infty$. Thus, $y_{n_k} \rightarrow x$.

Since F is closed, F contains its limit points, so $x \in F$. This implies $x \in K \cap F$, contradicting $K \cap F = \emptyset$. □

A.7 Week 10

Problem A.7.1.

(1) Find $\lim_{n \rightarrow \infty} \left((n + \log n)^{1/3} - n^{1/3} \right)$.

(2) Find $\lim_{x \rightarrow 0} \frac{\tan x (\sqrt{1 + \sin x} - 1)}{1 - \cos(\sin x)}$.

Solution.

(1) We first factor $n^{1/3}$ from the expression:

$$(n + \log n)^{1/3} - n^{1/3} = n^{1/3} \left[\left(1 + \frac{\log n}{n} \right)^{1/3} - 1 \right]$$

Recall that $\lim_{x \rightarrow 0} \frac{(1+x)^{1/3} - 1}{x} = \frac{1}{3}$. We multiply and divide by $\frac{\log n}{n}$:

$$= \frac{(1 + \frac{\log n}{n})^{1/3} - 1}{\frac{\log n}{n}} \cdot \frac{n^{1/3} \log n}{n} = \frac{(1 + \frac{\log n}{n})^{1/3} - 1}{\frac{\log n}{n}} \cdot \frac{\log n}{n^{2/3}}$$

As $n \rightarrow \infty$, the term $\frac{\log n}{n} \rightarrow 0$. Thus, the first factor approaches $\frac{1}{3}$ and the second factor $\frac{\log n}{n^{2/3}} \rightarrow 0$. Hence, the limit is 0.

(2) The idea is to use the known limits $\lim_{x \rightarrow 0} \frac{\sqrt{1+x}-1}{x} = \frac{1}{2}$ and $\lim_{x \rightarrow 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$. We decompose the expression:

$$\frac{\tan x (\sqrt{1 + \sin x} - 1)}{1 - \cos(\sin x)} = \frac{1}{\cos x} \cdot \frac{\sqrt{1 + \sin x} - 1}{\sin x} \cdot \frac{\sin^2 x}{1 - \cos(\sin x)}$$

As $x \rightarrow 0$, we have

$$\frac{1}{\cos x} \rightarrow 1, \quad \frac{\sqrt{1 + \sin x} - 1}{\sin x} \rightarrow \frac{1}{2}, \quad \frac{\sin^2 x}{1 - \cos(\sin x)} \rightarrow 2.$$

Hence, the limit is $1 \cdot \frac{1}{2} \cdot 2 = 1$.

□

Remark. Replace x with some function $u(x) \rightarrow 0$ as $x \rightarrow 0$.

Problem A.7.2.

- (1) Let $f: [0, 1] \rightarrow [0, 1]$ be a continuous function. Prove that there exists $x_0 \in [0, 1]$ such that $f(x_0) = x_0$.
- (2) Let $f(x) = x^3 - 3x^2 + 2x + 16$. Prove there exists $x_0 \in \mathbb{R}$ such that $f(x_0) = 0$.

In (1), x_0 is called a *fixed point*.

Solution.

- (1) Consider the function $g(x) = f(x) - x$, which is continuous on $[0, 1]$.

At the endpoints, that $g(0) = f(0) \geq 0$ and $g(1) = f(1) - 1 \leq 1 - 1 = 0$.

If $g(0) = 0$ or $g(1) = 0$, then we are done, since $x_0 = 0$ or 1 .

Otherwise, $g(0) > 0$ and $g(1) < 0$. By intermediate value theorem, there exists $x_0 \in (0, 1)$ such that $g(x_0) = 0$, so $f(x_0) = x_0$.

- (2) We examine the behavior of $f(x)$ for large x :

$$f(x) = x^3 \left(1 - \frac{3}{x} + \frac{2}{x^2} + \frac{16}{x^3} \right).$$

Denote $g(x) = 1 - \frac{3}{x} + \frac{2}{x^2} + \frac{16}{x^3}$. Since $\lim_{x \rightarrow \infty} g(x) = 1$, there exists sufficiently large M such that $g(M) > \frac{1}{2}$ and $g(-M) < \frac{1}{2}$.

Thus, $f(M) = M^3 \cdot g(M) > 0$ and $f(-M) = -M^3 \cdot g(M) < 0$.

By intermediate value theorem, there exists $x_0 \in (-M, M)$ such that $f(x_0) = 0$.

□

Remark. Idea: construct some function, use IVT on that function.

Problem A.7.3. For $x > 0$, define

$$a_n(x) = \left(1 + \frac{x}{n} \right)^n.$$

Prove that for any $n \in \mathbb{N}$, $a_n(x) < e^x$.

notice that if $x = 1$, then this reduces $\left(1 + \frac{1}{n} \right)^n < e$, which we proved using monotone convergence theorem by showing $\left(1 + \frac{1}{n} \right)^n$ is increasing. Thus, we proceed using a similar idea.

Solution. Fix $x \in \mathbb{R}$. Write

$$a_n(x) = \left[\left(1 + \frac{1}{n/x} \right)^{n/x} \right]^x.$$

Since power function is continuous, by sequential criterion, we see that $\lim_{n \rightarrow \infty} a_n(x) = e^x$.

By monotone convergence theorem, to prove $a_n(x)$ is less than the limit, it suffices to show that $a_n(x)$ is strictly increasing. We compare $a_n(x)$ and $a_{n+1}(x)$. By binomial expansion,

$$\begin{aligned} a_n(x) &= \left(1 + \frac{x}{n} \right)^n \\ &= \sum_{k=0}^n \frac{n(n-1) \cdots (n-k+1)}{k!} \left(\frac{x}{n} \right)^k \\ &= \sum_{k=0}^n \frac{x^k}{k!} \frac{n(n-1) \cdots (n-k+1)}{n^k} \\ &= \sum_{k=0}^n \frac{x^k}{k!} \underbrace{\left(1 - \frac{1}{n} \right) \cdots \left(1 - \frac{k-1}{n} \right)}_{y_{k,n}} \end{aligned}$$

and

$$\begin{aligned}
 a_{n+1}(x) &= \left(1 + \frac{x}{n+1}\right)^{n+1} \\
 &= \sum_{k=0}^{n+1} \frac{(n+1)(n) \cdots (n-k+2)}{k!} \left(\frac{x}{n+1}\right)^k \\
 &= \sum_{k=0}^{n+1} \frac{x^k}{k!} \frac{(n+1)(n) \cdots (n-k+2)}{(n+1)^k} \\
 &= \sum_{k=0}^{n+1} \frac{x^k}{k!} \underbrace{\left(1 - \frac{1}{n+1}\right) \cdots \left(1 - \frac{k-1}{n+1}\right)}_{y_{k,n+1}}.
 \end{aligned}$$

To compare $a_n(x)$ and $a_{n+1}(x)$, we compare $y_{k,n}$ and $y_{k,n+1}$. Note that

$$1 - \frac{1}{n} < 1 - \frac{1}{n+1}, \quad 1 - \frac{2}{n} < 1 - \frac{2}{n+1}, \quad \dots, \quad 1 - \frac{k-1}{n} < 1 - \frac{k-1}{n+1}.$$

This implies that $y_{k,n} < y_{k,n+1}$. Thus,

$$a_n(x) = \sum_{k=0}^n \frac{x^k}{k!} y_{k,n} < \sum_{k=0}^n \frac{x^k}{k!} y_{k,n+1} < \sum_{k=0}^{n+1} \frac{x^k}{k!} y_{k,n+1} = a_{n+1}(x).$$

Hence, $a_n(x)$ is strictly increasing, so $a_n(x) < \lim_{n \rightarrow \infty} a_n(x) = e^x$. $\square$

Note: proof is similar to $\left(1 + \frac{1}{n}\right)^n < e$, just factor out x in front.

Problem A.7.4. Let (X, d) be a metric space and (X^*, D) be the completion of (X, d) . Prove that X is dense in X^* , namely for any $\delta > 0$ and any $x^* \in X$, there exists $x \in X \subset X^*$ such that $D(x, x^*) < \delta$.

We write $X \subset X^*$ to mean X is embedded in X^* , where the embedding is given by identifying each $x \in X$ with the sequence $\{x_n = x\} \in X^*$.

Solution. Fix $x^* \in X$. If $x^* \in X$, then pick $x = x^*$ so that $D(x, x^*) = 0$.

If $x^* \notin X$, let $x^* = \{a_n\}$ be a Cauchy sequence in X .

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m > n \geq N, \quad d(a_m, a_n) < \varepsilon.$$

Take $x = a_N$. Then $D(x_N, x^*) = \lim_{n \rightarrow \infty} d(x_N, x_n) \leq \varepsilon$, so take $\varepsilon = \delta/2$ such that $D(x, x^*) \leq \delta/2 < \delta$. Hence, X is dense in X^* . $\square$

e.g. $\mathbb{Q}$ is dense in $\mathbb{R}$.

Problem A.7.5 (Baire category theorem). For every $n \in \mathbb{N}$, let $F_n \subseteq \mathbb{R}$ be a closed subset such that

$$\bigcup_{n=1}^{\infty} F_n = \mathbb{R}.$$

Prove that there exists $n \in \mathbb{N}$ such that F_n contains an open interval $(a, b) \subseteq F_n$.

special case in $\mathbb{R}$, see general case in functional analysis

Solution. Suppose, for the sake of contradiction, that no F_n contains an open interval. Let $G_n = \mathbb{R} \setminus F_n$. Since each F_n is closed, each G_n is open. Furthermore, since no F_n contains an open interval, every G_n must be dense in $\mathbb{R}$. We proceed by constructing a nested sequence of closed intervals $I_n = [a_n, b_n]$ to find a point that avoids all F_n .

- (1) Since G_1 is dense and open, there exists a closed interval $I_1 \subseteq G_1$ with length $|I_1| < 1$.
- (2) Since G_2 is dense, the intersection of its interior $I_1^\circ \cap G_2$ is a non-empty open set. Thus, we can choose a closed interval $I_2 \subseteq I_1^\circ \cap G_2$ such that $|I_2| < 1/2$.
- (3) Inductively, given I_{n-1} , the density of G_n ensures that $I_{n-1}^\circ \cap G_n$ is non-empty and open. We can therefore choose a closed interval I_n such that:

$$\begin{aligned} I_n &\subseteq I_{n-1}^\circ \cap G_n \\ |I_n| &< 1/n \end{aligned}$$

This construction yields a nested sequence of non-empty closed intervals $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$. By the Nested Interval Theorem, the intersection of these sets is non-empty. Let:

$$x \in \bigcap_{n=1}^{\infty} I_n$$

By our construction, $I_n \subseteq G_n$ for every $n \in \mathbb{N}$. It follows that:

$$\begin{aligned} x \in \bigcap_{n=1}^{\infty} G_n &\implies x \in G_n \text{ for all } n \in \mathbb{N} \\ &\implies x \notin F_n \text{ for all } n \in \mathbb{N} \\ &\implies x \notin \bigcup_{n=1}^{\infty} F_n \end{aligned}$$

This implies $x \notin \mathbb{R}$, which is a contradiction. Therefore, at least one F_n must contain an open interval. $\square$

A.8 Week 12

Problem A.8.1.

- (1) Find $\lim_{x \rightarrow +\infty} \left(\frac{x+1}{x-1} \right)^x$.
- (2) Find $\lim_{x \rightarrow 0} (2 - \cos x)^{\frac{1}{x^2}}$.

Solution. To evaluate the limits of a power function $f(x)^{g(x)}$, rewrite it as

$$e^{g(x) \log f(x)}.$$

Then it suffices to evaluate the limit of $g(x) \log f(x) = \frac{\log f(x)}{1/g(x)}$.

- (1) Write $\left(\frac{x+1}{x-1} \right)^x = e^{x \log \left(\frac{x+1}{x-1} \right)}$. Thus, it suffices to evaluate the limit of the power as $x \rightarrow +\infty$. By a change of variables, let $t = \frac{1}{x}$. Then $t \rightarrow 0$, so

$$\begin{aligned} \lim_{x \rightarrow +\infty} x \log \left(\frac{x+1}{x-1} \right) &= \lim_{t \rightarrow 0} \frac{1}{t} \log \left(\frac{1/t+1}{1/t-1} \right) = \lim_{t \rightarrow 0} \frac{1}{t} \log \left(\frac{1-t}{1+t} \right) \\ &= \lim_{t \rightarrow 0} \frac{\log(1+t) - \log(1-t)}{t} \\ &= \lim_{t \rightarrow 0} \frac{\log(1+t)}{t} - \frac{\log(1-t)}{t} \\ &= 1 - (-1) = 2, \end{aligned}$$

where we used the known limit $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$. Therefore, $\lim_{x \rightarrow +\infty} \left(\frac{x+1}{x-1} \right)^x = e^2$.

(2) Write $(2 - \cos x)^{\frac{1}{x^2}} = e^{\frac{1}{x^2} \log(2 - \cos x)}$. Evaluate

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\log(2 - \cos x)}{x^2} &= \lim_{x \rightarrow 0} \frac{\log(1 + (1 - \cos x))}{1 - \cos x} \cdot \frac{1 - \cos x}{x^2} \\ &= 1 \cdot \frac{1}{2} = \frac{1}{2}, \end{aligned}$$

where we used the known limits $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$. Therefore, the final limit is $e^{1/2}$.

□

Problem A.8.2. Let $f: [0, 1] \rightarrow \mathbb{R}$ be a continuous function on $[0, 1]$ such that

$$f(x) \in [0, 1] \quad \text{for any } x \in [0, 1],$$

and

$$|f(x) - f(y)| < |x - y| \quad \text{for any } x, y \in [0, 1], x \neq y.$$

Prove that there exists a unique $x_0 \in (0, 1]$ such that $f(x_0) = \frac{1-x_0}{x_0}$.

Solution. For existence, define $g(x) = f(x) - \frac{1-x}{x}$, which is continuous on $(0, 1]$.

Since $g(1) = f(1) - 0 \geq 0$ and $g(1/2) = f(1/2) - 2 \cdot 1/2 \leq 0$, by IVT, there exists $x_0 \in [1/2, 1]$ such that $g(x_0) = 0$, implying $f(x_0) = \frac{1-x_0}{x_0}$.

For uniqueness, for a contradiction, assume that there exist $x \neq y$ such that $f(x) = \frac{1-x}{x}$ and $f(y) = \frac{1-y}{y}$. Then

$$|f(x) - f(y)| = \left| \frac{1-x}{x} - \frac{1-y}{y} \right| = \frac{|y-x|}{xy}.$$

Since $0 < xy \leq 1$, it follows that $\frac{1}{xy} \geq 1$. Thus, $|f(x) - f(y)| \geq |x - y|$, which contradicts the given condition that $|f(x) - f(y)| < |x - y|$. □

Problem A.8.3. For $x > 0$, prove that

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x.$$

Solution. One can easily show that the series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges using the ratio test.

Fix $x > 0$. Let $s_N = \sum_{n=0}^N \frac{x^n}{n!}$ be the partial sum. Then

$$s_N = \sum_{n=0}^N \frac{x^n}{n!} \geq \sum_{n=0}^N \frac{N(N-1) \cdots (N-n+1)}{N^n} \cdot \frac{x^n}{n!} = \sum_{n=0}^N \binom{N}{n} \left(\frac{x}{N}\right)^n = \left(1 + \frac{x}{N}\right)^N.$$

Taking the limit $N \rightarrow \infty$ yields $\sum_{n=0}^{\infty} \frac{x^n}{n!} \geq e^x$.

On the other hand, fix $N \in \mathbb{N}$. Let

$$x_{n,k} = \frac{n(n-1) \cdots (n-k+1)}{n^k}.$$

As $n \rightarrow \infty$, $x_{n,k} \rightarrow 1$. Recall that $a_n = \left(1 + \frac{x}{n}\right)^n \rightarrow e^x$, so for any $n \geq N$,

$$a_n = \sum_{k=0}^n \binom{n}{k} \left(\frac{x}{n}\right)^k \geq \sum_{k=0}^N \binom{n}{k} \left(\frac{x}{n}\right)^k = \sum_{k=0}^N x_{n,k} \frac{x^k}{k!}.$$

Taking $n \rightarrow \infty$ yields $e^x \geq \sum_{k=0}^N \frac{x^k}{k!} = s_N$. Since this holds for any N , we have $e^x \geq \lim_{N \rightarrow \infty} s_N = \sum_{n=0}^{\infty} \frac{x^n}{n!}$.

Therefore, $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$. □

Problem A.8.4. Let sequences $\{a_n\}$ and $\{b_n\}$ satisfy that $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$. Prove that

$$\lim_{n \rightarrow \infty} \frac{a_1 b_n + a_2 b_{n-1} + \cdots + a_{n-1} b_2 + a_n b_1}{n} = AB.$$

Solution. Let $\varepsilon > 0$. Since $\{a_n\}$ and $\{b_n\}$ are convergent, there exists $N \in \mathbb{N}$ such that $|a_n - A| \leq \varepsilon_1$ and $|b_n - B| \leq \varepsilon_2$ for all $n \geq N$.

Since $\{a_n\}$ and $\{b_n\}$ are bounded, there exists $M > 0$ such that $|a_n| \leq C$ and $|b_n| \leq C$ for all $n \in \mathbb{N}$.

We claim that if both i, j are large, then $a_i b_j$ is close to AB .

Lemma. *There exists $N \in \mathbb{N}$ such that $|a_i b_j - AB| < \varepsilon$ for all $i, j \geq N$.*

Proof. By inserting a new term, we have

$$\begin{aligned} |a_i b_j - AB| &= |a_i b_j - Ab_j + Ab_j - AB| \\ &\leq |a_i b_j - Ab_j| + |Ab_j - AB| \\ &= |a_i - A||b_j| + |A||B - b_j| \\ &\leq C|a_i - A| + |A||B - b_j|. \end{aligned}$$

Then choose N such that $|a_i - A| < \frac{\varepsilon}{2C}$ for all $i \geq N$, and $|b_j - B| < \frac{\varepsilon}{2|A|}$ for all $j \geq N$. □

Split the summation into three parts based on this N :

$$\begin{aligned} \left| \frac{a_1 b_n + \cdots + a_n b_1}{n} - AB \right| &= \left| \frac{(a_1 b_n - AB) + \cdots + (a_n b_1 - AB)}{n} \right| \\ &\leq \sum_{k=1}^n \frac{|a_k b_{n+1-k} - AB|}{n} \\ &= \sum_{k=1}^N \frac{|a_k b_{n+1-k} - AB|}{n} + \sum_{k=n-N+1}^n \frac{|a_k b_{n+1-k} - AB|}{n} + \sum_{k=N+1}^{n-N} \frac{|a_k b_{n+1-k} - AB|}{n}. \end{aligned}$$

We bound the second term:

$$\sum_{k=N+1}^{n-N} \frac{|a_k b_{n+1-k} - AB|}{n} \leq \sum_{k=N+1}^{n-N} \frac{\varepsilon}{n} = \varepsilon \cdot \frac{n - 2N}{n} < \varepsilon.$$

For the first term,

$$\sum_{k=1}^N \frac{|a_k b_{n+1-k} - AB|}{n} \leq \sum_{k=1}^N \frac{C^2 + |A||B|}{n} = \sum_{k=1}^N \frac{L}{n} \leq \frac{L \cdot N}{n}.$$

Similarly, the third term is bounded as follows:

$$\sum_{k=N+1}^{n-N} \frac{|a_k b_{n+1-k} - AB|}{n} \leq \frac{L \cdot N}{n}.$$

Hence, we choose n large enough such that $\frac{LN}{n} < \frac{\varepsilon}{2}$, i.e., $n > \frac{2LN}{\varepsilon}$. Then the total difference becomes

$$\left| \frac{\sum_{k=1}^n a_k b_{n-k+1}}{n} - AB \right| < \varepsilon + \varepsilon = 2\varepsilon.$$

Since ε is arbitrary, the limit is AB . □

Problem A.8.5. Define a function sequence $\{f_n(x)\}$ on $[0, 1]$ as follows: $f_1(x) = x$, for $n \in \mathbb{N}$,

$$f_{n+1}(x) = f_n(x) + \frac{x - f_n^2(x)}{2}.$$

Prove that $f_n(x) \rightarrow \sqrt{x}$ uniformly on $[0, 1]$.

Proof. We first look at pointwise convergence. Fix $x \in [0, 1]$. We need to prove

$$\lim_{n \rightarrow \infty} f_n(x) = \sqrt{x}.$$

Since f_n is defined recursively, we shall use the monotone convergence theorem to prove the above limit. Consider the difference

$$f_{n+1}(x) - f_n(x) = \frac{x - f_n^2(x)}{2} = \frac{(\sqrt{x} - f_n(x))(\sqrt{x} + f_n(x))}{2}.$$

We first show by induction that $0 \leq f_n(x) \leq \sqrt{x}$ for all $n \in \mathbb{N}$. The base case where $n = 1$ is clear. Suppose $0 \leq f_n(x) \leq \sqrt{x}$. Then

$$\begin{aligned} f_{n+1}(x) &= f_n(x) + \frac{x - f_n^2(x)}{2} \\ &= f_n(x) + \frac{(\sqrt{x} - f_n(x))(\sqrt{x} + f_n(x))}{2} \geq 0 \end{aligned}$$

and

$$\begin{aligned} f_{n+1}(x) - \sqrt{x} &= f_n(x) - \sqrt{x} + \frac{(\sqrt{x} - f_n(x))(\sqrt{x} + f_n(x))}{2} \\ &= (\sqrt{x} - f_n(x)) \left(-1 + \frac{\sqrt{x} + f_n(x)}{2} \right) \leq 0. \end{aligned}$$

Hence, $f_{n+1}(x) - f_n(x) \geq 0$, so $\{f_n(x)\}$ is increasing.

Since the sequence $\{f_n(x)\}$ is increasing and bounded above (by $\sqrt{x}$), by monotone convergence theorem, $\{f_n(x)\}$ converges to a limit $f(x)$. To find the limit, take $n \rightarrow \infty$ in the recurrence relation:

$$f(x) = f(x) + \frac{x - f(x)^2}{2}$$

so $f(x) = \sqrt{x}$ (reject $f(x) = -\sqrt{x}$ because each $f_n \geq 0$ so $f \geq 0$). Hence, $f_n(x) \rightarrow \sqrt{x}$ pointwise.

Since $f_n(x) \rightarrow \sqrt{x}$ pointwise and $\{f_n\}$ is monotonically increasing, by Dini's theorem, the convergence is uniform. $\square$

Remark. Every $f_n(x)$ is a polynomial, but the limit function f is not a polynomial.

Let (X, d) be a metric space. A map $f: X \rightarrow X$ is called a *contraction mapping* on X if there exists $0 < K < 1$ such that

$$d(f(x), f(y)) \leq Kd(x, y) \quad \text{for all } x, y \in X.$$

Note that a contraction mapping is a Lipschitz function and thus continuous.

Problem A.8.6 (Banach fixed-point theorem). Let (X, d) be a complete metric space, and let $f: X \rightarrow X$ be a contraction mapping. Then f admits a unique fixed point $x^* \in X$, namely $f(x^*) = x^*$.

Solution. Let $x_0 \in X$ be arbitrary. Define a sequence $\{x_n\}$ by $x_n = f(x_{n-1})$. Since f is a contraction mapping, one can show by induction that

$$d(x_{n+1}, x_n) \leq K^n d(x_1, x_0).$$

The idea is that the limit of $\{x_n\}$ should be our desired fixed point. To show $\{x_n\}$ converges, we will show that

$\{x_n\}$ is a Cauchy sequence. If $m > n$, then

$$\begin{aligned} d(x_m, x_n) &\leq d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + \cdots + d(x_{n+1}, x_n) \\ &\leq K^{m-1}d(x_1, x_0) + K^{m-2}d(x_1, x_0) + \cdots + K^n d(x_1, x_0) \\ &= K^n d(x_1, x_0) \sum_{k=0}^{m-n-1} K^k \\ &\leq K^n d(x_1, x_0) \sum_{k=0}^{\infty} K^k \\ &= K^n d(x_1, x_0) \cdot \frac{1}{1-K}. \end{aligned}$$

Note that $\frac{K^n}{1-K}d(x_1, x_0) \rightarrow 0$ as $n \rightarrow \infty$, so $d(x_m, x_n)$ becomes small. More formally, let $\varepsilon > 0$. Since $K \in (0, 1)$, choose $N \in \mathbb{N}$ sufficiently large such that

$$K^n < \frac{\varepsilon(1-K)}{d(x_1, x_0)}.$$

If $m, n \geq N$, then $d(x_m, x_n) < \varepsilon$. This proves that $\{x_n\}$ is a Cauchy sequence. Since X is complete, the sequence has a limit $x^* \in X$. We show that x^* is a fixed point of f :

$$x^* = \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} f(x_{n-1}) = f\left(\lim_{n \rightarrow \infty} x_{n-1}\right) = f(x^*),$$

where we used the continuity of f to bring the limit inside f .

To show uniqueness, suppose $p_1 \neq p_2$ are fixed points. Then

$$d(p_1, p_2) = d(f(p_1), f(p_2)) \leq Kd(p_1, p_2) < d(p_1, p_2),$$

a contradiction. □

A.9 Week 13

Questions taken from previous examinations.

Problem A.9.1.

(1) Find $\lim_{n \rightarrow \infty} \left(n - (n-1) \sqrt[3]{1 - \frac{1}{n}} \right)$.

(2) Find $\lim_{x \rightarrow +\infty} \left(\frac{\sqrt{1+x^2}}{x} \right)^{x^2}$.

Solution.

(1) Rewrite the limit as

$$\lim_{n \rightarrow \infty} n \left[1 - \left(1 - \frac{1}{n} \right) \sqrt[3]{1 - \frac{1}{n}} \right] = \lim_{n \rightarrow \infty} \frac{1 - \left(1 - \frac{1}{n} \right)^{4/3}}{\frac{1}{n}}.$$

Since $\frac{1}{n} \rightarrow 0$, by sequential criterion, consider the limit

$$\lim_{x \rightarrow 0} \frac{1 - (1-x)^{4/3}}{x}.$$

But we know $\lim_{x \rightarrow 0} \frac{(1+x)^{4/3} - 1}{x} = \frac{4}{3}$, so $\lim_{x \rightarrow 0} \frac{(1-x)^{4/3} - 1}{-x} = \frac{4}{3}$.

(2) Evaluate

$$\lim_{x \rightarrow +\infty} \left(\frac{\sqrt{1+x^2}}{x} \right)^{x^2} = \lim_{x \rightarrow +\infty} \left(\sqrt{\frac{1+x^2}{x^2}} \right)^{x^2} = \lim_{x \rightarrow +\infty} \sqrt{\left(1 + \frac{1}{x^2} \right)^{x^2}} = \sqrt{\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x^2} \right)^{x^2}} = \sqrt{e}$$

where we used the continuity of $\sqrt{x}$ to bring the limit in.

□

Problem A.9.2. For every $n \in \mathbb{N}$, prove that there exists a unique number $a_n \in (0, 1)$ such that

$$a_n + a_n^n = 1.$$

Further prove that the sequence $\{a_n : n \in \mathbb{N}\}$ is convergent, and find the limit.

Solution.

1. Existence and Uniqueness

Consider $f(x) = x + x^n$. At the endpoints, $f(0) = 0 < 1$ and $f(1) = 2 > 1$. By IVT, there exists $a_n \in (0, 1)$ such that $f(a_n) = 1$.

For uniqueness, note that f is strictly increasing (since $x + x^n < y + y^n$ if $x < y$). Thus, if $x < a_n$, then $f(x) < f(a_n) = 1$; if $x > a_n$, then $f(x) > f(a_n) = 1$.

2. Monotonicity and Convergence

To determine the behavior of the sequence $\{a_n\}$, consider $f_{n+1}(a_n)$:

$$f_{n+1}(a_n) = a_n + a_n^{n+1} - 1.$$

Since $a_n \in (0, 1)$, we know $a_n^{n+1} < a_n^n$. Thus,

$$f_{n+1}(a_n) = a_n + a_n^{n+1} - 1 < a_n + a_n^n - 1 = f_n(a_n) = 0.$$

So $f_{n+1}(a_n) < 0$. Since we also know $f_{n+1}(a_{n+1}) = 0$ and f_{n+1} is strictly increasing, it must be that $a_n < a_{n+1}$. Thus, the sequence $\{a_n\}$ is strictly increasing. Since it is bounded above by 1, it converges by the Monotone Convergence Theorem. Let $\lim_{n \rightarrow \infty} a_n = L$.

3. Finding the Limit

Since each $a_n \leq 1$, we have $L \leq 1$. We will show $L = 1$. For a contradiction, assume $L < 1$. Since $a_n \leq L$, we have

$$1 = a_n + a_n^n \leq L + L^n.$$

Taking the limit as $n \rightarrow \infty$,

$$1 = \lim_{n \rightarrow \infty} (a_n + a_n^n) = \lim_{n \rightarrow \infty} (L + L^n) = L$$

since $L^n \rightarrow 0$ as $L < 1$. Thus, $L \geq 1$, which gives the desired contradiction.

□

Problem A.9.3. Prove that the following series

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{x}{n}\right)$$

converges for $x \in [-1, 1]$, and $f(x)$ is a continuous function on $[-1, 1]$.

Solution. Since $\|f_n\| \leq \frac{1}{n^2}$, by comparison test, $\sum \|f_n\|$ converges, so $f(x)$ converges.

Since every f_n is continuous, f is continuous.

□

Problem A.9.4. Let (X, d) be a compact metric space. Prove that there exists a constant $C > 0$ such that for any $x, y \in X$,

$$d(x, y) \leq C.$$

Solution. Consider the collection of unit balls centered at every point in X :

$$\mathcal{U} = \{V_1(x) : x \in X\}.$$

Clearly, $\mathcal{U}$ is an open cover for X since every $x \in X$ is contained in $V_1(x)$. Since X is compact, there exists a finite subcover $\{V_1(x_1), V_1(x_2), \dots, V_1(x_n)\}$.

For any $x, y \in X$, there exist $1 \leq i, j \leq n$ such that $x \in V_1(x_i)$ and $y \in V_1(x_j)$. By triangle inequality,

$$\begin{aligned} d(x, y) &\leq d(x, x_i) + d(x_i, x_j) + d(x_j, y) \\ &< 1 + d(x_i, x_j) + 1 = 2 + d(x_i, x_j). \end{aligned}$$

Let $M = \max_{1 \leq i, j \leq n} d(x_i, x_j)$, which exists because it is a maximum over a finite set. Then for all $x, y \in X$,

$$d(x, y) \leq 2 + M.$$

□

Alternative solution. Fix $x_0 \in X$. Consider the collection of open balls centered at x_0 with integer radii:

$$\mathcal{V} = \{V_n(x_0) : n \in \mathbb{N}\}.$$

For any point $x \in X$, the distance $d(x, x_0)$ is a finite real number, so there exists some $n > d(x, x_0)$, meaning $x \in V_n(x_0)$. Thus, $\mathcal{V}$ is an open cover of X .

By compactness, there exists a finite subcover. Since these balls are nested ($V_1 \subset V_2 \subset V_3 \dots$), a finite subcover simply implies that the entire space is contained in the largest ball of the subcover. That is, there exists some $N \in \mathbb{N}$ such that:

$$X \subset V_N(x_0).$$

This means that for all $x \in X$, $d(x, x_0) < N$. For any $x, y \in X$, the triangle inequality gives:

$$d(x, y) \leq d(x, x_0) + d(x_0, y) < N + N = 2N.$$

□

Problem A.9.5. Let $f: [0, +\infty) \rightarrow (0, +\infty)$ be a uniformly continuous function. Prove that $g(x) = \sqrt{f(x)}$ is uniformly continuous on $[0, +\infty)$.

Solution. Let $\varepsilon > 0$. Since f is uniformly continuous on $[0, +\infty)$, there exists $\delta > 0$ such that for all $x, y \in [0, +\infty)$,

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon^2.$$

Now, consider the difference $|g(x) - g(y)|$ for $|x - y| < \delta$. We have

$$|g(x) - g(y)| = |\sqrt{f(x)} - \sqrt{f(y)}|.$$

Case 1: $\sqrt{f(x)} + \sqrt{f(y)} \geq \varepsilon$. Multiplying by the conjugate,

$$|g(x) - g(y)| = \frac{|f(x) - f(y)|}{\sqrt{f(x)} + \sqrt{f(y)}} < \frac{\varepsilon^2}{\sqrt{f(x)} + \sqrt{f(y)}} < \frac{\varepsilon^2}{\varepsilon} = \varepsilon.$$

Case 2: $\sqrt{f(x)} + \sqrt{f(y)} < \varepsilon$. Then

$$|g(x) - g(y)| = |\sqrt{f(x)} - \sqrt{f(y)}| \leq \sqrt{f(x)} + \sqrt{f(y)} < \varepsilon.$$

□

Problem A.9.6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a uniformly continuous function. Suppose for any $x \in [0, 1]$,

$$\lim_{n \rightarrow \infty} f(x + n) = 0.$$

Here $\{f(x + n) : n \in \mathbb{N}\}$ is considered as a sequence. Prove that

$$\lim_{x \rightarrow +\infty} f(x) = 0.$$

Solution. Our goal is to show

$$\forall \varepsilon > 0, \quad \exists L > 0, \quad \forall x \geq L, \quad |f(x)| < \varepsilon.$$

Let $\varepsilon > 0$. Since f is uniformly continuous on $\mathbb{R}$, there exists $\delta > 0$ such that for all $x, x' \in \mathbb{R}$,

$$|x - x'| < \delta \implies |f(x) - f(x')| < \frac{\varepsilon}{2}.$$

Now, partition $[0, 1]$ into subintervals each of length $< \delta$. Choose some large $K \in \mathbb{N}$ such that $\frac{1}{K} < \delta$. Then the endpoints of the subintervals are

$$x_k = \frac{k}{K}, \quad k = 0, 1, \dots, K.$$

By the given condition, for each $k \in \{0, 1, \dots, K\}$, since $x_k \in [0, 1]$, $\lim_{n \rightarrow \infty} f(x_k + n) = 0$. Thus, there exists $N_k \in \mathbb{N}$ such that

$$\left| f\left(\frac{k}{K} + n\right) \right| < \frac{\varepsilon}{2} \quad \text{for all } n \geq N_k.$$

Let $L = \max\{N_0, N_1, \dots, N_K\}$. We will show that this is the desired K , namely for all $x \geq L$, $|f(x)| < \varepsilon$. For any $x \geq L$, write

$$x = \lfloor x \rfloor + \{x\},$$

where $\lfloor x \rfloor$ is the integer part, $\{x\}$ is the fractional part. Since the points $\{\frac{k}{K}\}_{k=0}^K$ partition $[0, 1]$ into sub-intervals of length $\frac{1}{K} < \delta$, then $\{x\}$ lies in one of the subintervals. Namely, there exists some $k \in \{0, 1, \dots, K-1\}$ such that:

$$\frac{k}{K} \leq \{x\} < \frac{k+1}{K}.$$

This implies $\left| \{x\} - \frac{k}{K} \right| < \frac{1}{K} < \delta$. By definition, $\lfloor x \rfloor \geq N_k$, so

$$\left| f\left(\frac{k}{K} + \lfloor x \rfloor\right) \right| < \frac{\varepsilon}{2}.$$

By triangle inequality,

$$|f(x)| \leq \left| f(x) - f\left(\lfloor x \rfloor + \frac{k}{K}\right) \right| + \left| f\left(\lfloor x \rfloor + \frac{k}{K}\right) \right| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, $\lim_{x \rightarrow +\infty} f(x) = 0$. □

A harder problem is as follows: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Suppose that for every $\delta > 0$, $\lim_{n \rightarrow \infty} f(n\delta) = 0$. Prove that $\lim_{x \rightarrow +\infty} f(x) = 0$.

Problem A.9.7. Let $\{a_n\}$ be a sequence such that $S = \sum_{n=1}^{\infty} a_n$ converges. Suppose the sequence of partial sums $\{S_n = \sum_{k=1}^n a_k\}$ satisfies that

$$|S - S_n| \leq \frac{1}{n} \quad \text{for any } n \in \mathbb{N}.$$

Prove that $\sum_{n=1}^{\infty} a_n \log n$ converges.

Remark. We cannot apply Abel/Dirichlet's test here.

Solution. Let R_n denote the remainder of the series S , defined by $R_n = S - S_n = \sum_{k=n+1}^{\infty} a_k$. By the given condition, $|R_n| \leq \frac{1}{n}$. We also note that $a_n = R_{n-1} - R_n$ for $n > 1$.

To show that $\sum a_n \log n$ converges, we use the Cauchy criterion. Consider the partial sum from $n+1$ to m . By Abel's summation formula,

$$\begin{aligned} \left| \sum_{k=n+1}^m a_k \log k \right| &= \left| \sum_{k=n+1}^m (R_{k-1} - R_k) \log k \right| \\ &= \left| R_n \log(n+1) - R_m \log m + \sum_{k=n+1}^{m-1} R_k (\log(k+1) - \log k) \right| \\ &\leq |R_n| \log(n+1) + |R_m| \log m + \sum_{k=n+1}^{m-1} |R_k| |\log(k+1) - \log k| \\ &\leq \frac{\log(n+1)}{n} + \frac{\log m}{m} + \sum_{k=n+1}^{m-1} \frac{1}{k} \log \left(1 + \frac{1}{k} \right) \\ &\leq \frac{\log(n+1)}{n} + \frac{\log m}{m} + \sum_{k=n+1}^{m-1} \frac{1}{k^2}, \end{aligned}$$

where we used the inequality $\log(k+1) - \log k = \log\left(1 + \frac{1}{k}\right) \leq \frac{1}{k}$ in the last line above.

Since $\frac{\log(n+1)}{n} \rightarrow 0$, there exists N_1 such that

$$\frac{\log(n+1)}{n} < \frac{\varepsilon}{3} \quad \text{for all } n \geq N.$$

Since $\frac{\log m}{m} \rightarrow 0$, there exists N_2 such that

$$\frac{\log m}{m} < \frac{\varepsilon}{3} \quad \text{for all } m \geq N.$$

Since $\sum \frac{1}{n^2}$ converges, by Cauchy criterion, there exists N_3 such that

$$\sum_{k=n+1}^{m-1} \frac{1}{k^2} < \frac{\varepsilon}{3} \quad \text{for all } m > n \geq N.$$

Take $N = \max\{N_1, N_2, N_3\}$. Then for all $m > n \geq N$,

$$\left| \sum_{k=n+1}^m a_k \log k \right| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

By the Cauchy Criterion, the series $\sum_{n=1}^{\infty} a_n \log n$ converges. □

Problem A.9.8. Let $f: [0, 1] \rightarrow [0, 1]$ be an increasing function. Prove that there exists $x_0 \in [0, 1]$ such that

$$f(x_0) = x_0.$$

Remark. We cannot use IVT, since f may not be continuous.

Solution. The idea is to construct a sequence of nested intervals such that their intersection is precisely x_0 .

Let $I_1 = [0, 1]$. If $f\left(\frac{1}{2}\right) \geq \frac{1}{2}$, set $I_2 = \left[\frac{1}{2}, 1\right]$; if $f\left(\frac{1}{2}\right) < \frac{1}{2}$, set $I_2 = \left[0, \frac{1}{2}\right]$. Inductively, suppose $I_n = [a_n, b_n]$ is defined. Let $m_n = \frac{a_n + b_n}{2}$. If $f(m_n) \geq m_n$, set $I_{n+1} = [m_n, b_n]$; if $f(m_n) < m_n$, set $I_{n+1} = [a_n, m_n]$.

This produces a nested decreasing sequence of closed intervals $\{I_n\}$. Since $|I_n| = 2^{1-n} \rightarrow 0$, by nested interval theorem, $\bigcap_{n=1}^{\infty} I_n = \{x_0\}$. Consequently, we have $\lim_{n \rightarrow \infty} a_n = x_0$ and $\lim_{n \rightarrow \infty} b_n = x_0$, with $a_n \nearrow x_0$ and

$b_n \searrow x_0$.

Claim: $f(x_0) = x_0$.

Since f is a monotone function, the one-sided limits always exist. By sequential criterion, since $a_n \nearrow x_0$, the left limit equals

$$\lim_{n \rightarrow \infty} f(a_n) = \lim_{x \rightarrow x_0^-} f(x) \leq f(x_0).$$

Note that by construction, we have $f(a_n) \geq a_n$ for all $n \in \mathbb{N}$, so

$$\lim_{n \rightarrow \infty} f(a_n) \geq \lim_{n \rightarrow \infty} a_n = x_0.$$

Hence, $f(x_0) \geq x_0$. Similarly, one can show that $f(x_0) \leq x_0$. Therefore, $f(x_0) = x_0$, as desired. $\square$

Alternative solution. Assume that $f(0) > 0$ and $f(1) < 1$ (for otherwise we are done). Then

$$x_0 := \sup\{x \in [0, 1] : f(x) > x\}$$

belongs to the open interval $(0, 1)$.

Claim: $f(x_0) = x_0$.

Since, by definition of x_0 , $f(x) \leq x$ for every $x \in (x_0, 1)$, by monotonicity it holds

$$f(x_0) \leq f(x) \leq x, \quad \text{for all } x \in (x_0, 1),$$

hence $f(x_0) \leq x_0$.

On the other hand, by definition of $\sup$, for every (small) $\varepsilon > 0$, $x_0 - \varepsilon$ is not an upper bound for the set above. Thus, there exists $x_\varepsilon \in (x_0 - \varepsilon, x_0)$ such that $f(x_\varepsilon) > x_\varepsilon$, so that

$$x_\varepsilon < f(x_\varepsilon) \leq f(x_0)$$

and, finally, $x_0 \leq f(x_0)$. $\square$

Problem A.9.9. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying that there exists a constant $M > 0$ such that for any $x, y \in \mathbb{R}$,

$$|xf(y) - yf(x)| \leq M(|x| + |y|).$$

Prove that $\lim_{x \rightarrow \infty} \frac{f(x)}{x}$ exists.

Solution. We are given the condition

$$\left| \frac{f(y)}{y} - \frac{f(x)}{x} \right| \leq M \left(\frac{1}{|x|} + \frac{1}{|y|} \right).$$

This is like a Cauchy condition.

Let $\{a_n\}$ be a sequence such that $a_n \rightarrow +\infty$, a_n increasing. Then $\left\{ \frac{f(a_n)}{a_n} \right\}$ is a Cauchy sequence. Let $L = \lim_{n \rightarrow \infty} \frac{f(a_n)}{a_n}$, which is a candidate for the limit. We will prove that $\lim_{x \rightarrow \infty} \frac{f(x)}{x} = L$.

Let $\varepsilon > 0$. Choose N large enough such that $\frac{2M}{N} < \frac{\varepsilon}{2}$. For x such that $|x| > N$, we want to show $\left| \frac{f(x)}{x} - L \right| < \varepsilon$.

$$\left| \frac{f(x)}{x} - L \right| \leq \left| \frac{f(x)}{x} - \frac{f(a_n)}{a_n} \right| + \left| \frac{f(a_n)}{a_n} - L \right|$$

second term is small. for first term, choose n large enough such that $\left| \frac{f(a_n)}{a_n} - L \right| < \frac{\varepsilon}{2}$, $|a_n| > N$.

Then above $\leq \frac{2M}{N} + \frac{\varepsilon}{2} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. $\square$

Problem A.9.10. Let $f: \mathbb{R} \rightarrow \mathbb{Q}$ be a continuous function. Prove that f is a constant function.

Solution. For a contradiction, assume f is not a constant function. Then there exist $y_1 < y_2$ such that $y_1, y_2 \in f(\mathbb{R})$. By IVT, $[y_1, y_2] \subseteq f(\mathbb{R})$. But there exists an irrational number $y \in [y_1, y_2]$. This contradicts the fact that $f(\mathbb{R}) \subseteq \mathbb{Q}$. $\square$

B Extra Problems

Selected problems from past year papers, Berkeley Problems in Mathematics, and online sources.

B.1 Preliminaries

B.2 Real and Complex Number Systems

B.3 Sequences and Series

Problem B.3.1 (MA2108S AY23/24). For each of the following statements, first answer whether it is true or false. If it is true, prove it. If it is false, find a counter example.

- (a) Let $\{a_n\}$ be a sequence in $\mathbb{R}$. Suppose that $\sum a_n$ is convergent. Then $\sum a_n^2$ is also convergent.
- (b) Let $\{a_n\}$ and $\{b_n\}$ be convergent sequences in $\mathbb{R}$. Suppose that $\lim_{n \rightarrow \infty} a_n \geq \lim_{n \rightarrow \infty} b_n$. Then there is an integer N such that

$$a_n \geq b_n \quad \text{for all } n \geq N.$$

Solution.

- (a) False. Let $a_n = \frac{(-1)^n}{\sqrt{n}}$. By Leibniz test, $\sum a_n$ converges, but $\sum a_n^2 = \sum \frac{1}{n}$ diverges.
- (b) False. Let $a_n = \frac{1}{2n}$ and $b_n = \frac{1}{n}$. Then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = 0$, but $a_n < b_n$ for all $n \in \mathbb{N}$.

□

Problem B.3.2 (MA2108S AY23/24). Determine whether the following series are convergent. Justify your answer.

- (a) $\sum \frac{1}{n^2} \left(2 + \frac{3}{n}\right)$
- (b) $\sum \frac{2^n}{n^n}$
- (c) $\sum \cos\left(\frac{1}{n}\right)$

Solution.

- (a) Apply Abel's Test. Let $a_n = \frac{1}{n^2}$, $b_n = 2 + \frac{3}{n}$. Then $\sum a_n = \frac{\pi^2}{6}$, and $2 + \frac{3}{n}$ is decreasing and bounded.
- (b) Apply Root Test. Let $a_n = \frac{2^n}{n^n}$. Then $|a_n|^{1/n} = \frac{2}{n} \leq \frac{2}{3} < 1$ for $n \geq 3$.
- (c) Apply Divergence Test. By continuity, $\lim_{n \rightarrow \infty} \cos\left(\frac{1}{n}\right) = \cos\left(\lim_{n \rightarrow \infty} \frac{1}{n}\right) = \cos(0) = 1 \neq 0$.

□

Problem B.3.3 (MA2108S AY22/23).

- (a) Let $\{a_n\}$ be a sequence of real numbers. Suppose that $\sum a_n$ is absolutely convergent. For a real number $p \geq 1$, show that $\sum a_n^p$ is also absolutely convergent.
- (b) Suppose that $\sum a_n$ is convergent. Is $\sum a_n^2$ necessarily convergent? If it has to be convergent, prove the statement. Otherwise, find a counterexample.

Solution.

- (a) Since $\sum a_n$ is absolutely convergent, by Divergence Test, $|a_n| \rightarrow 0$. Thus, there exists $N \in \mathbb{N}$ such that for $n \geq N$,

$$|a_n| < 1 \implies |a_n^p| = |a_n|^p \leq |a_n|.$$

By Comparison Test, $\sum a_n^p$ is absolutely convergent.

(b) False. See above for counterexample.

□

Problem B.3.4 (MA2108S AY21/22). Let (a_n) and (b_n) be sequences in $\mathbb{R}$ defined recursively by

$$a_1 = 3, \quad b_1 = 2, \quad a_{n+1} = a_n + 2b_n, \quad b_{n+1} = a_n + b_n \quad \text{for all } n \in \mathbb{N}.$$

Moreover, for each $n \in \mathbb{N}$, let $c_n = \frac{a_n}{b_n}$.

(a) Show that for each $n \in \mathbb{N}$, one has

$$\left| c_{n+1} - \sqrt{2} \right| < \frac{1}{2} \left| c_n - \sqrt{2} \right|.$$

(b) Show that the sequence (c_n) converges in $\mathbb{R}$, and determine the value of $\lim_{n \rightarrow \infty} c_n$.

Problem B.3.5. Compute

$$L = \lim_{n \rightarrow \infty} \left(\frac{n^n}{n!} \right)^{1/n}.$$

Solution. Let $a_1 = 1$, $a_2 = (2/1)^2$, $a_3 = (3/2)^3$, ..., $a_n = (n/n(n-1))^n$. Then

$$\frac{n^n}{n!} = \frac{a_1 a_2 \cdots a_n}{n}.$$

Since $a_n \rightarrow e$, we have $\lim_{n \rightarrow \infty} (n^n/n!)^{1/n} = e$ as well (using the fact that $n^{1/n} \rightarrow 1$).

□

Problem B.3.6. Let $x_1 = 1$ and

$$x_{n+1} = \frac{3 + 2x_n}{3 + x_n}$$

for $n \geq 1$. Prove that $\{x_n\}$ converges, and find its limit.

Solution. Show $\{x_n\}$ increasing and bounded above (by 2), then apply MCT.

□

Problem B.3.7. Let $\alpha \in (0, 1)$. Prove that any sequence $\{x_n\}$ of real numbers satisfying the recurrence relation

$$x_{n+1} = \alpha x_n + (1 - \alpha)x_{n-1}$$

converges, and find an expression for the limit in terms of α , x_0 and x_1 .

Solution. Since this recurrence is linear, consider the differences between consecutive terms. Let $d_n = x_n - x_{n-1}$. By the given recurrence,

$$d_{n+1} = (\alpha x_n + (1 - \alpha)x_{n-1}) - x_n = (1 - \alpha)(x_{n-1} - x_n) = (\alpha - 1)(x_n - x_{n-1}) = (\alpha - 1)d_n.$$

This gives a geometric progression $d_n = d_1(\alpha - 1)^{n-1}$. Now write x_n as $x_n = x_0 + \sum_{k=1}^n d_k$, and the geometric series converges.

□

Problem B.3.8. Show that the series

$$\sum_{n=0}^{\infty} \frac{(n!)^2}{(2n)!}$$

converges.

Solution. Let $a_n = \frac{(n!)^2}{(2n)!}$. To prove convergence, use the Ratio Test:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{((n+1)!)^2}{(2(n+1))!} \cdot \frac{(2n)!}{(n!)^2} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{(n+1)!(n+1)!}{(2n+2)!} \cdot \frac{(2n)!}{(n!)(n!)} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+1)}{(2n+1)(2n+2)} \right|. \end{aligned}$$

As you notice, the leading degree of the denominator is the same as the leading degree of the numerator. Therefore, we look at the coefficients, and so the limit goes to $\frac{1}{4}$, and the series converges absolutely by the Ratio Test. $\square$

Problem B.3.9. Determine the convergence of the series

$$\sum_{n=1}^{\infty} n^{-1/2} \sin(n) \sin(n^2).$$

Solution. We can use the Dirichlet test; here, since

$$\sin(n) \sin(n^2) = \frac{1}{2} (\cos n(n-1) - \cos n(n+1)),$$

we have a telescopic sum

$$\sum_{n=1}^N \sin(n) \sin(n^2) = \frac{1}{2} - \frac{1}{2} \cos N(N) = \sin^2 \left(\frac{N(N+1)}{2} \right)$$

that does not exceed 1 in absolute value. $\square$

Problem B.3.10. Let $k \in \mathbb{N}$ and let $x_1 \geq x_2 \geq \cdots \geq x_k \geq 0$. Evaluate

$$\lim_{n \rightarrow \infty} (x_1^n + \cdots + x_k^n)^{1/n}.$$

Solution. Squeeze

$$\max\{x_i\}^n \leq x_1^n + \cdots + x_k^n \leq k \max\{x_i\}^n,$$

so that

$$\max\{x_i\} \leq (x_1^n + \cdots + x_k^n)^{1/n} \leq k^{1/n} \max\{x_i\}.$$

Then take $n \rightarrow \infty$ and squeeze. $\square$

Problem B.3.11. If $\{a_n\}$ is a sequence of non-negative real numbers satisfying

$$a_{n+1} \leq a_n + \frac{1}{n^2} \quad \text{for all } n \in \mathbb{N},$$

prove that $\{a_n\}$ converges.

Solution. Let $n > 0$. For $m \geq n$, we have

$$a_m \leq a_n + \sum_{k=n}^{m-1} \frac{1}{k^2} \leq a_n + \xi_n,$$

where $\xi_n = \sum_{k=n}^{\infty} \frac{1}{k^2}$. Taking limsup with respect to m , we have

$$x_n \geq \limsup_{m \rightarrow \infty} a_m - \xi_n.$$

Since $\sum \frac{1}{k^2}$ converges, we have $\xi_n \rightarrow 0$. Then take $\liminf$ with respect to n , which gives

$$\liminf_{n \rightarrow \infty} x_n \geq \limsup_{m \rightarrow \infty} x_m - \liminf_{n \rightarrow \infty} \xi_n \geq \limsup_{m \rightarrow \infty} x_m.$$

The reverse inequality also holds, so $\lim x_n$ exists. $\square$

Problem B.3.12. Let A be the set of positive integers that do not contain the digit 9 in their decimal expansions. Prove that

$$\sum_{a \in A} \frac{1}{a} < \infty;$$

that is, A defines a convergent subseries of the harmonic series.

Solution. Group terms by the number of digits. For $n = 1, 2, \dots$, the number of terms in A that are less than 10^n is $9^n - 1$, so we have

$$\begin{aligned} \sum_{a \in A} \frac{1}{a} &= \sum_{n \geq 1} \sum_{\substack{10^{n-1} \leq a < 10^n \\ a \in A}} \frac{1}{a} \\ &\leq \sum_{n \geq 1} \frac{9^n}{10^n - 1} \\ &= 10 \sum_{n \geq 1} \left(\frac{9}{10} \right)^n < \infty. \end{aligned}$$

$\square$

B.4 Continuity

Problem B.4.1. Let $f, g: [0, 1] \rightarrow [0, \infty)$ be continuous functions satisfying

$$\sup_{0 \leq x \leq 1} f(x) = \sup_{0 \leq x \leq 1} g(x).$$

Prove that there exists $t \in [0, 1]$ such that $f(t)^2 + 3f(t) = g(t)^2 + 3g(t)$.

Solution. The given expression $f(t)^2 + 3f(t) = g(t)^2 + 3g(t)$ is not really relevant; it suffices to find $t \in [0, 1]$ such that $f(t) = g(t)$.

Denote the common supremum of f and g by M . By extreme value theorem, there exist $\alpha, \beta \in [0, 1]$ such that $f(\alpha) = g(\beta) = M$.

Consider the function $h(x) = f(x) - g(x)$ on $[\alpha, \beta]$. Since $h(\alpha) = M - g(\alpha) \geq 0$, $h(\beta) = f(\beta) - M \leq 0$, by IVT, h has a zero $t \in [\alpha, \beta]$. $\square$

Problem B.4.2.

- (1) Prove that there is no continuous map from $[0, 1]$ onto $(0, 1)$.
- (2) Find a continuous surjective map from $(0, 1)$ onto $[0, 1]$.
- (3) Prove that no map in part (2) can be bijective.

Solution.

- (1) Easy.
- (2) $|\sin 2\pi x|$
- (3) For a contradiction, suppose $g: (0, 1) \rightarrow [0, 1]$ is a continuous bijection. Let $x_0 = g^{-1}(0)$ and $x_1 = g^{-1}(1)$. WLOG assume $x_0 < x_1$. By IVT, we have $g([x_0, x_1]) = [0, 1]$. As $x_0, x_1 \in (0, 1)$, g is not an injection, which contradicts our assumption.

□

Problem B.4.3. A function $f: [0, 1] \rightarrow \mathbb{R}$ is said to be *upper semicontinuous* if given $x \in [0, 1]$, for any $\varepsilon > 0$, there exists $\delta > 0$ such that if $|y - x| < \delta$, then $f(y) < f(x) + \varepsilon$. Prove that an upper semicontinuous function f on $[0, 1]$ is bounded above and attains its maximum value.

Solution. Fix $\varepsilon > 0$. For each $x \in [0, 1]$, let δ_x be as in the hypothesis and $I_x = (x - \delta_x, x + \delta_x)$. The open intervals $\{I_x\}$ is an open cover of $[0, 1]$. Since $[0, 1]$ is compact, we can choose a finite subcover

$$[0, 1] \subseteq I_{x_1} \cup \cdots \cup I_{x_n}.$$

Choose $M = \max\{f(x_i) + \varepsilon\}$. If $x \in [0, 1]$, then $f(x) < M$, so f is bounded from above.

Let $N = \sup_{x \in [0, 1]} f(x)$. Then there is a sequence of points $\{x_n\}$ such that $\{f(x_n)\}$ tends to N from below¹. Since $\{x_n\}$ is bounded, by Bolzano-Weierstrass, it has a convergent subsequence, so (by passing to a subsequence) we may assume that $\{x_n\}$ converges to some $x^* \in [0, 1]$.

We claim that $f(x^*) = N$. Evidently $f(x^*) \leq N$, so it remains to show that $f(x^*) \geq N$.

By the upper semicontinuity of f and the convergence of $\{f(x_n)\}$, we have, for n sufficiently large, $f(x_n) < f(x^*) + \varepsilon$ and $N < f(x_n) + \varepsilon$. Combining these, we get $f(x^*) \leq N < f(x^*) + 2\varepsilon$. □

B.5 Basic Topology

Problem B.5.1. Let E be a subset of a metric space X . Let $L(E)$ denote the set of limit points of E . Let $L^2(E) = L(L(E))$, $L^3(E) = L(L(L(E)))$, and so on. Let $L^0(E) = E$. The set $L^n(E)$ is called the n -th derived set of E . Show that for every $n \in \mathbb{N}$, there exists $E \subseteq \mathbb{R}$ such that $L^{n-1}(E) \neq \emptyset$ but $L^n(E) = \emptyset$.

Solution. We start small. If $n = 1$, take $E_1 = \{0\}$. Then $L^0(E_1) \neq \emptyset$ but $L^1(E_1) = \emptyset$, since a singleton has no limit points.

For $n = 2$, let $E_2 = \{\frac{1}{k} : k \in \mathbb{N}\}$, so that $L(E_2) = \{0\}$. The idea is that we construct such sets “layer by layer”, so that each round of taking limit points removes exactly one level.

We now define recursively

$$E_{n+1} = \bigcup_{k=1}^{\infty} \left(E_n + \frac{1}{k} \right),$$

so that $L(E_{n+1}) = E_n$. □

Problem B.5.2. Prove that a finite union of compact sets is compact, but that a countable union need not be.

Solution. Let $K_n = [\frac{1}{n}, 1]$ for $n \geq 1$. Each K_n is compact, but their countable union is $(0, 1]$, which is not compact by Heine-Borel, because it is not closed. □

Problem B.5.3. Let K be a compact metric space, and $\{U_\alpha\}_{\alpha \in \mathcal{A}}$ an open cover of K . Prove that there exists $\varepsilon > 0$ such that for every $x \in K$, there exists $\alpha \in \mathcal{A}$ such that $V_\varepsilon(x) \subseteq U_\alpha$.

Solution. For a contradiction, suppose that no such $\varepsilon > 0$ exists. For each $n \in \mathbb{N}$, take $\varepsilon = \frac{1}{n}$. Then there exists a point $x_n \in K$ such that $V_{1/n}(x_n) \not\subseteq U_\alpha$ for every $\alpha \in \mathcal{A}$.

Since K is compact, K is sequentially compact, so the sequence $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$ that converges to some $x \in K$.

Since $\{U_\alpha\}$ covers K , there exists $\alpha_0 \in \mathcal{A}$ such that $x \in U_{\alpha_0}$. Since U_{α_0} is open, there exists $m \in \mathbb{N}$ such that $V_{2/m}(x) \subseteq U_{\alpha_0}$.

Since $x_{n_k} \rightarrow x$, there exists $K \in \mathbb{N}$ such that $d(x_{n_k}, x) < \frac{1}{m}$ for all $k \geq K$, or equivalently, $x_{n_k} \in V_{1/m}(x)$.

¹For every $n \in \mathbb{N}$, $N - \frac{1}{n}$ is not an upper bound, so there exists $x_n \in [0, 1]$ such that $f(x_n) > N - \frac{1}{n}$. Thus, $N - \frac{1}{n} < f(x_n) \leq N$. Taking $n \rightarrow \infty$, we see that $f(x_n) \rightarrow N$ from below.

Now, choose k large enough such that $k \geq K$ and $n_k \geq m$. Let $s = n_k$. We claim that $V_{1/s}(x_s) \subseteq V_{2/m}(x)$. To see this, let $y \in V_{1/s}(x_s)$. Then $d(y, x_s) < 1/s$. By triangle inequality,

$$d(y, x) \leq d(y, x_s) + d(x_s, x) < \frac{1}{s} + \frac{1}{m} \leq \frac{1}{m} + \frac{1}{m} = \frac{2}{m}.$$

Thus, $y \in V_{2/m}(x)$, which implies $V_{1/s}(x_s) \subseteq V_{2/m}(x)$. Since $V_{2/m}(x) \subseteq U_{a_0}$, we have

$$V_{1/s}(x_s) \subset U_{a_0}.$$

This contradicts our initial assumption that $V_{1/n}(x_n)$ is not contained in any U_α for any n . Therefore, such an ε must exist. $\square$

Problem B.5.4. Let X be a metric space and $\{x_n\}$ a convergent sequence in X with limit x_0 . Prove that the set $C = \{x_0, x_1, x_2, \dots\}$ is compact.

Solution. Let $\mathcal{U}$ be an open cover of C . Then there is a set U_0 in $\mathcal{U}$ that contains x_0 .

Since $\lim_{n \rightarrow \infty} x_n = x_0$, there exists $N \in \mathbb{N}$ such that $x_n \in U_0$ for all $n \geq N$. For each $n < N$, there exist $U_n \in \mathcal{U}$ such that $x_n \in U_n$.

Then $\{U_0, U_1, \dots, U_N\}$ is a finite subcover of C . This proves that C is compact. $\square$

Problem B.5.5. Let X be a compact metric space and $f: X \rightarrow X$ an isometry. Show that $f(X) = X$.

Solution. For any $x_0 \in X$, consider the sequence $\{x_n\}$ defined by $x_{n+1} = f(x_n)$. Since X is sequentially compact, let $x_n \rightarrow x$.

Let $\varepsilon > 0$. For some $n, m \in \mathbb{N}$ with $n < m$, we have $d(x_n, x) < \frac{\varepsilon}{2}$ and $d(x_m, x) < \frac{\varepsilon}{2}$, so $d(x_n, x_m) < \varepsilon$.

But then $d(x_0, f(X)) \leq d(x_0, x_{m-n}) = d(x_n, x_m) < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, we conclude $d(x_0, f(X)) = 0$, i.e., $x_0 \in f(X)$. $\square$

B.6 Sequences and Series of Functions

Problem B.6.1. Suppose that $\{f_n\}$ is a sequence of non-decreasing functions which map the unit interval into itself. Suppose that

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)$$

pointwise and that f is a continuous function. Prove that $f_n \rightarrow f$ uniformly on $[0, 1]$. Note that the functions f_n are not necessarily continuous.

Solution. Let $\varepsilon > 0$. Since f is uniformly continuous, there exists $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Now divide $[0, 1]$ into N subintervals of length $< \delta$. That is, take N satisfying $1/N < \delta$. For $0 \leq k \leq N$, let $\xi_k = \frac{k}{N}$, and divide $[0, 1]$ into subintervals $[\xi_{k-1}, \xi_k]$, $1 \leq k \leq N$.

Since $f_n \rightarrow f$ pointwise, there exists M such that if $n \geq M$, then

$$|f_n(\xi_k) - f(\xi_k)| < \varepsilon$$

for every $0 \leq k \leq N$. Since each f_n is non-decreasing, we have, for $x \in [\xi_{k-1}, \xi_k]$,

$$f(\xi_{k-1}) - \varepsilon < f_n(x) < f(\xi_k) + \varepsilon,$$

or

$$|f_n(x) - f(\xi_{k-1})| < 2\varepsilon.$$

Therefore,

$$|f_n(x) - f(x)| \leq |f_n(x) - f(\xi_{k-1})| + |f(\xi_{k-1}) - f(x)| < 3\varepsilon.$$

Since this bound does not depend on x , the convergence is uniform.

□